

# Distinct multiplicity values in integer partitions

(OEIS A373271 and A373273)

**Part I:** the number of distinct multiplicity values, matched asymptotics and a shrinking inverse

**Part II:** the sum of distinct multiplicity values, five asymptotic scales and a shrinking inverse

**Part III:** fluctuations, Gaussian and non-Gaussian limits and a logarithmic window

Two research reports of one external research session  
(bundle Reports 197 and 198) and a continuation manuscript,  
bound into one ProveIt report

Parts I and II: 4 October 2026    Part III: 7 October 2026    merged: 7 October 2026

## Abstract

For a partition  $\lambda = (1^{M_1} 2^{M_2} \dots)$  of  $n$ , consider the set of positive values taken by the multiplicities  $M_j$ . Part I counts this set:  $a(n) = \sum_{\lambda \vdash n} D(\lambda)$  is OEIS A373271, and it proves  $a(n)/p(n) = (6n)^{1/4} - \frac{23}{16} + Kn^{-1/4} + O(n^{-1/2})$  with an explicit  $K$ , the constant  $-\frac{23}{16}$  needing an exact telescoping tail  $-\frac{1}{2}$  that a local expansion misses; it derives two relative corrections for  $a(n)$ , eventual strict increase and a two-ceiling inverse whose width before rounding tends to zero. Part II sums the set:  $a_n = \sum_{\lambda \vdash n} W(\lambda)$  is OEIS A373273, and it proves an expansion of  $a_n/p(n)$  through the five scales  $\sqrt{n} \log n$ ,  $\sqrt{n}$ ,  $n^{1/4}$ ,  $\log n$ ,  $1$  with error  $O(n^{-1/4})$ , the leading term being  $\sqrt{6n} \log n / (4\pi)$ ; there the square of a nonlocal correction and a finite-difference tail enter the constant, and zeta derivatives cancel. Both Parts rest on one exact identity,  $\sum_n a_n q^n = P(q) \sum_{m \geq 1} w(m)(1 - Q_m(q))$  with  $w = 1$  and  $w(m) = m$ , on a whole-circle bound for positive products that forbid one multiplicity, and on local saddle transfer. Part II answers the case  $w(m) = m$  of Part I's question on weighted occupied values. Part III, a continuation manuscript added on 7 October 2026, studies  $D_\alpha = \sum_m m^\alpha \mathbf{1}_{\{N_m > 0\}}$  for every  $\alpha \geq 0$ , so that  $D_0 = D$  and  $D_1 = W$ : Gaussian fluctuations with an explicit covariance kernel for  $\alpha < \frac{1}{2}$ , in particular  $\text{Var}_n D \sim v_0 n^{1/4}$  with  $v_0 = 6^{1/4}(\sqrt{2} - \frac{3}{4} - \frac{3\sqrt{2}}{8} \log(1 + \sqrt{2})) = 0.3080002444\dots$  and a central limit theorem, which answers Part I's variance question; a factor  $\log n$  in the variance at  $\alpha = \frac{1}{2}$ , with a Gaussian field in a logarithmic window of exponents; non-Gaussian series of exponentials above  $\frac{1}{2}$ , among them a Gumbel law for  $W$  with  $\text{Var}_n W \sim n$ ; mutual independence of the three regimes; and the leading means of every  $D_\alpha$ . Mean expansions of the precision of Parts I and II remain open for other weights. The write adds the OEIS entries, the b-file of A373271 that Part I could not open (all 200 terms agree), a comparison of the inverses with the repository's transseries volume, and, for Part III, recomputations of its constants and tables and of the geometric-law means against Parts I and II. The report is unrefereed, and nothing in it is formalized.

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## Guide to this report

Parts I and II are two manuscripts of the session bundle of Reports 1–243 (arrival commit [60f54ea06](#)), placed in this directory by commit [d451ef3d8](#) (batch 111); Part III, a later continuation manuscript, was placed by commit [9a10a617f](#) (below).

| Part | Bundle report | Delivered title                                                                                                                                                              | Labels                |
|------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| I    | 197 (base)    | Distinct positive multiplicity values in integer partitions: Matched asymptotics and a shrinking inverse for OEIS A373271 (4 October 2026)                                   | <code>dmv:cnt:</code> |
| II   | 198           | The sum of distinct multiplicity values in integer partitions: Five asymptotic scales and a shrinking inverse for OEIS A373273 (4 October 2026)                              | <code>dmv:sum:</code> |
| III  | none          | Fluctuations of Distinct Multiplicities in Random Integer Partitions: Exact variance, Gaussian and non-Gaussian limits, and a logarithmic transition window (7 October 2026) | <code>dmv:fl:</code>  |

*One identity, two weights.* Write  $N_m(\lambda)$  for the number of part sizes of  $\lambda$  with multiplicity exactly  $m$ . Part I studies  $D = \sum_{m \geq 1} \mathbf{1}_{\{N_m > 0\}}$  and Part II studies  $W = \sum_{m \geq 1} m \mathbf{1}_{\{N_m > 0\}}$ . Both start from the same exact identity,

$$\sum_{n \geq 0} a_n q^n = P(q) \sum_{m \geq 1} w(m)(1 - Q_m(q)), \quad Q_m(q) = \prod_{j \geq 1} (1 - (1 - q^j)q^{mj}),$$

with  $w \equiv 1$  in Proposition 2.1 and  $w(m) = m$  in (13.3), and both control the coefficient circle through the positive products  $P_m = PQ_m$  that forbid one multiplicity. The OEIS records the link: A373271 comments that summing the distinct multiplicities instead gives A373273 (Remark 1.4).

|                  | Part I (Report 197)                                                                 | Part II (Report 198)                                                                                                                           |
|------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Statistic        | $D$ , the number of distinct multiplicity values (A373271)                          | $W$ , their sum (A373273)                                                                                                                      |
| Mean             | $(6n)^{1/4} - \frac{23}{16} + Kn^{-1/4} + O(n^{-1/2})$                              | five scales $\sqrt{n} \log n, \sqrt{n}, n^{1/4}, \log n, 1$ ; error $O(n^{-1/4})$                                                              |
| Radial expansion | $\sqrt{\pi} t^{-1/2} - \frac{23}{16} + \sqrt{\pi} d t^{1/2} + O( t )$ (Theorem 5.1) | $(\frac{1}{2} \log \frac{1}{t} + c)/t - \frac{7\sqrt{\pi}}{16\sqrt{t}} + \frac{1}{24} \log \frac{1}{t} + K + O(\sqrt{ t })$ (Proposition 18.1) |
| Decisive tail    | the telescoping $\sum \delta_m = -\frac{1}{2}$                                      | the delta square $E$ and the finite-difference tail $-\frac{1}{16}$ ; $\zeta'(-1)$ cancels                                                     |
| Tools            | matched sums, Euler–Maclaurin profiles                                              | the same, plus Mellin continuation and convergent regularization                                                                               |
| Transfer         | $J_s/p(n)$ for real $s$ (Lemma 7.1)                                                 | the same formula, analytic in $s$ , differentiated for logarithms                                                                              |
| Inverse          | two ceilings about $Z(Y)$ , width $O(x_0^{-1/2})$                                   | two ceilings about $x_Y^2 + \frac{1}{24}$ , width $O((\log Y)^{-1/2} / \log \log Y)$                                                           |
| Exact range      | $n \leq 2000$                                                                       | $n \leq 2500$                                                                                                                                  |

*Why one report, and how the Parts relate.* The two manuscripts treat one family of statistics by one method. Report 198’s package was adapted from Report 197’s (its README: the infrastructure “adapted from the preceding Report197 reproducibility package”), and Report 198 answers, for the weight  $w(m) = m$ , the question on weighted occupied values that Part I leaves open (Section 11,

Question 4); the general weight  $m^\alpha$  stays open (Part II, Question 2), and dated notes in both question lists say so. **Neither Part uses a theorem of the other.** Report 198’s README states that “No A373271 theorem or coefficient is used as a mathematical premise”, and the write confirmed that Part II cites nothing of Part I. The constants differ, so neither result is a special case of the other.

*Part III, added 7 October 2026.* A third manuscript, *Fluctuations of Distinct Multiplicities in Random Integer Partitions* (7 October 2026, archive `ProveIt_Research_2026-10-07.zip`), is printed as Part III (Sections 25–34, labels `dmv:f1:`). It was written from Part I alone, as the continuation of Part I’s Questions 2 and 4, and studies  $D_\alpha = \sum_{m \geq 1} m^\alpha \mathbf{1}_{\{N_m > 0\}}$  for every  $\alpha \geq 0$ , so that  $D_0 = D$  of Part I and  $D_1 = W$  of Part II. It proves their fluctuation laws (Gaussian below  $\alpha = \frac{1}{2}$ , a logarithmic variance factor at  $\frac{1}{2}$  with a Gaussian window field, non-Gaussian series of exponentials above, mutually independent), the variance and central limit theorem for  $D$ , the Gumbel law of  $W$ , and the leading means of every  $D_\alpha$ . It thereby answers Part I’s Question 2 and Part II’s Question 4, and, for the weights  $m^\alpha$  as to fluctuations and leading means, Part I’s Question 4 and part of Part II’s Question 2; the sentence above saying that  $m^\alpha$  stays open, and the earlier notes in both question lists, were written before Part III arrived and are kept, with dated notes after them. Its means at  $\alpha = 0, 1$  re-derive, more weakly, terms of Parts I and II (Section 25.2). Its provenance, checks, non-claims and reading conventions are in its Section 25; it is numbered with the shift  $k \mapsto k + 25$  (Section 25.1).

*Arrangement and duplication.* The base of the merge is Report 197: its `Report197.tex` was staged as this `article.tex`, it arrived first, and Report 198 builds on its package; it is printed first, as Part I. Report 198 is printed second, as Part II. Part II re-derives, for its weighted sums, machinery that Part I also proves: the zero-free sector, the power-sum and small-multiplicity estimates (its Section 14 against Part I’s Sections 3 and 4), the whole-circle bound (Section 19 against Section 6) and the monomial transfer (Section 20 against Lemma 7.1). These are kept, each with a dated note naming Part I’s counterpart, because Part II’s later statements cite them by their own labels and because they are stated for the weighted sums (the weight changes the polynomial factor in the whole-circle bound and the transfer is extended to complex  $s$ ); no result is printed twice in the same form for the same object.

*Numbering.* Sections are numbered continuously; statements and equations are numbered within sections, as delivered. Part I keeps all its numbers. Report 198’s Section  $k$  is Section  $k + 11$  here; its statements and equations move with their sections, and its Table 1 keeps its number. The labels of the two Parts carry the prefixes `dmv:cnt:` and `dmv:sum:`; the labels added by the write carry the Part prefix, or `dmv:` for this Guide. The bibliography is merged at the end: the two OEIS entries are `oeisA373271` and `oeisA373273` (both delivered as `oeis`), and the four shared entries (Ralaivaosaona; Grabner, Knopfmacher and Wagner; Corteel, Pittel, Savage and Wilf; Lugo) are printed once, with each Part’s own note on what it inspected.

[write] **Provenance, added 7 October 2026, batch 111.** This report was built from two manuscripts. *Part I:* Report 197, archive `Report197_TeX_and_Reproducible_Code.zip` (793,611 bytes, SHA-256 `192aeca1a0b8...bc038875d653adc`, 14 files, no wrapper directory); text `Report197.tex` (759 lines, 18 pages as delivered). *Part II:* Report 198, archive `Report198_TeX_and_Reproducible_Code.zip` (909,536 bytes, SHA-256 `8d10255b48c3...8eacce24e871492`, 17 files, no wrapper directory); text `Report198.tex` (663 lines, 17 pages as delivered). Both are dated 4 October 2026, arrived in `60f54ea06` and were placed in `d451ef3d8`. Their author lines and PDF author fields read “Report 197” and “Report 198”: they name no person, tool or addressee; neither package carries a “prepared for private review” line, an e-mail address or personal data, and neither records a ProveIt commit. Each Part’s programs are shipped under `code/` and its data and recorded outputs under `data/`, with the prefixes `197-count-` and `198-sum-`; Report 197’s

delivered `README.txt` was staged as the report README and is replaced by it. Not shipped, and retrievable from the arrival commit: the two delivered PDFs, Report 198's `Report198.tex` and `README.txt` (the former is Part II of this file), and the four checksum manifests (Report 197: `MANIFEST.json` 13 entries, `SOURCE_MANIFEST.json` 7; Report 198: 16 and 9), all verified at the write. Where the merge had to choose: the base and the order of the Parts; the two OEIS citation keys; the merged bibliography; keeping Part II's re-derivations (above). The write prefixed the 89 labels of Report 197 and the 78 of Report 198 and updated the 50 and 44 references to them (36 and 38 `\eqref`, 14 and 6 `\ref`); it added this Guide, the two Part headers, Remarks 1.4, 9.1, 12.4 and 21.1, and the notes marked [write]. The added remarks are the last statements of their sections and the added displays are unnumbered, so every delivered number of Part I, and every delivered number of Part II up to the shift by eleven sections, is unchanged (label tables of builds of the two delivered texts and of this one compared).

*The sources, as the write read them* (7 October 2026). OEIS A373271 and its b-file, A373273 (in the internal format) and its synthesized b-file, and the comment lines of A373272 and A373269 (Remarks 1.4 and 12.4); the transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex` (p0:def:model, p0:def:core, p0:thm:lambert-core, p0:thm:lambert-centered, p0:prop:two-grid, p0:thm:backward-error, p0:def:three-inverses, p0:thm:staircase). Not read by the write: Ralaivaosaona, Grabner, Knopfmacher and Wagner, Corteel, Pittel, Savage and Wilf, Lugo, Archibald et al., Fill, Janson and Ward; the Parts' accounts of what they inspected are reported, not confirmed.

*What was checked at intake.* The batch-111 dossier (7 October 2026) read both manuscripts, found no error, and reran both delivered suites on copies: their outputs reproduce, the only failures being Windows artifacts (line-ending bytes in hashed outputs, a recorded Python version string, a console code page), none mathematical. *What the write checked.* Every proof of both Parts line by line, with no error found; in particular, by hand: Part I's table of contributions (constant  $-1 - \frac{1}{2} + \frac{1}{16} = -\frac{23}{16}$ , and  $\frac{1}{16} + \frac{1}{12} + \frac{1}{162} - \frac{3}{1024} = \frac{12365}{82944}$ ), the centered profiles (5.11), the transfer correction  $5/(16A)$  and its conversion to  $K$ , the total-sequence constants (1.5) and both forms of (8.2), the inverse (9.3) and  $Z(Y)$ ; Part II's leading terms ( $L$  and  $C_0$  by their Mellin poles, including  $Z(0) = \frac{1}{6}$  and the residue  $-\frac{1}{4}$ ), the rational constant  $\frac{1}{4} + \frac{1}{6} + \frac{19}{144} - \frac{3}{32} + \frac{1}{81} - \frac{1}{256} = \frac{9607}{20736}$ , the profile integrals of  $H_2$ ,  $H_3$ ,  $H_{22}$ , the logarithmic transfer (20.7), the mean, the absolute expansion and the initializer (21.4). Numerically (mpmath at 40 digits) it confirmed the two regularized integrals  $\int_0^\infty yb'(y)^2 dy = -\frac{13}{72} - 2\zeta'(-1) - \frac{\gamma}{6} = 0.05408412102842415943\dots$  and  $R_D = -\zeta'(-1) - \frac{\gamma}{12} = 0.11731983829198985749\dots$  (25 digits each), and  $\int_0^\infty yb''(2y) dy = -\frac{1}{8}$ ; it recomputed  $K$  of Part I (0.680582149566835918851...; the printed digits are a truncation), every entry of Part I's table of  $T(n)$  and of Part II's Table 1 from the shipped exact data (all printed digits agree), and the residuals  $(a(n)/p(n) - \text{three terms})/\tau = -0.068, -0.061, -0.066$  (Part I, at  $n = 500, 1000, 2000$ ) and  $(a_n/C_N - \text{bracket})/\sqrt{\tau} = -0.055, -0.052, -0.052$  (Part II, at  $n = 500, 1000, 2500$ ), which are bounded as the theorems require. With its own code it enumerated all partitions of  $n \leq 55$  by multiplicity vectors and computed both statistics (all agree with the shipped tables), checked the shipped partition numbers against Euler's pentagonal recurrence to 2500, and found both tables strictly increasing from  $n = 1$ . The delivered verifiers and symbolic checks of both Parts were rerun from the shipped files: all receipts and term tables byte-identical, except the sector Boltzmann rows of Part II's floating diagnostics, whose 15 residual values agree with the record only to about ten significant digits (floating library and platform; its README promises no cross-platform identity for floating bytes; the other diagnostic rows are identical). Finite and floating computations are observations, not certificates.

*Relation to the repository.* Before batch 111 no file of the repository named A373271 or A373273. No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. The nearest reports are in the same directory `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/`: `a239964-sizes-equal-max-multiplicity` (batch 111: the number of distinct sizes equal to the maximum multiplicity; the placement kept it apart because that event is not an occupied-value functional) and `a239950-maximal-schreier-supports` (batch 111: least part equal to the number of distinct sizes). They share the Boltzmann occupancy setting and the whole-circle damping argument, not a result, so no reciprocal note is proposed. The inverses are compared with the transseries volume in Remarks 9.1 and 21.1.

[write] **What is not claimed, collected from both Parts.** *Part I.* The result is finite-order: no all-orders expansion, no explicit numerical remainder constants or onset, no variance or fluctuation theorem, no central limit theorem, no exact single-ceiling inverse (the two ceilings can differ by one; no values of  $C_1$ ,  $Y_0$ ), no interpolation of  $a(n)$  assumed and no error asserted for piecewise-linear interpolation, and no absolute novelty, complete-library coverage, global absence or historical priority: the leading mean is “naturally suggested by the Poisson transition” of Ralaivaosaona and the transfer mechanism is classical (Grabner, Knopfmacher and Wagner); Corteel, Pittel, Savage and Wilf was read only at the abstract level and Lugo’s thesis only at targeted passages, so overlap with them is not excluded. The b-file comparison was not made by the source. Finite checks support the implementation, not the analytic remainders. *Part II.* No explicit numerical onset, certified error constant, all-orders expansion, distributional theorem, single exact ceiling for every  $Y$ , numerical  $X, M, n_0$ , exponentially small final coefficient remainder, new general saddle-transfer theorem or absolute novelty; the comparisons with Archibald et al., Fill–Janson–Ward, Corteel–Pittel–Savage–Wilf and Lugo distinguish the objects and are not exclusion certificates; the numerical residuals prove no next coefficient; the diagnostics and the inverse root calculation are not interval-certified. *The write adds:* its checks are floating, finite or hand computations; Remarks 9.1 and 21.1 claim no novelty for any inversion; the merge claims no result relating the two statistics beyond the shared identity.

[write] **Independent check of the write, 7 October 2026.** An independent adversarial check re-derived every [write] statement of this Guide and of both Parts with its own code. Confirmed: the provenance figures of both archives (bytes, SHA-256, 14 and 17 files, 759 and 663 lines, 18 and 17 delivered pages, manifests 13 and 7, 16 and 9 entries, the not-shipped sizes); both OEIS remarks against the live entries (A373271 #15 and its b-file, SHA-256 as quoted; A373273 #14, line for line equal to the shipped record); both exact tables, by brute force over all partitions for  $n \leq 50$  and by the identity of the Guide with the check’s own product code through  $n = 2000$  and  $n = 2500$  (and the partition-number columns); strict increase from  $n = 1$ ; the rational constants  $-\frac{23}{16}, \frac{12365}{82944}, \frac{9607}{20736}, \frac{19}{144}$ ;  $K$  of Part I in both forms (the printed digits are a truncation); every entry of Part I’s  $T(n)$  table and of Part II’s Table 1; the residuals; the two regularized integrals to 25 digits by quadrature and  $\int_0^\infty yb''(2y) dy = -\frac{1}{8}$ ; the label tables (Part I unchanged, Part II shifted by eleven sections, Table 1 unchanged; 89, 78 and 7 labels; 50 and 44 references); the merged bibliography against both delivered ones; the counterpart correspondences in the duplication notes; the transseries classifications; and the reruns of both verifiers and symbolic checks from the shipped files (byte-identical, PASS; Part II’s diagnostics differ only in the 15 sector residuals, at about the tenth significant digit). One precision is recorded after Remark 9.1. No claim of either source was found wrong or unproved.

[write] **Part III, 7 October 2026.** The checks above and the collected non-claims concern Parts I and II. Part III’s provenance, its non-claims and what the write checked in it are in Sections 25.1, 25.4 and 25.3, and its reading conventions, against Parts I and II, in Section 25.5.

*Reading conventions* (letters that the two Parts use differently, or reuse inside one Part, with the tempting false readings; no symbol of either source was renamed; symbols of the transseries volume carry the subscript “vol” in the two transseries remarks).

| Symbol           | Part I                                                                        | Part II                                                                                                              |
|------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| $a(n), a_n$      | $\sum_\lambda D(\lambda)$ (A373271)                                           | $\sum_\lambda W(\lambda)$ (A373273)                                                                                  |
| $D, W$           | $D(\lambda)$ , the statistic                                                  | $D(t)$ , the linear delta correction (15.1);<br>$W(\lambda)$ the statistic; $W_{-1}$ (Part I) is<br>Lambert’s        |
| $K$              | the $n^{-1/4}$ coefficient (1.4)                                              | $\frac{9607}{20736} - \frac{\gamma}{24}$ (12.2)                                                                      |
| $d, c, k$        | $d = \frac{12365}{82944}$ ; $c$ a generic constant                            | $d = \frac{1}{24} + \frac{1}{4A}$ ; $c = \frac{3}{2} - \frac{\gamma}{2}$ ; $k = \sqrt{A}$ (and a<br>summation index) |
| $B, \mathcal{B}$ | $B = 2\sqrt{A}$ ; $\mathcal{B}(z) = (e^z - 1)^{-1}$                           | $B(z) = (e^z - 1)^{-1}$ in Sections 14–18;<br>$B = 2k$ in Section 21                                                 |
| $C, C_N$         | $C = 6^{1/4}/(4\sqrt{3})$ , the total constant; $C_1$<br>the inverse constant | $C = 1/(8\sqrt{3}k)$ in Section 21; $C_N$ the<br>absolute scale; $C_0(t)$ a matched sum                              |

| Symbol            | Part I                                                                                                     | Part II                                                                                                                |
|-------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| $\nu, N$          | $\nu = n - \frac{1}{24}$ ; $N(Y)$ the threshold; $N$ a cutoff (Section 10); $N_m$ in Section 11            | $N = n - \frac{1}{24}$ ; $\nu(Y)$ the threshold; $N_m$ in Section 23                                                   |
| $\ell$            | $\ell_1, \ell_2$ , logarithmic coefficients                                                                | $\ell = \log(1/\tau)$                                                                                                  |
| $\Phi, H_{\pm}$   | $\Phi(x)$ a template for $\log a(n)$ ;<br>$H_{\pm} = \Phi \pm Mx^{-3/2}$                                   | $\Phi(x)$ a template for $a_n$ itself;<br>$H_{\pm} = \log \Phi \pm Mx^{-3/2}/\log x$ ; $H_2, H_3, H_{22}$ matched sums |
| $x, x_0, x_{\pm}$ | $x = \sqrt{n - 1/24}$ ; $x_0(Y)$ the Lambert root;<br>$x_{\pm}$ envelope roots                             | $x_n, x_Y, x_{\pm}(Y)$ ; $x = hm$ in Section 18                                                                        |
| $Z$               | $Z(Y)$ the inverse centre; $Z_m(z)$ a factor                                                               | $Z(u)$ the Dirichlet series (16.2); $Z_{m,j}$ a factor                                                                 |
| $L, L_m$          | $L_m = -\log Q_m$ ; $L = \log(1/\tau)$                                                                     | $L(t) = \sum_m B(tm)$                                                                                                  |
| $E$               | $E_{2,m}, E_{3,m}$ regular parts                                                                           | $E(t)$ , the delta square; $E_{2,m}$ ; $E_1$ the exponential integral                                                  |
| $G, F$            | $G(t)$ the main exponential sum; $F_n$                                                                     | $G = F^2 - 2F/y^2$ , $F(y) = e^y/(e^y - 1)^2$ ; $F_n$                                                                  |
| $T$               | $T_{2,m}, T_{3,m}$ ; $T(n)$ the diagnostic                                                                 | $T_{2,m}$ ; $T_n$ the template; $T$ a cutoff                                                                           |
| $M, X, Y$         | $M$ a fixed cutoff and the envelope constant; $M_j$ multiplicities                                         | the same $M, M_j$ ; $X > 1$ a constant (not the volume's $X_{\text{vol}}$ )                                            |
| $a, \beta, s$     | $\beta = s + \frac{1}{2}$ in Lemma 7.1                                                                     | $a = s + \frac{1}{2}$ in Section 20 (not $a_n$ , not $a_{\text{vol}}$ )                                                |
| $h, r, \tau, A_m$ | the same in both Parts: $h = \sqrt{t}$ , $r =  t $ ,<br>$\tau = \sqrt{A/(n - 1/24)}$ , $A_m = 1/(tm(m+1))$ | (same)                                                                                                                 |

## Part I

# Distinct positive multiplicity values in integer partitions: matched asymptotics and a shrinking inverse for OEIS A373271

Bundle Report 197, archive `Report197_TeX_and_Reproducible_Code.zip`, delivered as `Report197.tex` (18 pages), the base of the merge. Delivered title block “Distinct positive multiplicity values in integer partitions / Matched asymptotics and a shrinking inverse for OEIS A373271 / Report 197 / 4 October 2026”. Printed as delivered, with `[write]` notes and remarks; section, statement and equation numbers are those of the delivery, and its labels carry the prefix `dmv:cnt:.` Its citation key `oeis` is `oeisA373271` here. This Part is self-contained.

**Abstract of this Part (as delivered).** For a partition  $\lambda$ , let  $D(\lambda)$  be the number of different positive integers that occur as multiplicities of its part sizes. If  $a(n) = \sum_{\lambda \vdash n} D(\lambda)$  and  $p(n)$  is the partition number, we prove

$$\frac{a(n)}{p(n)} = (6n)^{1/4} - \frac{23}{16} + \frac{\pi}{6^{1/4}} \left( \frac{12365}{82944} + \frac{15}{8\pi^2} \right) n^{-1/4} + O(n^{-1/2}).$$

The constant correction requires matching the entire multiplicity range: an exact telescoping tail contributes  $-1/2$ , which a purely local expansion misses. A zero-free sector argument, explicit summed remainder bounds, and centered Euler–Maclaurin summation give the radial expansion. A uniform whole-circle bound for positive generating functions that forbid one multiplicity then permits coefficient extraction; the saddle monomials are used only locally. We derive two relative quarter-power corrections for the total sequence, eventual strict increase, and an integer inverse bracket whose width before rounding tends to zero. The accompanying code recomputes exact terms through 2000 and checks independent models at smaller cutoffs. The result is finite-order; no explicit numerical remainder constants or onset are supplied. No all-orders expansion, fluctuation theorem, or absolute novelty claim is asserted.

## 1 The statistic and the main results

Write a partition of  $n$  as  $\lambda = (1^{M_1} 2^{M_2} \dots)$ , so that  $M_j \geq 0$  and  $\sum_j j M_j = n$ . Define

$$D(\lambda) = \#\{m \geq 1 : M_j = m \text{ for some } j \geq 1\}, \quad a(n) = \sum_{\lambda \vdash n} D(\lambda). \quad (1.1)$$

The empty partition has  $D = 0$ , hence  $a(0) = 0$ . This is OEIS A373271 [1]. For example,  $\lambda = (5, 5, 3, 3, 3, 1)$  has positive multiplicities 2, 3, 1, so  $D = 3$ ; the partition  $(5, 5, 3, 3, 1)$  has positive multiplicities 2, 2, 1, so  $D = 2$ . Repeated multiplicity values are counted once. Zero multiplicity is never counted.

Thus  $D$  is neither the number of occupied part sizes, nor the maximum multiplicity, nor an indicator that all nonzero multiplicities are different. If  $\lambda$  is uniform among the  $p(n)$  ordinary partitions, then  $a(n)/p(n) = \mathbb{E}_n D$ .

Throughout the report, put

$$A = \frac{\pi^2}{6}, \quad d = \frac{12365}{82944}, \quad \nu = n - \frac{1}{24}, \quad \tau = \sqrt{\frac{A}{\nu}}. \quad (1.2)$$

The symbol  $O$  has its usual asymptotic meaning, with a fixed implied constant once a sector or a derivative order has been fixed. In complex estimates,  $O(t^\alpha)$  always means a bound in modulus by a constant times  $|t|^\alpha$ .

**Theorem 1.1** (Three-term mean). *As  $n \rightarrow \infty$ ,*

$$\frac{a(n)}{p(n)} = \sqrt{\pi} \tau^{-1/2} - \frac{23}{16} + \sqrt{\pi} \left( d + \frac{5}{16A} \right) \tau^{1/2} + O(\tau). \quad (1.3)$$

*Equivalently,*

$$\frac{a(n)}{p(n)} = (6n)^{1/4} - \frac{23}{16} + Kn^{-1/4} + O(n^{-1/2}), \quad K = \frac{\pi}{6^{1/4}} \left( \frac{12365}{82944} + \frac{15}{8\pi^2} \right). \quad (1.4)$$

*Numerically,  $K = 0.68058214956683591885 \dots$*

The quarter-power scale is consistent with the classical transition for the number of part sizes with one prescribed multiplicity [4]. The new work of the proof presented here is the complete whole-range matching and justified coefficient extraction for this occupied-value statistic; no historical priority claim is attached to that description.

**Theorem 1.2** (Total sequence). *Let*

$$x = \sqrt{n - \frac{1}{24}}, \quad B = 2\sqrt{A}, \quad C = \frac{6^{1/4}}{4\sqrt{3}}, \quad b_1 = -\frac{23}{16 \cdot 6^{1/4}}, \quad b_2 = \sqrt{A}d - \frac{3}{16\sqrt{A}}. \quad (1.5)$$

*Then*

$$a(n) = Ce^{Bx} x^{-3/2} \left( 1 + b_1 x^{-1/2} + b_2 x^{-1} + O(x^{-3/2}) \right). \quad (1.6)$$

*In unshifted variables this is*

$$a(n) = Cn^{-3/4} e^{B\sqrt{n}} \left( 1 + b_1 n^{-1/4} + \left( b_2 - \frac{B}{48} \right) n^{-1/2} + O(n^{-3/4}) \right). \quad (1.7)$$

*The sequence  $a(n)$  is eventually strictly increasing.*

**Theorem 1.3** (Shrinking integer inverse). *Set  $\ell_1 = b_1$  and  $\ell_2 = b_2 - b_1^2/2$ . For large positive  $Y$ , define*

$$\begin{aligned} x_0(Y) &= -\frac{3}{2B} W_{-1} \left( -\frac{2B}{3} (C/Y)^{2/3} \right), \\ Z(Y) &= x_0^2 - \frac{2\ell_1}{B} x_0^{1/2} - \frac{2\ell_2}{B} + \frac{1}{24}. \end{aligned} \quad (1.8)$$

*Here  $W_{-1}$  is the real branch of the Lambert function with values at most  $-1$ . There are constants  $C_1 > 0$  and  $Y_0$  such that the actual integer threshold*

$$N(Y) = \min\{n \geq 0 : a(n) \geq Y\} \quad (1.9)$$

*satisfies, for  $Y \geq Y_0$ ,*

$$\left\lceil Z(Y) - C_1 x_0^{-1/2} \right\rceil \leq N(Y) \leq \left\lceil Z(Y) + C_1 x_0^{-1/2} \right\rceil. \quad (1.10)$$

The width  $2C_1x_0^{-1/2}$  before rounding tends to zero. The two ceilings can nevertheless differ by one; this theorem is not an eventual exact single-ceiling formula. No explicit numerical values for  $C_1$  or  $Y_0$  are supplied. The proof does not assume an interpolation of  $a(n)$ , nor assert an error estimate for piecewise-linear interpolation.

*Remark 1.4* ([write] The OEIS entry, quoted, 7 October 2026). OEIS A373271, read on 7 October 2026 in the internal format, is at revision #15 (2 June 2024), author Olivier Gérard (29 May 2024), offset 1. Its name is “a(n) = sum for all integer partitions of n of the number of distinct multiplicities in each partition.”; its comments give it as the row sums of A373269 and A373270 and add “If all distinct multiplicities of all parts of all integer partitions are summed, one gets A373273 (1, 3, 5, 11, 18, 29, 48, 74, 107, 161, ...).” (the statistic of Part II); it has an example at  $n = 20$  (“contributes 3 to  $a(20) = 1358$ ”), Gérard’s Mathematica program, and a b-file by Alois P. Heinz for  $1 \leq n \leq 200$ . It states no generating function, asymptotic formula or conjecture. The source could not open the b-file (Section 10); the write fetched it (SHA-256 b30df65d4321...ae401d), and all 200 terms agree with the shipped table `data/197-count-exact_A373271.txt`, as do the 45 data terms. The write’s own enumeration of all partitions of  $n \leq 55$  by multiplicity vectors agrees with the table. The name quoted in the bibliography entry [1], “Sum of number of distinct multiplicities over all partitions of  $n$ .”, is a paraphrase, not the entry’s name at the revision the write saw; the delivered text is kept. Nothing was submitted to the OEIS.

## 2 An exact occupancy generating function

Let

$$P(q) = \prod_{j \geq 1} (1 - q^j)^{-1} = \sum_{n \geq 0} p(n)q^n, \quad |q| < 1. \quad (2.1)$$

For each positive integer  $m$ , define

$$Q_m(q) = \prod_{j \geq 1} (1 - (1 - q^j)q^{mj}), \quad P_m(q) = P(q)Q_m(q). \quad (2.2)$$

The product

$$P_m(q) = \prod_{j \geq 1} \left( \frac{1}{1 - q^j} - q^{mj} \right) = \prod_{j \geq 1} \sum_{\substack{k \geq 0 \\ k \neq m}} q^{jk} \quad (2.3)$$

counts the partitions in which no part size has multiplicity  $m$ . In particular, its coefficients are nonnegative and no greater than the corresponding coefficients of  $P$ .

**Proposition 2.1** (Exact identity). *As a formal power series and for  $|q| < 1$ ,*

$$\sum_{n \geq 0} a(n)q^n = P(q) \sum_{m \geq 1} (1 - Q_m(q)). \quad (2.4)$$

*For each fixed  $n$ , the finite generating function*

$$F_n(q) = \sum_{m=1}^n (P(q) - P_m(q)) \quad (2.5)$$

*satisfies  $[q^n]F_n(q) = a(n)$  exactly.*

*Proof.* For each partition, write  $D = \sum_{m \geq 1} \mathbf{1}_{\{\exists j: M_j = m\}}$  and sum over partitions. The summand  $P - P_m$  starts in degree  $m$ , so coefficientwise only finitely many  $m$  contribute, proving both formal claims. For fixed  $\rho = |q| < 1$ ,

$$\sum_{j, m \geq 1} |(1 - q^j)q^{mj}| \leq 2 \sum_{j \geq 1} \frac{\rho^j}{1 - \rho^j} < \infty.$$

The associated products and sums therefore converge absolutely and locally uniformly. For example,  $|1 - Q_m| \leq \exp(\sum_j |(1 - q^j)q^{mj}|) - 1$ , and the right side is summable in  $m$ .  $\square$

Put  $q = e^{-t}$  and abbreviate

$$g(t) = \sum_{m \geq 1} (1 - Q_m(e^{-t})). \quad (2.6)$$

At real  $t > 0$ , independent geometric random variables  $M_j$  with

$$\mathbb{P}_t\{M_j = m\} = (1 - e^{-tj})e^{-tmj}, \quad m \geq 0, \quad (2.7)$$

give the usual Boltzmann distribution on partitions. Accordingly  $g(t) = \mathbb{E}_t D$  and  $Q_m(e^{-t})$  is the probability that the value  $m$  is absent. This probabilistic interpretation motivates the analytic products, but complex  $t$  will be handled by explicit estimates.

### 3 Uniform sector estimates and the small-multiplicity range

Fix a sufficiently small closed sector  $|\arg t| \leq \theta_0$ , initially with  $\theta_0 \leq \pi/12$ , and let  $t \rightarrow 0$  within it. Write

$$r = |t|, \quad h = \sqrt{t} \text{ on the principal branch}, \quad p_{jm} = (1 - e^{-tj})e^{-tmj}, \quad S_{k,m} = \sum_{j \geq 1} p_{jm}^k, \quad \mu_m = S_{1,m}. \quad (3.1)$$

All constants below are independent of  $m$  and  $t$ ; dependence on the fixed sector and fixed  $k$  is allowed.

#### 3.1 A zero-free logarithm

For  $z$  in the sector,

$$1 - e^{-z} = z \int_0^1 e^{-uz} du, \quad |1 - e^{-z}| \leq |z|.$$

Consequently

$$|(1 - e^{-z})e^{-mz}| \leq |z|e^{-m \operatorname{Re} z} \leq \frac{1}{em \cos \theta_0} < 1. \quad (3.2)$$

This supplies one uniform disk of radius strictly less than 1 for all  $p_{jm}$ . The power-series logarithm  $-\log(1 - p)$  is therefore analytic and single-valued here, agrees with the principal logarithm, and encounters no zero of  $1 - p$ .

For every fixed integer  $k \geq 1$ ,

$$|S_{k,m}| \leq \sum_{j \geq 1} (rj)^k e^{-kmj \operatorname{Re} t} \leq \frac{C_k}{rm^{k+1}}, \quad \sup_j |p_{jm}| \leq \frac{C}{m}. \quad (3.3)$$

To see that the exponential-sum bound holds over the entire range of  $m$ , note that  $a^{k+1} \sum_{j \geq 1} j^k e^{-aj}$  is bounded for  $a > 0$ . At zero this follows from an integral comparison; at infinity it decays exponentially.

### 3.2 The matched first moment

Use the notation

$$\mathcal{B}(z) = \frac{1}{e^z - 1}, \quad b(z) = \mathcal{B}(z) - \frac{1}{z}, \quad A_m = \frac{1}{tm(m+1)}, \quad \delta_m = \mu_m - A_m. \quad (3.4)$$

The different symbols  $\mathcal{B}$  and  $B = 2\sqrt{A}$  distinguish the Bose factor from the exponential-growth constant. Geometric summation gives

$$\mu_m = \mathcal{B}(tm) - \mathcal{B}(t(m+1)), \quad \delta_m = -t \int_0^1 b'(t(m+u)) du. \quad (3.5)$$

The removable singularity of  $b$  at zero has expansion

$$b(z) = -\frac{1}{2} + \frac{z}{12} + O(z^3).$$

For each fixed  $\ell \geq 1$ , analyticity on a bounded sector and  $b(z) = -z^{-1} + O(e^{-\operatorname{Re} z})$  at infinity show that

$$|b^{(\ell)}(z)| \leq \frac{C_\ell}{1 + |z|^{\ell+1}}. \quad (3.6)$$

It follows that

$$|\delta_m| \leq \frac{Cr}{1 + (rm)^2}, \quad \delta_m = -\frac{t}{12} + O(r^3 m^2) \quad (rm \leq 1). \quad (3.7)$$

In particular,

$$\sum_{m \geq 1} |\delta_m| = O(1), \quad \sum_{m \geq 1} |\delta_m|^2 = O(r). \quad (3.8)$$

Most importantly, exact telescoping before expansion gives

$$\sum_{m \geq 1} \delta_m = \mathcal{B}(t) - \frac{1}{t} = -\frac{1}{2} + O(r). \quad (3.9)$$

The global  $O(1)$  mass in this identity cannot be recovered by summing only the pointwise approximation  $\delta_m \approx -t/12$  near  $m \asymp r^{-1/2}$ .

### 3.3 Why fixed small multiplicities are harmless

Let

$$L_m = -\log Q_m(e^{-t}) = \sum_{j \geq 1} f_m(jt), \quad f_m(z) = -\log(1 - (1 - e^{-z})e^{-mz}). \quad (3.10)$$

Here  $-\log Q_m$  denotes the sum of the factor logarithms; it need not be the principal logarithm of the full product. For fixed  $m$ ,  $f_m$  is analytic on the sector, is  $O(z)$  at zero, and decays exponentially at infinity. Its derivatives of every fixed order are integrable along each ray, uniformly in the closed angular interval. Euler–Maclaurin along a ray, using  $f_m(0) = 0$ , yields

$$L_m = \frac{1}{t} \int_{\arg z = \arg t} f_m(z) dz + O(r) = \frac{I_m}{t} + O(r), \quad I_m = \int_0^\infty f_m(y) dy > 0. \quad (3.11)$$

For the contour deformation, the small circular arc contributes  $O(\epsilon^2)$  because  $f_m(z) = O(z)$ , and the large arc tends to zero exponentially. No positivity off the real axis is needed. Since  $\operatorname{Re}(1/t) \geq \cos \theta_0/r$ , for each fixed finite range of  $m$  there is  $c > 0$  with  $\operatorname{Re} L_m \geq c/r$  for small  $r$ . Thus  $Q_m = e^{-L_m}$  is exponentially small uniformly on that range. The same is true of  $e^{-A_m}$  times any fixed power of  $r^{-1}$ . This will justify removing finitely many small  $m$  in the algebraic expansions below.

## 4 A summed logarithmic expansion with controlled remainder

The following elementary estimate records the effective scale of multiplicities:

**Lemma 4.1** (Weighted power sums). *For  $k > 1$  and  $c > 0$ ,*

$$\sum_{m \geq 1} m^{-k} \exp\left(-\frac{c}{rm^2}\right) = O(r^{(k-1)/2}). \quad (4.1)$$

*Proof.* The integral has the asserted order after  $m = u/\sqrt{r}$ . The summand is unimodal on the positive real axis, so its sum is bounded by its integral plus a constant times its maximum. The maximum is  $O(r^{k/2})$ , smaller than the displayed bound for  $r \leq 1$ .  $\square$

Choose a sufficiently large fixed  $M$ . For  $m \geq M$ , (3.3) and the convergent logarithm give

$$L_m = \mu_m + \frac{1}{2}S_{2,m} + \frac{1}{3}S_{3,m} + R_{4,m}, \quad |R_{4,m}| \leq \frac{C}{rm^5}. \quad (4.2)$$

Indeed, the tail of the logarithm is bounded by  $C \sum_j |p_{jm}|^4$ , and (3.3) applies also to the sum of absolute powers. The whole correction  $H_m = L_m - \mu_m$  obeys  $|H_m| \leq C/(rm^3)$ . On the other hand  $\operatorname{Re} A_m \geq c/(rm^2)$ . Enlarging  $M$  makes  $|H_m|$  a fixed small fraction of  $\operatorname{Re} A_m$ .

The first-moment error  $\delta_m$  need not be relatively small compared with  $A_m$  when  $m \gg r^{-1}$ , but its global absolute bound  $|\delta_m| \leq Cr$  suffices. Every exponential remainder is consequently dominated by a fixed multiple of  $\exp(-c'/(rm^2))$  times its polynomial estimate.

**Proposition 4.2** (Whole-range matched identity). *Uniformly in the fixed sector,*

$$\begin{aligned} g(t) &= G(t) + \sum_m \delta_m + \sum_m \delta_m (e^{-A_m} - 1) + \frac{1}{2} \sum_m e^{-A_m} S_{2,m} \\ &\quad + \frac{1}{3} \sum_m e^{-A_m} S_{3,m} - \frac{1}{8} \sum_m e^{-A_m} S_{2,m}^2 + O(r), \\ G(t) &= \sum_{m \geq 1} (1 - e^{-A_m}). \end{aligned} \quad (4.3)$$

*Every unqualified sum in this identity runs over all positive integers  $m$ .*

*Proof.* Expand  $1 - e^{-\mu_m - H_m}$  through  $S_{3,m}$  and  $S_{2,m}^2$ . Terms from the logarithmic tail, the mixed quadratic term, higher quadratic products and the cubic exponential remainder are bounded, after summing, by a constant times

$$\sum_m e^{-c/(rm^2)} \left( \frac{1}{rm^5} + \frac{1}{r^2 m^7} + \frac{1}{r^3 m^9} \right) = O(r), \quad (4.4)$$

by Lemma 4.1. Expanding  $\mu_m = A_m + \delta_m$  adds a second-order error  $O(\sum |\delta_m|^2) = O(r)$ . Its linear term is written in the exactly summable form

$$\sum_m e^{-A_m} \delta_m = \sum_m \delta_m + \sum_m \delta_m (e^{-A_m} - 1).$$

Replacing  $e^{-\mu_m}$  by  $e^{-A_m}$  in the retained nonlinear terms costs at most

$$Cr \sum_m e^{-c/(rm^2)} (|S_{2,m}| + |S_{3,m}| + |S_{2,m}|^2) = O(r).$$

All these estimates were established for  $m \geq M$ . On the finitely many omitted  $m$ , the original absence probabilities and each weighted approximation term are exponentially small by (3.11). This completes the genuinely summed expansion.  $\square$

## 5 Evaluation of the matched sums

We use centered coordinates

$$y = h(m + \tfrac{1}{2}), \quad A_m = (y^2 - h^2/4)^{-1}. \quad (5.1)$$

They eliminate an apparent  $h$  correction in the quadratic occupancy term and make the cancellation transparent.

### 5.1 Euler–Maclaurin profiles and their errors

The profiles needed below are  $f_0(y) = 1 - e^{-1/y^2}$  and functions of the form  $y^{-k}e^{-1/y^2}$  with  $k > 1$ . Their derivatives of every fixed order are integrable along rays  $|\arg y| \leq \theta_0/2$ . At zero, the exponential suppresses every negative power;  $f_0(0) = 1$  and every positive-order endpoint derivative vanishes. At infinity, rational decay gives the required integrability.

For completeness, Euler–Maclaurin with four derivatives bounds the unscaled sum remainder by

$$C|h|^3 \int_0^\infty |f^{(4)}(ue^{i \arg h})| du.$$

All boundary corrections on the full midpoint grid vanish, because the odd positive-order endpoint derivatives vanish. Its integral is initially along the same ray as  $h$ . Rotating it to the positive axis is legitimate: the small arc is exponentially small for the weighted profiles and  $O(\epsilon)$  for  $f_0$ , and the large arc tends to zero by rational decay. Therefore

$$\begin{aligned} \sum_{m \geq 1} f_0(h(m + \tfrac{1}{2})) &= h^{-1} \int_0^\infty f_0(y) dy - 1 + O(|h|^3), \\ \sum_{m \geq 1} f(h(m + \tfrac{1}{2})) &= h^{-1} \int_0^\infty f(y) dy + O(|h|^3) \end{aligned} \quad (5.2)$$

for each of the weighted profiles. The  $-1$  in the first line is the omitted midpoint  $h/2$ , whose value is 1 up to an exponentially small error. The omitted midpoint is exponentially small for the other profiles. Applying more derivatives would strengthen these summation errors, but is unnecessary here.

The elementary integrals are

$$\int_0^\infty (1 - e^{-1/y^2}) dy = \sqrt{\pi}, \quad \int_0^\infty y^{-k} e^{-1/y^2} dy = \frac{1}{2} \Gamma\left(\frac{k-1}{2}\right) \quad (k > 1). \quad (5.3)$$

The second follows from  $u = y^{-2}$ ; the first follows from integration by parts.

### 5.2 The main exponential sum

For  $m \geq M$ , expansion of the rational function in (5.1) gives

$$A_m = y^{-2} + \frac{h^2}{4y^4} + O(|h|^4|y|^{-6}).$$

The error in expanding  $1 - e^{-A_m}$  through the displayed correction is bounded by

$$C e^{-c/|y|^2} |h|^4 (|y|^{-6} + |y|^{-8}).$$

After summing, Lemma 4.1 bounds this by  $O(|h|^3)$ , hence by  $O(r)$ . The finitely many small  $m$  are exponentially harmless as before. Equations (5.2)–(5.3) yield

$$G(t) = \frac{\sqrt{\pi}}{h} - 1 + \frac{\sqrt{\pi}}{16}h + O(r). \quad (5.4)$$

### 5.3 The weighted first-moment correction

Uniformly in  $m$ ,

$$|e^{-Am} - 1| \leq C \min\left(1, \frac{1}{rm^2}\right). \quad (5.5)$$

For  $m \leq r^{-1}$  the error in replacing  $\delta_m$  by  $-t/12$  is bounded by (3.7). Splitting once more at  $r^{-1/2}$ , its weighted sum is at most

$$Cr^3 \sum_{m \leq r^{-1}} m^2 \min(1, (rm^2)^{-1}) = O(r). \quad (5.6)$$

For  $m > r^{-1}$ ,  $|\delta_m + t/12| \leq Cr$  and the same weight gives  $O(\sum_{m > r^{-1}} m^{-2}) = O(r)$ . Thus the replacement is valid over the whole range, and

$$\sum_m \delta_m (e^{-Am} - 1) = \frac{t}{12} G(t) + O(r) = \frac{\sqrt{\pi}}{12} h + O(r). \quad (5.7)$$

This weighted correction is separate from the unweighted  $-1/2$  in (3.9).

### 5.4 The nonlinear occupancy terms

Expanding  $(1 - e^{-tj})^k$  and summing in  $j$  gives the exact finite-difference formula

$$S_{k,m} = \sum_{i=0}^k (-1)^i \binom{k}{i} \mathcal{B}(t(km + i)). \quad (5.8)$$

For  $k = 2, 3$ , separate the rational pole and regular part:

$$\begin{aligned} S_{2,m} &= T_{2,m} + E_{2,m}, & T_{2,m} &= \frac{1}{2tm(m+1)(2m+1)}, \\ S_{3,m} &= T_{3,m} + E_{3,m}, & T_{3,m} &= \frac{6}{t(3m)(3m+1)(3m+2)(3m+3)}. \end{aligned} \quad (5.9)$$

The regular finite differences can be represented as integrals of  $t^2 b''$  and  $-t^3 b'''$  over unit cubes. Hence (3.6) implies

$$|E_{2,m}| \leq \frac{Cr^2}{1 + (rm)^3}, \quad |E_{3,m}| \leq \frac{Cr^3}{1 + (rm)^4}. \quad (5.10)$$

In particular  $\sum |E_{2,m}| = O(r)$  and  $\sum |E_{3,m}| = O(r^2)$ . The  $E_2$  error in the square term also causes no difficulty: its cross term is bounded by  $O(r^2) \sum_m e^{-c/(rm^2)} |T_{2,m}| = O(r^2)$ , and  $\sum |E_{2,m}|^2 = O(r^3)$ .

In the centered coordinate  $y = h(m + 1/2)$ ,

$$T_{2,m} = \frac{h}{4y^3} + O(|h|^3 |y|^{-5}), \quad T_{3,m} = \frac{2h^2}{27y^4} + O(|h|^4 |y|^{-6}). \quad (5.11)$$

These estimates hold for  $m \geq M$ ; the corresponding weighted remainders sum to the required order by Lemma 4.1. There is no  $h^2$  term in  $T_2$  and no term of order  $h$  in  $A_m - y^{-2}$ . Therefore there can be no  $h$  correction after summing  $e^{-A_m} S_{2,m}/2$ ; explicitly,

$$\frac{1}{2} \sum_m e^{-A_m} S_{2,m} = \frac{1}{8} \int_0^\infty y^{-3} e^{-1/y^2} dy + O(r) = \frac{1}{16} + O(r). \quad (5.12)$$

Similarly,

$$\begin{aligned} \frac{1}{3} \sum_m e^{-A_m} S_{3,m} &= \frac{2h}{81} \int_0^\infty y^{-4} e^{-1/y^2} dy + O(r) = \frac{\sqrt{\pi}}{162} h + O(r), \\ -\frac{1}{8} \sum_m e^{-A_m} S_{2,m}^2 &= -\frac{h}{128} \int_0^\infty y^{-6} e^{-1/y^2} dy + O(r) = -\frac{3\sqrt{\pi}}{1024} h + O(r). \end{aligned} \quad (5.13)$$

## 5.5 The sector expansion and its essential tail

Combining Proposition 4.2 with (3.9), (5.4), (5.7), (5.12) and (5.13) proves the following.

**Theorem 5.1** (Uniform radial expansion). *In a fixed sufficiently small closed sector about the positive real axis,*

$$g(t) = \sqrt{\pi} t^{-1/2} - \frac{23}{16} + \sqrt{\pi} d t^{1/2} + O(|t|), \quad d = \frac{12365}{82944}. \quad (5.14)$$

The bookkeeping, with each coefficient of  $h$  divided by  $\sqrt{\pi}$ , is

| Contribution                           | Constant | Coefficient of $\sqrt{\pi}h$ |
|----------------------------------------|----------|------------------------------|
| $G(t)$                                 | −1       | 1/16                         |
| $\sum \delta_m$                        | −1/2     | 0                            |
| $\sum \delta_m(e^{-A_m} - 1)$          | 0        | 1/12                         |
| $\frac{1}{2} \sum e^{-A_m} S_{2,m}$    | 1/16     | 0                            |
| $\frac{1}{3} \sum e^{-A_m} S_{3,m}$    | 0        | 1/162                        |
| $-\frac{1}{8} \sum e^{-A_m} S_{2,m}^2$ | 0        | −3/1024                      |
| Total                                  | −23/16   | 12365/82944                  |

In the third column, the entry  $c$  means a contribution  $c\sqrt{\pi}h$ . A calculation confined to  $m \asymp t^{-1/2}$  and omitting (3.9) would give  $-15/16$ . That value is incorrect: the missing  $-1/2$  is a whole-range tail effect. This is why pointwise occupancy limits are not enough for the constant term.

## 6 Whole-circle localization from forbidden-value products

A radial expansion alone does not prove a coefficient asymptotic. We now establish a full-torus bound without assuming a global growth hypothesis for  $g$ .

Take  $q = e^{-\tau+i\theta}$  with  $|\theta| \leq \pi$ , where  $\tau$  is defined in (1.2). A factor in (2.3) is

$$Z_m(z) = \sum_{\substack{k \geq 0 \\ k \neq m}} z^k.$$

For  $j \in [\tau^{-1}, 2\tau^{-1}]$ , put  $\rho_j = e^{-\tau j} \in [e^{-2}, e^{-1}]$ . The normalized coefficients of  $Z_m(\rho_j z)/Z_m(\rho_j)$  form a probability distribution. Every fixed permitted value  $k$  has a probability bounded below uniformly in  $m, j$ , because  $Z_m(\rho_j) \leq (1 - \rho_j)^{-1}$ .

If  $m \neq 1$ , the permitted values 0 and 1 form an adjacent pair. If  $m = 1$ , use 2 and 3. A characteristic function  $\phi(\alpha) = \sum_k \pi_k e^{ik\alpha}$  satisfies

$$|\phi(\alpha)|^2 = 1 - 4 \sum_{k < \ell} \pi_k \pi_\ell \sin^2((k - \ell)\alpha/2). \quad (6.1)$$

Keeping the appropriate adjacent pair and using  $\sqrt{1 - u} \leq e^{-u/2}$  gives

$$\frac{|Z_m(e^{-\tau j + i j \theta})|}{Z_m(e^{-\tau j})} \leq \exp(-c \sin^2(j\theta/2)), \quad (6.2)$$

uniformly in  $m$ . The unrestricted geometric factors satisfy the same estimate, using their values 0, 1.

**Lemma 6.1** (Consecutive-block damping). *For all sufficiently small  $\tau > 0$  and  $|\theta| \leq \pi$ ,*

$$\sum_{j=\lceil 1/\tau \rceil}^{\lfloor 2/\tau \rfloor} \sin^2(j\theta/2) \geq \frac{c}{\tau} \min\left(1, (\theta/\tau)^2\right). \quad (6.3)$$

*Proof.* When  $|\theta| \leq a\tau$  for a fixed small  $a > 0$ , the small-angle lower bound for sine and  $\sum j^2 \asymp \tau^{-3}$  give  $c\theta^2\tau^{-3}$ . On  $a\tau \leq |\theta| \leq C\tau$ , set  $u = \theta/\tau$ . The Riemann sum, multiplied by  $\tau$ , converges uniformly on this compact  $u$  range to  $\int_1^2 \sin^2(uy/2) dy$ , which has a strictly positive minimum there. Finally, for  $C\tau \leq |\theta| \leq \pi$ ,

$$\left| \sum_{j=J}^K \cos(j\theta) \right| \leq \frac{1}{|\sin(\theta/2)|} \leq \frac{\pi}{|\theta|}.$$

Since  $\sin^2(j\theta/2) = (1 - \cos(j\theta))/2$ , a sufficiently large fixed  $C$  makes the block-length term dominate this geometric-sum error. These three ranges cover the torus.  $\square$

All normalized factors outside the block have modulus at most 1. Also  $P_m(e^{-\tau}) \leq P(e^{-\tau})$ . From (2.5), (6.2) and Lemma 6.1,

$$|F_n(e^{-\tau + i\theta})| \leq 2nP(e^{-\tau}) \exp\left[-\frac{c}{\tau} \min\left(1, (\theta/\tau)^2\right)\right]. \quad (6.4)$$

The same damping estimate holds for  $P$ , without the factor  $2n$ .

Let

$$L = \log(1/\tau), \quad \theta_* = \tau^{3/2}L. \quad (6.5)$$

On the complementary arc  $\theta_* \leq |\theta| \leq \pi$ , the exponential factor in (6.4) is at most  $e^{-cL^2}$ . Since the classical partition saddle has scale

$$p(n) \asymp e^{n\tau} P(e^{-\tau}) \tau^{3/2}, \quad (6.6)$$

and  $n = O(\tau^{-2})$ , the outer Cauchy integral for  $F_n$  is  $p(n)$  times a quantity smaller than every fixed power of  $\tau$ . Equation (6.6) also follows directly from the local saddle calculation in the next section combined with this same bound for  $P$ .

On the central arc  $|\theta| \leq \theta_*$ , put  $t = \tau - i\theta$ . Then  $t$  lies in the fixed sector of Theorem 5.1 for small  $\tau$ . For  $m > n$ ,

$$\sum_j |p_{jm}| \leq 2 \sum_j e^{-\tau m j} = O(e^{-\tau m}), \quad |1 - Q_m| = O(e^{-\tau m}).$$

The sum over  $m > n$  is  $O(\tau^{-1}e^{-\tau n}) = O(e^{-c\tau n})$  after reducing  $c > 0$ . As  $\tau n \asymp \tau^{-1}$ , replacing the finite  $F_n$  by  $P(e^{-t})g(t)$  only on this arc causes an exponentially small relative coefficient error, even after the harmless polynomial factors from integration. The finite truncation is exact globally; the infinite extension is justified locally. No mild global bound on  $g$  has been presumed.

## 7 Local saddle transfer and the mean

The classical modular expansion of the partition product is

$$P(e^{-t}) = \left(\frac{t}{2\pi}\right)^{1/2} \exp\left(\frac{A}{t} - \frac{t}{24}\right) \left(1 + O(e^{-c/\tau})\right) \quad (7.1)$$

on the shrinking central arc. It follows, for example, from the Dedekind eta transformation and the absolutely convergent transformed product. The same formula gives

$$|P(e^{-t})| \leq CP(e^{-\tau}) \exp(-c\theta^2/\tau^3) \quad (|\theta| \leq \theta_*). \quad (7.2)$$

Indeed,  $\operatorname{Re}(A/t) - A/\tau = -A\theta^2/(\tau(\tau^2 + \theta^2))$  and  $\theta/\tau = o(1)$ . Thus a uniform  $O(|t|)$  error in  $g$  contributes  $O(\tau)p(n)$  after absolute integration.

### 7.1 A local monomial lemma

For any fixed real  $s$ , define the local integral

$$J_s = \frac{e^{n\tau}}{2\pi} \int_{-\theta_*}^{\theta_*} e^{-in\theta} P(e^{-\tau+i\theta})(\tau - i\theta)^s d\theta. \quad (7.3)$$

It is important that  $J_s$  is defined by a local contour integral. We do not regard  $(-\log q)^s$  as a globally analytic generating function at  $q = 0$ .

**Lemma 7.1** (Normalized monomial transfer). *For each fixed real  $s$ ,*

$$\frac{J_s}{p(n)} = \tau^s \left(1 - \frac{s(s+3)}{4A}\tau + O(\tau^2)\right). \quad (7.4)$$

*Proof.* Set  $\lambda = A/\tau = \nu\tau$  and change orientation if necessary so that  $t = \tau(1 + iv/\sqrt{\lambda})$ . The new interval is symmetric with  $|v| \leq \sqrt{AL}$ . The amplitude exponent in (7.1) is  $\beta = s + 1/2$ . Uniform Taylor expansion gives

$$\begin{aligned} \nu t + A/t &= 2\lambda - v^2 + \frac{iv^3}{\sqrt{\lambda}} + \frac{v^4}{\lambda} + \cdots, \\ (1 + iv/\sqrt{\lambda})^\beta &= 1 + \frac{i\beta v}{\sqrt{\lambda}} - \frac{\beta(\beta-1)v^2}{2\lambda} + \cdots. \end{aligned} \quad (7.5)$$

Odd terms have zero integral on the symmetric interval. Expanding through degree  $\lambda^{-3/2}$ , the remaining integrated error is  $O(\lambda^{-2})$ : on the logarithmic interval, analytic Taylor remainders are bounded by  $\lambda^{-2}$  times fixed polynomials in  $v$ , multiplied by  $e^{-cv^2}$  after slightly weakening the Gaussian. These polynomials have finite Gaussian moments. Extending the moments to the real line costs  $O(e^{-cL^2})$ .

Under the normalized density  $e^{-v^2}/\sqrt{\pi}$ , the moments needed at order  $\lambda^{-1}$  are  $\mathbb{E}v^2 = 1/2$ ,  $\mathbb{E}v^4 = 3/4$ , and  $\mathbb{E}v^6 = 15/8$ . The first correction is therefore

$$\begin{aligned} c(\beta) &= \frac{3}{4} - \frac{15}{16} - \frac{3\beta}{4} - \frac{\beta(\beta-1)}{4} \\ &= -\frac{4(\beta+1)^2 - 1}{16}. \end{aligned} \tag{7.6}$$

For  $s = 0$ ,  $\beta = 1/2$  and  $c(1/2) = -1/2$ . The already proved outer bound for  $P$  shows that  $J_0/p(n) = 1 + O(\tau^R)$  for every fixed  $R$ . Dividing the two saddle expansions and observing

$$c(s+1/2) - c(1/2) = -\frac{s(s+3)}{4}$$

gives (7.4), since  $\lambda^{-1} = \tau/A$ . □

This is the first-order form of the classical Bessel-ratio transfer mechanism; see Grabner, Knopfmacher and Wagner [5]. Their global hypothesis is not imported without verification. Section 6 supplies the necessary independent outer-arc argument for the present statistic.

## 7.2 Completion of the mean proof

Apply Theorem 5.1 only in the central integral. Section 6 treats the complementary integral and the extension of the finite  $m$  sum. Equation (7.2) transfers the radial  $O(|t|)$  remainder to  $O(\tau)p(n)$ . Thus

$$a(n) = \sqrt{\pi}J_{-1/2} - \frac{23}{16}J_0 + \sqrt{\pi}dJ_{1/2} + O(\tau p(n)).$$

In Lemma 7.1,  $s = -1/2$  supplies the extra  $5\tau/(16A)$ ; the correction for  $s = 1/2$  is already below the retained order. This proves (1.3). Since

$$\sqrt{\pi}\tau^{-1/2} = 6^{1/4}(n - 1/24)^{1/4},$$

replacing  $n - 1/24$  by  $n$  changes the leading term by  $O(n^{-3/4})$ , smaller than the claimed remainder. The corresponding changes in later terms are even smaller. This proves Theorem 1.1.

## 8 Total asymptotics and eventual strict increase

The same  $s = 0$  saddle calculation, including its leading normalization, gives

$$p(n) = \frac{e^{Bx}}{4\sqrt{3}x^2} \left( 1 - \frac{1}{Bx} + O(x^{-2}) \right), \quad x = \sqrt{n - 1/24}. \tag{8.1}$$

For clarity, the leading saddle factor before substituting  $\tau = \sqrt{A}/x$  is

$$\frac{1}{2\pi} \left( \frac{\tau}{2\pi} \right)^{1/2} \frac{\tau}{\sqrt{\lambda}} e^{2\lambda} \sqrt{\pi} = \frac{\tau^2}{2\pi\sqrt{2A}} e^{2A/\tau},$$

which equals  $e^{Bx}/(4\sqrt{3}x^2)$ . Also  $-1/(2\lambda) = -1/(Bx)$ .

Inserting  $\tau = \sqrt{A}/x$  into (1.3), multiplying by (8.1), and factoring the leading term yields (1.6). The coefficient arithmetic is

$$b_2 = \sqrt{A} \left( d + \frac{5}{16A} \right) - \frac{1}{2\sqrt{A}} = \sqrt{A}d - \frac{3}{16\sqrt{A}} = \frac{\pi}{\sqrt{6}} \left( d - \frac{9}{8\pi^2} \right). \quad (8.2)$$

The relative mean error is  $O(x^{-3/2})$ , so all omitted cross terms fall within that error. Expanding  $x = \sqrt{n} - 1/(48\sqrt{n}) + O(n^{-3/2})$  gives (1.7); the exponential contributes  $-B/48$  to the  $n^{-1/2}$  coefficient.

To prove eventual strict increase without differentiating an unknown remainder, put

$$\Phi(x) = Bx - \frac{3}{2} \log x + \log C + \ell_1 x^{-1/2} + \ell_2 x^{-1}, \quad \ell_1 = b_1, \quad \ell_2 = b_2 - b_1^2/2. \quad (8.3)$$

Taking logarithms in (1.6), there are  $M > 0$  and  $n_0$  such that

$$\log a(n) = \Phi(x_n) + R_n, \quad x_n = \sqrt{n - 1/24}, \quad |R_n| \leq Mx_n^{-3/2} \quad (n \geq n_0). \quad (8.4)$$

Since  $x_{n+1} - x_n = 1/(2x_n) + O(x_n^{-3})$ ,

$$\Phi(x_{n+1}) - \Phi(x_n) = \frac{B}{2x_n} + O(x_n^{-2}).$$

The two independent remainder bounds give  $|R_{n+1} - R_n| = O(x_n^{-3/2}) = o(x_n^{-1})$ . The adjacent logarithmic increment is therefore positive for all sufficiently large  $n$ . This proves the last statement of Theorem 1.2.

## 9 A rigorous two-ceiling inverse

We prove Theorem 1.3 using monotone logarithmic envelopes, so no unspecified smooth interpolation enters the argument.

For large  $Y$ , the leading logarithmic equation

$$Bx_0 - \frac{3}{2} \log x_0 + \log C = \log Y \quad (9.1)$$

has a unique large solution. Rearranging  $x_0 e^{-2Bx_0/3} = (C/Y)^{2/3}$  gives exactly the  $W_{-1}$  expression in (1.8). The argument tends to 0 from below, so the branch is real and  $x_0 \rightarrow \infty$ .

With  $M$  as in (8.4), define

$$H_+(x) = \Phi(x) + Mx^{-3/2}, \quad H_-(x) = \Phi(x) - Mx^{-3/2}. \quad (9.2)$$

For all sufficiently large  $x$ , both derivatives exceed  $B/2$ . Let  $x_+$  and  $x_-$  be their unique large roots at height  $\log Y$ :

$$H_+(x_+) = \log Y = H_-(x_-), \quad x_+ \leq x_-.$$

Consider

$$\hat{x} = x_0 - \frac{\ell_1}{B} x_0^{-1/2} - \frac{\ell_2}{B} x_0^{-1}.$$

Substitution using (9.1) gives  $H_{\pm}(\hat{x}) - \log Y = O(x_0^{-3/2})$ . For example, the logarithmic change is  $O((\hat{x} - x_0)/x_0) = O(x_0^{-3/2})$ , while the leading correction terms have already been canceled. The derivative lower bound and the mean-value theorem prove

$$x_{\pm} = x_0 - \frac{\ell_1}{B}x_0^{-1/2} - \frac{\ell_2}{B}x_0^{-1} + O(x_0^{-3/2}). \quad (9.3)$$

Squaring gives

$$x_{\pm}^2 + \frac{1}{24} = x_0^2 - \frac{2\ell_1}{B}x_0^{1/2} - \frac{2\ell_2}{B} + \frac{1}{24} + O(x_0^{-1/2}) = Z(Y) + O(x_0^{-1/2}). \quad (9.4)$$

The square of the correction is  $O(x_0^{-1})$ , which is smaller than the stated error.

Choose  $Y$  larger than all exceptional early values  $a(n)$  for which (8.4) is unavailable. If an integer  $n < x_+^2 + 1/24$ , then

$$\log a(n) \leq H_+(x_n) < \log Y.$$

If  $n \geq x_-^2 + 1/24$ , then

$$\log a(n) \geq H_-(x_n) \geq \log Y.$$

It follows, including equality cases at integral endpoints, that

$$\left\lceil x_+^2 + \frac{1}{24} \right\rceil \leq N(Y) \leq \left\lfloor x_-^2 + \frac{1}{24} \right\rfloor. \quad (9.5)$$

A single fixed  $C_1$  dominating the errors in (9.4) yields (1.10). This is a genuine shrinking error bound for the threshold location before integer rounding. At values of  $Y$  for which  $Z(Y)$  is very near an integer, the theorem alone need not decide between the two adjacent possibilities.

*Remark 9.1* ([write] The inverse and the transseries volume, 7 October 2026). Statement by statement, against the volume named in the Guide (its symbols carry the subscript “vol”).

- (a) *Instance.* For  $Y$  beyond the finitely many early values excluded in the proof,  $N(Y)$  of (1.9) is the integer staircase  $N_*(Y)$  of `p0:def:three-inverses` with  $A_n = a(n)$  and  $n_1$  the monotonicity onset, so `p0:thm:staircase(1)` applies. In the shipped table  $a(n+1) > a(n)$  for  $1 \leq n < 2000$ ; that is a finite observation, not a proof of the onset.
- (b) *Instance.* The root  $x_0(Y)$  of (9.1) is `p0:thm:lambert-core(3)`, case  $b_{\text{vol}} < 0$ , in the variable  $X_{\text{vol}} = x = \sqrt{n - 1/24}$ :  $a_{\text{vol}} = B$ ,  $b_{\text{vol}} = -\frac{3}{2}$ ,  $L_{\text{vol}} = \log(Y/C)$ , and the large solution  $(|b_{\text{vol}}|/a_{\text{vol}})(-W_{-1}(-(a_{\text{vol}}/|b_{\text{vol}}|)e^{-L_{\text{vol}}/|b_{\text{vol}}|}))$  is exactly (1.8); it exists once  $L_{\text{vol}}$  exceeds the fold value  $L_c$ .
- (c) *Formal instance, and an instance of the residual step.* The envelopes  $H_{\pm}$  of (9.2) are exactly of exponential–power type in  $x$  (`p0:def:model`) with data  $(C, B, 1, -\frac{3}{2}, \frac{1}{2}, \ell_1 t + \ell_2 t^2 \pm M t^3)$ , so  $\delta_{\text{vol}} = \frac{1}{2}$ . Their formal reversion about  $x_0$  is `p0:prop:two-grid` with  $\delta_{\text{vol}} = \frac{1}{2}$ , whose terms through  $X_{\text{vol}}^{-1}$  are  $-(\ell_1/B)x_0^{-1/2} - (\ell_2/B)x_0^{-1}$ , the centre of (9.3). The volume states its analytic reversion (`p0:thm:lambert-centered(5)`) for  $\delta_{\text{vol}} = 1$  only, so the remainder in (9.3) is the source’s own; its residual-over-slope step (residual  $O(x_0^{-3/2})$ , slopes above  $B/2$ ) is `p0:thm:backward-error`.  $Z(Y)$  then follows by squaring.
- (d) *Analogues.* The brackets (1.10) and (9.5) are, as stated and proved, analogues of `p0:thm:staircase(2)`: two-ceiling brackets proved directly at the integers from smooth envelopes, not deduced from an admissible interpolation.

No novelty is claimed for any of these inversions.

[write] **Independent check, a precision, 7 October 2026.** In (a), `p0:thm:staircase(1)` is stated for  $Y \geq A_{n_1} = a(n_1)$ , and  $N(Y) = N_*(Y)$  needs  $Y > \max_{n < n_1} a(n)$ ; the early values excluded in the proof are those where (8.4) is unavailable, not necessarily those before the monotonicity onset. Both conditions hold for all sufficiently large  $Y$ , which is how (a) is to be read. Everything else in this remark was re-derived: the core data and branch of (b), the two-grid coefficients  $c_{0,1} = -\ell_1/B$ ,  $c_{1,0} = 0$ ,  $c_{0,2} = -\ell_2/B$  of (c) (the envelope term  $\pm Mt^3$  first enters at  $x_0^{-3/2}$ ), the restriction of `p0:thm:lambert-centered(5)` to  $\delta_{\text{vol}} = 1$  (`p0:rem:alpha-one`), and the residual-to-root step.

## 10 Exact computation, diagnostics, and reproducibility

The code archive is an executable companion to the proof, not a numerical substitute for its analytic estimates. It contains the TeX source, pinned exact data, exact verification code, a deterministic builder, adversarial path-safety tests, and source and artifact manifests. See its `README.txt` for exact commands and toolchain requirements.

### 10.1 An exact signed-product algorithm

For a cutoff  $N$ , work in  $\mathbb{Z}[q]/(q^{N+1})$ . Compute  $p(0), \dots, p(N)$  by ordinary partition dynamic programming. For each  $1 \leq m \leq N$ , form

$$Q_m(q) \equiv \prod_{j=1}^{\lfloor N/m \rfloor} (1 - q^{mj} + q^{(m+1)j}) \pmod{q^{N+1}}. \quad (10.1)$$

Every factor is multiplied in descending coefficient order, so the in-place operation uses only coefficients from before that factor. Form

$$R(q) = \sum_{m=1}^N (1 - Q_m(q)) \pmod{q^{N+1}}, \quad a(n) = \sum_{k=1}^n p(n-k)[q^k]R(q).$$

All arithmetic is exact integer arithmetic. No coefficient of the asymptotic expansion is used in this calculation. The package recomputes every  $a(n)$  and  $p(n)$  for  $0 \leq n \leq 2000$  and compares them with the included table.

### 10.2 Independent finite models

An independent positive dynamic program computes  $[q^n]P_m$  directly from the permitted multiplicities. If  $c_j(n)$  counts choices for part sizes at most  $j$  avoiding  $m$ , then

$$c_j(n) = \sum_{\substack{k \geq 0, jk \leq n \\ k \neq m}} c_{j-1}(n - jk). \quad (10.2)$$

The unrestricted inner sum is implemented by the usual unbounded-knapsack recurrence, followed by subtraction of the single forbidden contribution from the previous row. This neither constructs  $Q_m$  nor uses its convolution with  $P$ . Summing  $p(n) - [q^n]P_m$  for  $1 \leq m \leq n$  agrees with every exact term through  $n = 400$ .

A second check enumerates partitions  $\mu$  in descending order. Under Ferrers conjugation, the numbers  $\mu_i - \mu_{i+1}$ , with the final part followed by zero, are the multiplicities of the conjugate partition. Counting their distinct positive values and summing therefore computes the same statistic. This check agrees through  $n = 45$ . The partition numbers are also independently recomputed by Euler’s pentagonal recurrence through 2000.

The first 45 displayed terms of the OEIS entry agree with the exact table. Its linked b-file was inaccessible during the source check, so no b-file comparison is claimed. The small cases begin

| $n$    | 0 | 1 | 2 | 3 | 4 | 5  | 6  | 7  | 8  | 9  |
|--------|---|---|---|---|---|----|----|----|----|----|
| $a(n)$ | 0 | 1 | 2 | 3 | 6 | 10 | 14 | 24 | 34 | 49 |
| $p(n)$ | 1 | 1 | 2 | 3 | 5 | 7  | 11 | 15 | 22 | 30 |

### 10.3 Numerical illustrations are not remainder certificates

Let

$$T(n) = n^{1/4} \left( \frac{a(n)}{p(n)} - (6n)^{1/4} + \frac{23}{16} \right).$$

Theorem 1.1 proves  $T(n) = K + O(n^{-1/4})$ . Values from exact integers, rounded for display, are

| $n$  | $a(n)/p(n)$  | $T(n)$    |
|------|--------------|-----------|
| 50   | 2.9741854612 | 0.6645051 |
| 100  | 3.7209846117 | 0.6617148 |
| 200  | 4.6225550376 | 0.6558233 |
| 500  | 6.1031958784 | 0.6613941 |
| 1000 | 7.4820754099 | 0.6661385 |
| 2000 | 9.1286746210 | 0.6675582 |

These moderate-size values are consistent with the theorem; they neither establish it nor give an effective bound for the remainder. In particular, slow approach and a nonmonotone diagnostic at small  $n$  are compatible with the stated error.

The verification scripts check exact rational identities for  $-23/16$ ,  $d$ , the centered cancellation, the normalized saddle coefficients, and the total-sequence coefficient algebra. They use explicit exceptions rather than Python `assert`, so correctness checks remain active under `python3 -O`. Rebuilding requires the same documented toolchain for byte-identical PDFs; the runtime versions are recorded. Exact integer results and the mathematical claims do not depend on byte-level PDF reproducibility. The manifests verify file completeness and hashes, not the truth of a theorem.

## 11 Prior work, scope, and further questions

### 11.1 What the source screen establishes

Ralaivaosaona [4] studies the number of part sizes with a prescribed multiplicity. Theorem 1 is a fixed-multiplicity count limit theorem, while Theorem 2 gives the transition at multiplicities of order  $n^{1/4}$  between Gaussian, Poisson, and vanishing regimes. The occupied-value sum  $D = \sum_m \mathbf{1}_{\{N_m > 0\}}$  is a different statistic. Its leading mean is naturally suggested by the Poisson transition and can be

obtained after a suitable tail bound; no independent new Poisson phenomenon is claimed here. The matched constant and next mean term require the whole-range calculations above.

Grabner, Knopfmacher and Wagner [5] provide a general asymptotic scheme for partition statistics, including given-multiplicity applications and Bessel transfer. Their Theorem 2.3 has a global growth condition; the present proof explicitly supplies localization instead of assuming that condition automatically holds for  $g$ . Their transfer mechanism is classical prior work.

Cortee, Pittel, Savage and Wilf [6] is an important earlier source on multiplicities, including one prescribed multiplicity, a randomly selected occupied size, and related averages. Its publisher abstract was inspected, but the full text could not be retrieved from the attempted legitimate publisher/author routes. Therefore this report does not assert that every theorem of that article has been excluded as overlap.

Lugo’s thesis [7], Section 6.1 (printed pages 193–197), treats expected counts for a fixed multiplicity and finite allowed multiplicity sets; Proposition 6.1.3 and its surrounding discussion are relevant context. This targeted passage check is not a full-thesis exclusion of all possible connections.

The inspected OEIS entry gave the definition, elementary examples, and brute-force code, but no generating function or asymptotic for this statistic. A bounded search over relevant primary sources and available repository/artifact material did not locate an exact prior result for the displayed corrections. Such negative search evidence is limited. This report makes no absolute novelty, complete-library coverage, global absence, or historical-priority claim. The proofs are presented for their mathematical content independently of that unresolved bibliographic question.

## 11.2 Useful next problems

1. **One further matched order.** Determine the coefficient of  $t$  in  $g(t)$  with a uniform  $O(t^{3/2})$  remainder, explicitly tracking the far-multiplicity tail. This would sharpen the mean and reduce the inverse uncertainty further. An all-orders pattern should not be inferred before possible logarithmic resonances are checked.
2. **Variance and fluctuations.** The indicators that different multiplicity values occur are dependent: at one part size, two distinct multiplicity values are mutually exclusive. A rigorous covariance or cumulant analysis is needed before any variance constant or central limit theorem can be stated. Conditioning the Boltzmann model on total size also requires a separate argument.
3. **Effective error bounds.** Replace the unspecified constants in the sector, Fourier, and saddle estimates with explicit constants and a verified onset. Only then can the inverse bracket become a finite numerical certificate at a prescribed threshold.
4. **Weighted occupied values.** For a specified weight  $w(m)$ , analyze  $\sum_{m \geq 1} w(m) \mathbf{1}_{\{N_m > 0\}}$ . The weight can change the tail scale, and the present unweighted theorem cannot simply be multiplied by  $w$  or transferred term by term without new bounds.
5. **Complete the literature comparison.** Obtain and inspect the full 1999 article and trace subsequent work on occupied multiplicity values and distinct Ferrers gaps. This is necessary for a stronger priority assessment, separate from checking the present proof.

[write] **7 October 2026.** Question 4 is answered for the weight  $w(m) = m$  by Part II (Report 198), whose Theorem 12.1 gives the five-scale mean of  $\sum_m m \mathbf{1}_{\{N_m > 0\}}$ ; as this question warns, the weight changes the tail scale (a leading  $\sqrt{n} \log n$  instead of  $n^{1/4}$ ). Other weights, in particular  $w(m) = m^\alpha$ , remain open; Part II’s

Question 2 asks for that family. Under Vladimir’s standing rule of 4 October 2026 (unproved claims are moved to questions, wrong ones are refuted on the record), the write found no claim of this Part that is unproved as stated and none that is wrong, so nothing was moved here or refuted. For Question 3 note only that the shipped table increases strictly for  $1 \leq n < 2000$ , which proves no onset.

**[write] Part III, 7 October 2026.** Part III answers Question 2 and, for the weights  $w(m) = m^\alpha$ , Question 4; the note above, which calls those weights open, was written before Part III arrived and is kept. *Question 2:* Part III’s Corollary 26.2 gives  $\text{Var}_n D \sim v_0 n^{1/4}$ ,  $v_0 = 6^{1/4}(\sqrt{2} - \frac{3}{4} - \frac{3\sqrt{2}}{8} \log(1 + \sqrt{2})) = 0.3080002444\dots$ , and a central limit theorem for  $(D - (6n)^{1/4})/(\sqrt{v_0} n^{1/8})$ , through the cumulant analysis of its Section 29 that this question asks for; the conditioning on total size is its Lemmas 28.2 and 28.3 with Proposition 28.4. *Question 4:* for every fixed  $\alpha \geq 0$  it determines the fluctuations of  $\sum_m m^\alpha \mathbf{1}_{\{N_m > 0\}}$  (Theorem 26.1: Gaussian below  $\frac{1}{2}$ , a factor  $\log n$  at  $\frac{1}{2}$ , non-Gaussian series above; Theorem 26.5, the window  $\frac{1}{2} + \theta/\log(1/\tau_n)$ ; Theorem 33.1, independence of the three families) and the leading mean (Proposition 30.1 and (31.8)); as this question warns, the weight changes the tail scale. Open: other weights (Part III’s Question 4), and means of this Part’s precision for  $\alpha \neq 0$  (Part II gives  $\alpha = 1$ ). The sentence “Not established are . . . a variance theorem, a central limit theorem” of Section 11 describes this Part alone. Questions 1, 3 and 5 are untouched; Question 5 is also Part III’s Question 8.

### 11.3 Summary of established and unestablished claims

Established here are the exact occupancy generating function, its uniform three-term radial expansion with  $O(|t|)$  error, the full-torus localization, the three-term mean, two relative corrections for the total sequence, eventual strict increase, and the shrinking two-ceiling inverse. Finite exact checks support the implementation and the identification of the sequence. Not established are all fixed asymptotic orders, a variance theorem, a central limit theorem, explicit error constants or onset, an exact single-ceiling inverse, or absolute novelty.

## Part II

# The sum of distinct multiplicity values in integer partitions: five asymptotic scales and a shrinking inverse for OEIS A373273

Bundle Report 198, archive Report198\_TeX\_and\_Reproducible\_Code.zip, delivered as Report198.tex (17 pages). Delivered title block “The sum of distinct multiplicity values in integer partitions / Five asymptotic scales and a shrinking inverse for OEIS A373273 / Report 198 / 4 October 2026”. Printed as delivered, with **[write]** notes and remarks; Report 198’s Section  $k$  is Section  $k + 11$  here, its statements move with their sections, and its equation  $(k.m)$  is equation  $(k+11.m)$ ; its Table 1 keeps its number. Its labels carry the prefix `dmv:sum:`, and its citation key `oeis` is `oeisA373273` here. This Part is self-contained: its package states that no A373271 theorem or coefficient is used as a premise.

**Abstract of this Part (as delivered).** For an integer partition, sum each positive multiplicity value once, however many part sizes have that multiplicity. Let  $a_n$  be the total of this statistic over the partitions of  $n$ , and let  $p(n)$  be the partition number. We prove an expansion of  $a_n/p(n)$  through its five scales  $\sqrt{n} \log n$ ,  $\sqrt{n}$ ,  $n^{1/4}$ ,  $\log n$ , and 1, with error  $O(n^{-1/4})$ . A weighted matched decomposition is essential: the square of a nonlocal correction and a finite-difference tail both contribute to the constant. Mellin continuation evaluates the logarithmic leading

terms; convergent regularized integrals produce constants whose zeta derivatives cancel. A positive-product estimate localizes the full coefficient integral on the entire circle, and local analytic differentiation of saddle monomials transfers logarithmic amplitudes. An explicit smooth template gives eventual strict increase and two ceiling bounds for the first crossing of a large threshold, with pre-rounding uncertainty  $O((\log Y)^{-1/2}/\log \log Y)$ . Exact computations through 2500, independent positive dynamic programming through 450, and partition enumeration through 43 accompany the proof. These finite checks do not certify asymptotic error constants. No explicit numerical onset, all-orders expansion, distributional theorem, or absolute novelty assertion is supplied.

## 12 The statistic and the results

Write  $\lambda = (1^{M_1} 2^{M_2} \dots)$  for a partition of  $n$ , where  $\sum_j j M_j = n$ . Define

$$W(\lambda) = \sum_{m \geq 1: M_j=m \text{ for some } j} m, \quad a_n = \sum_{\lambda \vdash n} W(\lambda), \quad a_0 = 0. \quad (12.1)$$

This is OEIS A373273 [2]. For example,  $(5, 5, 3, 3, 3, 1)$  has multiplicity values 2, 3, 1, so  $W = 6$ , whereas  $(5, 5, 3, 3, 1)$  has values 2, 2, 1, so  $W = 3$ . Zero is excluded, and repeated positive values are summed only once. Thus  $W$  is different from the number of distinct multiplicities and from the total number of parts. Under the uniform measure on partitions of  $n$ ,  $\mathbb{E}_n W = a_n/p(n)$ .

Use the constants and scaled variables

$$A = \frac{\pi^2}{6}, \quad N = n - \frac{1}{24}, \quad \tau = \sqrt{\frac{A}{N}}, \quad \ell = \log \frac{1}{\tau}, \quad c = \frac{3}{2} - \frac{\gamma}{2}, \quad K = \frac{9607}{20736} - \frac{\gamma}{24}, \quad (12.2)$$

where  $\gamma$  is Euler's constant. All asymptotic statements concern  $n \rightarrow \infty$  or  $t \rightarrow 0$  in a specified fixed sector. Complex error terms are bounds in modulus. Implied constants exist but are not assigned numerical values here.

**Theorem 12.1** (Five-scale mean). *With (12.2),*

$$\begin{aligned} \frac{a_n}{p(n)} &= \frac{\ell/2 + c}{\tau} - \frac{7\sqrt{\pi}}{16\sqrt{\tau}} + \left(\frac{1}{24} + \frac{1}{4A}\right)\ell \\ &\quad + K + \frac{7/8 - \gamma/4}{A} + O(\sqrt{\tau}). \end{aligned} \quad (12.3)$$

Equivalently, putting  $k = \sqrt{A}$  and  $d = 1/24 + 1/(4A)$ ,

$$\begin{aligned} \frac{a_n}{p(n)} &= \frac{\sqrt{n}}{4k} \log n + \frac{\sqrt{n}}{k} \left(c - \frac{1}{2} \log k\right) - \frac{7\sqrt{\pi}}{16\sqrt{k}} n^{1/4} + \frac{d}{2} \log n \\ &\quad + K + \frac{7/8 - \gamma/4}{A} - d \log k + O(n^{-1/4}). \end{aligned} \quad (12.4)$$

Replacing  $N$  by  $n$  in (12.4) changes the first term by  $O(n^{-1/2} \log n)$ , which is smaller than the stated error. In particular the leading mean is  $\sqrt{6n} \log n / (4\pi)$ .

**Theorem 12.2** (Absolute expansion). *Let  $C_N = e^{2A/\tau} / (4\sqrt{3}N)$ . Then*

$$a_n = C_N \left[ \frac{\ell/2 + c}{\tau} - \frac{7\sqrt{\pi}}{16\sqrt{\tau}} + \frac{\ell}{24} + K + \frac{1}{8A} + O(\sqrt{\tau}) \right]. \quad (12.5)$$

*The sequence  $(a_n)$  is eventually strictly increasing.*

The entire constant in (12.5), including  $1/(8A)$ , is retained for the inverse error rate stated below. To state the inverse precisely, let

$$\Phi(x) = \frac{e^{2kx}}{4\sqrt{3}x^2} \left[ \frac{x}{k} \left( \frac{1}{2} \log \frac{x}{k} + c \right) - \frac{7\sqrt{\pi}}{16\sqrt{k}} \sqrt{x} + \frac{1}{24} \log \frac{x}{k} + K + \frac{1}{8A} \right]. \quad (12.6)$$

This is positive and strictly increasing for sufficiently large real  $x$ .

**Theorem 12.3** (Shrinking first-crossing bracket). *For large  $Y > 0$ , let  $x_Y$  be the unique large solution of  $\Phi(x_Y) = Y$ , and put  $v_Y = x_Y^2 + 1/24$ . There exist constants  $X > 1$ ,  $M > 0$ , and an integer  $n_0$  such that the following construction is valid. Set*

$$H_{\pm}(x) = \log \Phi(x) \pm M \frac{x^{-3/2}}{\log x} \quad (x \geq X). \quad (12.7)$$

Both functions are strictly increasing, and for all sufficiently large  $Y$  the roots

$$H_+(x_-(Y)) = \log Y, \quad H_-(x_+(Y)) = \log Y \quad (12.8)$$

exist beyond  $\sqrt{n_0 - 1/24}$ . Then, for  $\nu(Y) = \min\{n \geq 0 : a_n \geq Y\}$ ,

$$\left[ x_-(Y)^2 + \frac{1}{24} \right] \leq \nu(Y) \leq \left[ x_+(Y)^2 + \frac{1}{24} \right], \quad (12.9)$$

where each unrounded endpoint is

$$v_Y + O\left(\frac{(\log Y)^{-1/2}}{\log \log Y}\right). \quad (12.10)$$

The two ceilings are eventually equal or adjacent. The statement does not specify numerical  $X, M, n_0$ , and does not assert a single exact ceiling for every  $Y$ .

*Remark 12.4* ([write] The OEIS entry, quoted, 7 October 2026). OEIS A373273, read on 7 October 2026 in the internal format, is still at revision #14 (1 June 2024), the revision the source read; its text is identical, line for line, to the shipped record `data/198-sum-A373273.seq`. Author Olivier Gérard (29 May 2024), offset 1; name “a(n) = sum of all distinct multiplicities in every integer partition of n.”; one comment, “Sum of the rows of triangle A373272.”; Gérard’s Mathematica program; 43 data terms (its b-file is synthesized from them). It states no formula, asymptotic or conjecture. All 43 terms agree with the shipped table `data/198-sum-exact_A373273.txt`, and the write’s own enumeration of all partitions of  $n \leq 55$  by multiplicity vectors agrees with the table. A373271 (Part I) points to this entry (Remark 1.4). Nothing was submitted to the OEIS.

### 13 Exact generating functions and finite models

Let  $P(q) = \prod_{j \geq 1} (1 - q^j)^{-1} = \sum_{n \geq 0} p(n) q^n$ . Under the Boltzmann partition measure at real  $q \in (0, 1)$ , the  $M_j$  are independent with

$$\mathbb{P}_q(M_j = m) = (1 - q^j) q^{mj}. \quad (13.1)$$

For  $q = e^{-t}$  define

$$p_{jm} = (1 - e^{-tj})e^{-tmj}, \quad Q_m(q) = \prod_{j \geq 1} (1 - (1 - q^j)q^{mj}). \quad (13.2)$$

Thus  $Q_m$  is the probability that multiplicity  $m$  is absent. Linearity of expectation, applied to  $W = \sum_{m \geq 1} m 1_{\{M_j=m \text{ for some } j\}}$ , gives

$$\sum_{n \geq 0} a_n q^n = P(q)w(t), \quad w(t) = \sum_{m \geq 1} m(1 - Q_m(e^{-t})). \quad (13.3)$$

The identity also holds formally: the  $m$ th summand starts in degree  $m$ . On a compact subset of  $\text{Re } t > 0$ ,  $\sum_{m,j} m|p_{jm}| < \infty$ , and

$$|1 - Q_m| \leq \left( \sum_j |p_{jm}| \right) \exp \left( \sum_j |p_{jm}| \right),$$

which proves local absolute convergence and analyticity.

The product

$$P_m(q) = P(q)Q_m(q) = \prod_{j \geq 1} \left( \frac{1}{1 - q^j} - q^{mj} \right) \quad (13.4)$$

has nonnegative coefficients: it counts partitions avoiding the multiplicity  $m$ . Hence

$$F_n(q) = \sum_{m=1}^n m(P(q) - P_m(q)), \quad [q^n]F_n(q) = a_n. \quad (13.5)$$

This positive interpretation later controls the full coefficient circle without imposing a global hypothesis on  $w$ .

For a cutoff  $T$ , an exact integer algorithm first computes  $p(0), \dots, p(T)$ , then for each  $1 \leq m \leq T$  multiplies the finite factors

$$Q_m(q) \equiv \prod_{1 \leq j \leq T/m} (1 - q^{mj} + q^{(m+1)j}) \pmod{q^{T+1}}. \quad (13.6)$$

Terms of degree greater than  $T$  are discarded. Form the polynomial  $w = \sum_{m=1}^T m(1 - Q_m)$  and convolve with  $p$ . Descending coefficient updates, or separate old and new arrays, avoid accidentally reusing a factor. The construction uses integer arithmetic only. Its straightforward upper cost is  $O(T^2 \log T)$  arithmetic operations for the products plus  $O(T^2)$  for the final convolution; this counts integer operations, not bit complexity.

A genuinely different check constructs each  $P_m$  by positive allowed-occupancy dynamic programming: for part size  $j$ , convolve with  $\sum_{r \geq 0, r \neq m} q^{rj}$ , truncated at the cutoff. This never computes  $Q_m$  or a signed collision correction. A third check enumerates descending partitions and sums the set of positive adjacent gaps, including the last part minus zero. Under Ferrers conjugation these gaps are the positive multiplicities of the conjugate partition; summing over all partitions therefore gives the same total. The accompanying computations compare these models in Section 22.

## 14 Uniform estimates in a zero-free sector

Fix the closed sector  $|\arg t| \leq \theta_0 = \pi/12$ , and write  $r = |t|$ ,  $h = \sqrt{t}$ . All powers and  $\log t$  use their principal branches in this sector. Define

$$\begin{aligned} B(z) &= (e^z - 1)^{-1}, & b(z) &= B(z) - z^{-1}, \\ A_m &= \frac{1}{tm(m+1)}, & \mu_m &= B(tm) - B((m+1)t), & \delta_m &= \mu_m - A_m, \\ S_{k,m} &= \sum_{j \geq 1} p_{jm}^k. \end{aligned} \tag{14.1}$$

Regularity at zero and exponential decay of  $B$  at sector infinity give

$$b'(0) = \frac{1}{12}, \quad b''(z) = O(z) \quad (z \rightarrow 0), \quad |b^{(k)}(z)| \leq \frac{C_k}{1 + |z|^{k+1}} \quad (k \geq 1). \tag{14.2}$$

In particular,

$$\begin{aligned} \delta_m &= -t \int_0^1 b'(t(m+u)) du, & |\delta_m| &\leq \frac{Cr}{1 + (rm)^2}, \\ \delta_m &= -tb'(tm) + O\left(\frac{r^2}{1 + (rm)^3}\right). \end{aligned} \tag{14.3}$$

For  $z$  in the sector, integration of  $e^{-vz}$  over  $0 \leq v \leq 1$  yields

$$|(1 - e^{-z})e^{-mz}| \leq |z|e^{-m \operatorname{Re} z} \leq \frac{1}{em \cos \theta_0} < 1. \tag{14.4}$$

Thus each factor has a logarithm given by its convergent power series. The sum of these factor logarithms defines an analytic logarithm of  $Q_m$ . It need not coincide with the principal logarithm of the whole product; no such identification is used.

For fixed  $k \geq 1$ , the bound  $|p_{jm}| \leq rje^{-mj \operatorname{Re} t}$  and boundedness of  $a^{k+1} \sum_{j \geq 1} j^k e^{-aj}$  imply

$$|S_{k,m}| \leq \frac{C_k}{rm^{k+1}}, \quad \sup_j |p_{jm}| \leq C/m. \tag{14.5}$$

We repeatedly use, for  $k > 1$  and fixed  $c_1 > 0$ ,

$$\sum_{m \geq 1} m^{-k} e^{-c_1/(rm^2)} = O(r^{(k-1)/2}). \tag{14.6}$$

An integral comparison after  $m = x/\sqrt{r}$  proves this; the mesh-end pieces satisfy the same bound.

For each of finitely many small  $m$ , put

$$f_m(z) = -\log(1 - (1 - e^{-z})e^{-mz}).$$

Its chosen logarithm is analytic by (14.4). The function vanishes at zero, decays at infinity, and has integrable derivatives on the sector rays. Ray Euler–Maclaurin summation, followed by rotation to the positive ray, gives

$$-\log Q_m = I_m/t + O(r), \quad I_m = \int_0^\infty f_m(y) dy > 0. \tag{14.7}$$

Hence  $Q_m$  and the approximants  $e^{-A_m}$  times fixed inverse powers of  $r$  are exponentially small for fixed  $m$ . This disposes of the range where a nonlinear exponential expansion alone is not uniformly small.

[write] **Part I's counterpart, 7 October 2026.** This section repeats, for the same products  $Q_m$ , estimates that Part I proves in Section 3 and Lemma 4.1: (14.4) is (3.2), (14.5) is (3.3), (14.6) is (4.1), (14.7) is (3.11), and the first line of (14.3) is (3.5)–(3.7); the second-order form  $\delta_m = -tb'(tm) + O(r^2/(1 + (rm)^3))$  and the bound on  $b''$  near zero are new here. It is kept because the later sections of this Part cite these displays by their own labels; the two proofs are the same argument.

## 15 A convergent weighted matched decomposition

Define the seven separately convergent sums

$$\begin{aligned} L(t) &= \sum_{m \geq 1} B(tm), & C_0(t) &= \sum_{m \geq 1} m(1 - e^{-A_m} - A_m), \\ D(t) &= \sum_{m \geq 1} m\delta_m(e^{-A_m} - 1), & E(t) &= -\frac{1}{2} \sum_{m \geq 1} me^{-A_m}\delta_m^2, \\ H_2(t) &= \frac{1}{2} \sum_{m \geq 1} me^{-A_m}S_{2,m}, & H_3(t) &= \frac{1}{3} \sum_{m \geq 1} me^{-A_m}S_{3,m}, \\ H_{22}(t) &= -\frac{1}{8} \sum_{m \geq 1} me^{-A_m}S_{2,m}^2. \end{aligned} \tag{15.1}$$

The individual sum  $\sum m\delta_m$  diverges and is never separated from its convergent combinations.

**Lemma 15.1** (Weighted matching). *Uniformly in the fixed sector,*

$$w = L + C_0 + D + E + H_2 + H_3 + H_{22} + O(\sqrt{r}). \tag{15.2}$$

*Proof.* The absolutely convergent telescoping identity  $\sum_m m\mu_m = \sum_m B(tm) = L$  makes  $L + C_0$  the convergent sum  $\sum_m m(1 - e^{-A_m} + \delta_m)$ . Adding  $D$  changes the linear delta contribution to its exponentially damped version.

For  $m$  above a fixed sufficiently large  $M$ , logarithmic expansion gives

$$-\log Q_m = A_m + \delta_m + \frac{1}{2}S_{2,m} + \frac{1}{3}S_{3,m} + O((rm^5)^{-1}).$$

The nonlinear logarithmic part is  $O((rm^3)^{-1})$ , at most a small fixed fraction of  $\text{Re } A_m$  for large  $M$ . The delta part is  $O(r)$  absolutely; it need not be relatively small compared with  $A_m$  in the far tail. Thus integral Taylor remainders of the exponential are dominated by a polynomial prefactor times  $e^{-c_1/(rm^2)}$ . Retain the linear delta,  $S_2/2$ , and  $S_3/3$  terms, and the negative half-squares of delta and  $S_2/2$ . They give exactly (15.1).

Here is the weighted accounting for all omitted terms:

$$\begin{aligned} \text{logarithmic tail: } & Cr^{-1} \sum_m m^{-4} e^{-c_1/(rm^2)} = O(\sqrt{r}), \\ \text{quadratic nonlinear cross terms: } & Cr^{-2} \sum_m m^{-6} e^{-c_1/(rm^2)} = O(\sqrt{r}), \\ \text{cubic nonlinear terms: } & Cr^{-3} \sum_m m^{-8} e^{-c_1/(rm^2)} = O(\sqrt{r}), \\ \text{delta-}S_2 \text{ term: } & Cr \cdot r^{-1} \sum_m m^{-2} e^{-c_1/(rm^2)} = O(\sqrt{r}). \end{aligned}$$

Extra delta factors or replacement of  $S_2$  by  $S_3$  only improve these estimates. The pure delta-cubic remainder is bounded by

$$C \sum_m m |\delta_m|^3 \leq Cr^3 \sum_m \frac{m}{(1 + (rm)^2)^3} = O(r).$$

Higher terms are bounded by these same expressions through the Taylor integral remainder and the damping just established. Finally (14.7) handles the finite range  $m < M$  exponentially. All bounds refer to convergent combinations, not a rearrangement of  $\sum m \delta_m$ .  $\square$

## 16 Mellin evaluation of the leading terms

The classical Mellin kernel of  $L$  is  $\Gamma(s)\zeta(s)^2$ . Its double pole at  $s = 1$  and simple pole at zero give

$$L(t) = \frac{\log(1/t) + \gamma}{t} + \frac{1}{4} + O(r). \quad (16.1)$$

For instance, moving to  $\operatorname{Re} s = -3/2$  crosses also the residue  $-t/144$ , while the new line is  $O(r^{3/2})$ . Gamma decay dominates the polynomial vertical growth of zeta uniformly in our sector.

For  $C_0$ , the Mellin transform of  $1 - e^{-x} - x$  is  $-\Gamma(s)$  on  $-2 < \operatorname{Re} s < -1$ . Consequently

$$C_0(t) = \frac{1}{2\pi i} \int_{(a)} [-\Gamma(-u)] Z(u) t^{-u} du, \quad Z(u) = \sum_{m \geq 1} \frac{m}{[m(m+1)]^u}, \quad 1 < a < 2. \quad (16.2)$$

Absolute convergence justifies the initial interchange. We spell out the continuation needed for moving this contour, because formal expansion of  $Z$  at zero would give a wrong constant.

Separate  $m = 1$  and expand  $(1 + 1/m)^{-u}$  to  $J$  terms:

$$Z(u) = 2^{-u} + \sum_{j=0}^{J-1} \frac{(-1)^j (u)_j}{j!} [\zeta(2u + j - 1) - 1] + R_J(u). \quad (16.3)$$

The Taylor remainder is obtained from

$$\begin{aligned} (1+x)^{-u} - \sum_{j < J} \frac{(-1)^j (u)_j}{j!} x^j \\ = \frac{(-1)^J (u)_J x^J}{(J-1)!} \int_0^1 (1-v)^{J-1} (1+vx)^{-u-J} dv. \end{aligned} \quad (16.4)$$

Here  $x = 1/m$  is positive real. Its modulus has no exponential dependence on  $\operatorname{Im} u$ . Hence  $R_J$  converges normally for  $\operatorname{Re} u > (2 - J)/2$  and has polynomial vertical growth on closed substrips. Taking  $J = 4$  permits the line  $\operatorname{Re} u = -3/4$ . Gamma decay, even after multiplication by  $t^{-u}$ , is  $e^{-(\pi/2 - \theta_0)|\operatorname{Im} u|}$  up to polynomial factors; the shifted integral is  $O(r^{3/4})$ .

At  $u = 1$  the finite part is

$$Z(u) = \frac{1}{2(u-1)} + \gamma - 1 + O(u-1).$$

Indeed subtracting  $\zeta(2u-1)$  leaves at  $u = 1$  the convergent sum  $\sum_m (1/(m+1) - 1/m) = -1$ . Combining this with  $-\Gamma(-u) = -1/(u-1) + 1 - \gamma + O(u-1)$  gives the contribution

$$t^{-1} \left[ \frac{1}{2} \log t + \frac{3}{2}(1 - \gamma) \right].$$

At  $u = 1/2$ , the residue of  $Z$  is  $-1/4$ , producing  $-\sqrt{\pi}/(2\sqrt{t})$ . At zero,

$$Z(0) = \zeta(-1) + \frac{1}{4} = \frac{1}{6}.$$

The  $1/4$  is the removable value of  $(u)_2\zeta(2u+1)/2$ ; it cannot be dropped by substituting  $u = 0$  termwise into the original divergent series. The pole of  $-\Gamma(-u)$  thus contributes  $1/6$ . Finally at  $u = -1/2$  the residue of  $Z$  is  $1/32$ , producing  $-\sqrt{\pi}\sqrt{t}/32$ , inside our retained error. We have proved

$$C_0(t) = \frac{-\frac{1}{2}\log(1/t) + \frac{3}{2}(1-\gamma)}{t} - \frac{\sqrt{\pi}}{2\sqrt{t}} + \frac{1}{6} + O(\sqrt{r}). \quad (16.5)$$

## 17 Nonlocal matching and convergent regularization

### 17.1 The linear delta correction

In  $D$ , replace  $\delta_m$  by  $-tb'(tm)$ . The error is  $O(r \log(1/r))$ , by (14.3) and

$$|1 - e^{-A_m}| \leq C \min(1, (rm^2)^{-1}),$$

splitting the sum at  $r^{-1/2}$  and  $r^{-1}$ . The replacement of  $A_m$  by  $(tm^2)^{-1}$  next has error bounded by

$$C \sum_m m^{-2} e^{-c_1/(rm^2)} = O(\sqrt{r}).$$

Set  $f_t(y) = yb'(y)(1 - e^{-t/y^2})$ . We obtain

$$D(t) = \sum_{m \geq 1} f_t(tm) + O(\sqrt{r}) = \frac{1}{t} \int_0^\infty yb'(y)(1 - e^{-t/y^2}) dy + O(\sqrt{r}). \quad (17.1)$$

The integral initially runs along the  $t$  ray. Its rotation to the positive axis is valid:  $b'$  has no intervening poles,  $\arg(t/y^2)$  stays in  $[-\theta_0, \theta_0]$ , and the small and large arcs vanish.

Here is a uniform mesh-error justification. The profile  $f_t$  vanishes at both endpoints and  $\int |f_t''(y)| |dy| = O(1)$  along the sector ray. Split at  $|y| = \sqrt{r}$  and 1. Transition derivatives contain  $r|y|^{-3}e^{-cr/|y|^2}$  and  $r^2|y|^{-5}e^{-cr/|y|^2}$ , each with bounded integral. On  $\sqrt{r} \leq |y| \leq 1$ , the worst term is  $O(r|y|^{-3})$ , again with bounded integral. At infinity it is  $O(r|y|^{-5})$ ; the remaining terms use  $b''(y) = O(y)$  at zero. Second-order Euler–Maclaurin therefore bounds the mesh error by  $O(r)$ .

Let  $b_0 = 1/12$ . On the principal sector branch, define the exponential integral by

$$E_1(t) = -\gamma - \log t - \sum_{j \geq 1} \frac{(-t)^j}{j j!}.$$

For positive  $t$ , this equals  $\int_t^\infty e^{-u} du/u$ . The singular part is explicitly integrable:

$$\frac{1}{t} \int_0^1 y(1 - e^{-t/y^2}) dy = \frac{1}{2} \left( \frac{1 - e^{-t}}{t} + E_1(t) \right) = \frac{1}{2} [\log(1/t) + 1 - \gamma] + O(r). \quad (17.2)$$

After subtracting  $b_0$  on  $(0, 1)$ , replace the kernel  $y(1 - e^{-t/y^2})/t$  by  $1/y$ . The error is  $O(r \log(1/r))$ : its difference is bounded by  $C \min(|y|^{-1}, r|y|^{-3})$ , while  $b'(y) - b_0 = O(y^2)$  at zero. The remaining convergent integral is

$$R_D = \int_0^1 \frac{b'(y) - 1/12}{y} dy + \int_1^\infty \frac{b'(y)}{y} dy = -\zeta'(-1) - \frac{\gamma}{12}. \quad (17.3)$$

To justify this value without integrating divergent pieces, set  $F(y) = e^y/(e^y - 1)^2$ . The transform of  $-F$  is  $-\Gamma(s)\zeta(s-1)$  initially on  $\operatorname{Re} s > 2$ . Its singular-subtraction continuation is

$$\int_0^1 y^{s-1} b'(y) dy - \int_1^\infty y^{s-1} F(y) dy - \frac{1}{s-2}.$$

For  $0 < \operatorname{Re} s < 2$ , the last term equals  $\int_1^\infty y^{s-3} dy$ , so this is the convergent Mellin transform of  $b'$ . The expansion

$$-\Gamma(s)\zeta(s-1) = \frac{1}{12s} - \zeta'(-1) - \frac{\gamma}{12} + O(s)$$

then gives (17.3) as its finite part at zero. Consequently,

$$D(t) = \frac{1}{24} \log(1/t) + \frac{1}{24} - \frac{\gamma}{8} - \zeta'(-1) + O(\sqrt{r}). \quad (17.4)$$

## 17.2 The delta square cannot be discarded

Replacing  $\delta_m$  by  $-tb'(tm)$  in  $E$  costs  $O(r)$ . Replacing  $e^{-A_m}$  by 1 costs  $O(r \log(1/r))$ , bounded by

$$Cr^2 \sum_m \frac{m}{(1 + (rm)^2)^2} \min(1, (rm^2)^{-1}).$$

A ray Riemann sum and rotation yield

$$E(t) = -\frac{1}{2} \int_0^\infty y b'(y)^2 dy + O(r \log(1/r)). \quad (17.5)$$

This integral is absolutely convergent, since  $b'$  is bounded at zero and  $O(y^{-2})$  at infinity. Its Mellin identity is

$$\int_0^\infty y^{s-1} b'(y)^2 dy = \frac{\Gamma(s)}{6} [\zeta(s-3) - \zeta(s-1)] - 2\Gamma(s-2)\zeta(s-3), \quad (17.6)$$

for  $0 < \operatorname{Re} s < 4$ , with removable values interpreted by continuation. For a convergent proof let  $G = F^2 - 2F/y^2$ . The expansion

$$F(y)^2 = \sum_{j \geq 1} \frac{j^3 - j}{6} e^{-jy}$$

proves that the right side is the transform of  $G$  for  $\operatorname{Re} s > 4$ . At zero  $G = -y^{-4} + O(1)$ ; the  $y^{-2}$  terms cancel. Its continuation is

$$\int_0^1 y^{s-1} (G + y^{-4}) dy + \int_1^\infty y^{s-1} G dy - \frac{1}{s-4}.$$

For  $0 < \operatorname{Re} s < 4$  this is exactly the convergent transform of  $G + y^{-4} = b'^2$ , proving (17.6). At  $s = 2 + \epsilon$ , the two right-hand terms respectively equal

$$-\frac{1}{6\epsilon} - \frac{13}{72} + O(\epsilon), \quad \frac{1}{6\epsilon} - 2\zeta'(-1) - \frac{\gamma}{6} + O(\epsilon).$$

Therefore

$$\int_0^\infty y b'(y)^2 dy = -\frac{13}{72} - 2\zeta'(-1) - \frac{\gamma}{6}, \quad E(t) = \frac{13}{144} + \zeta'(-1) + \frac{\gamma}{12} + O(r \log(1/r)). \quad (17.7)$$

The  $\zeta'(-1)$  terms cancel between  $D$  and  $E$ , and their combined gamma contribution is  $-\gamma/24$ . This constant-order square term is particular to the weighted calculation.

## 18 Collision terms and the far-tail constant

### 18.1 The quadratic collision term

Finite differences give the exact decomposition

$$\begin{aligned} S_{2,m} &= T_{2,m} + E_{2,m}, & T_{2,m} &= \frac{1}{2tm(m+1)(2m+1)}, \\ E_{2,m} &= b(2tm) - 2b((2m+1)t) + b((2m+2)t) \\ &= t^2 b''(2tm) + O\left(\frac{r^3}{1+(rm)^4}\right). \end{aligned} \quad (18.1)$$

Its weighted remainder sums to  $O(r)$ . Replacing  $e^{-A_m}$  by 1 in its leading finite-difference term also costs  $O(r)$ , using  $b''(y) = O(y)$  near zero and  $O(y^{-3})$  at infinity, with the usual splits at  $r^{-1/2}$  and  $r^{-1}$ . Thus

$$\frac{1}{2} \sum_m m e^{-A_m} E_{2,m} = \frac{1}{2} \int_0^\infty y b''(2y) dy + O(r) = -\frac{1}{16} + O(r). \quad (18.2)$$

Indeed this integral is  $\frac{1}{8} \int_0^\infty u b''(u) du = -\frac{1}{8} [b(\infty) - b(0)] = -1/16$ . The rational approximation  $T_{2,m}$  alone misses this constant.

For its rational part use the centered coordinate  $z = h(m + 1/2)$ . Exactly,

$$\frac{1}{2} m T_{2,m} e^{-A_m} = e^{-1/(z^2 - h^2/4)} \left[ \frac{1}{8(z^2 - h^2/4)} - \frac{h}{16z(z^2 - h^2/4)} \right]. \quad (18.3)$$

For the required summed precision this is

$$e^{-1/z^2} \left( \frac{1}{8z^2} - \frac{h}{16z^3} \right) + O\left(|h|^2 e^{-c_1/|z|^2} (|z|^{-4} + |z|^{-6})\right).$$

One first separates finitely many small  $m$  as in (14.7); elsewhere Taylor remainders preserve the damping. The last expression sums to  $O(|h|)$ . The profiles and all derivatives vanish at zero and have integrable tails. Ray Euler–Maclaurin gives their integral divided by  $h$ ; the half-step displacement has no constant endpoint correction. Since

$$\int_0^\infty z^{-2} e^{-1/z^2} dz = \frac{\sqrt{\pi}}{2}, \quad \int_0^\infty z^{-3} e^{-1/z^2} dz = \frac{1}{2},$$

the rational part is  $\sqrt{\pi}/(16\sqrt{t}) - 1/32 + O(\sqrt{r})$ . Together with (18.2),

$$H_2(t) = \frac{\sqrt{\pi}}{16\sqrt{t}} - \frac{3}{32} + O(\sqrt{r}). \quad (18.4)$$

### 18.2 The remaining local constants

The third finite difference similarly yields

$$S_{3,m} = \frac{6}{t(3m)(3m+1)(3m+2)(3m+3)} + O\left(\frac{r^3}{1+(rm)^4}\right). \quad (18.5)$$

The weighted error sums to  $O(r)$ . With  $x = hm$ , the leading rational profiles are

$$\frac{m}{3}S_{3,m} = \frac{2h}{81x^3} + O(|h|^2|x|^{-4}), \quad -\frac{m}{8}T_{2,m}^2 = -\frac{h}{128x^5} + O(|h|^2|x|^{-6}), \quad (18.6)$$

where the first expansion refers to the rational part after its already bounded finite-difference error. In  $H_{22}$  the replacement of  $S_2$  by  $T_2$  has weighted error  $O(r^2 \log(1/r))$ : use

$$|E_{2,m}| \leq Cr^2 \min(rm, (rm)^{-3})$$

with comparable bounds near  $rm = 1$ , then split at  $r^{-1}$  and use exponential damping below it. This is smaller than the required error.

Applying the same damped Euler–Maclaurin argument to (18.6) gives

$$H_3(t) = \frac{1}{81} + O(\sqrt{r}), \quad H_{22}(t) = -\frac{1}{256} + O(\sqrt{r}). \quad (18.7)$$

All profile integrals follow from

$$\int_0^\infty x^{-j} e^{-1/x^2} dx = \frac{1}{2} \Gamma\left(\frac{j-1}{2}\right), \quad j > 1.$$

Ray rotation is harmless since  $|\arg h| \leq \pi/24$ , where the zero-end exponential decays.

**Proposition 18.1** (Sector expansion). *Uniformly for  $|\arg t| \leq \pi/12$  as  $t \rightarrow 0$ ,*

$$w(t) = \frac{\frac{1}{2} \log(1/t) + c}{t} - \frac{7\sqrt{\pi}}{16\sqrt{t}} + \frac{1}{24} \log(1/t) + K + O(\sqrt{|t|}). \quad (18.8)$$

*Proof.* Combine (15.2), (16.1), (16.5), (17.4), (17.7), (18.4), and (18.7). The rational part of the constant is exactly

$$\frac{1}{4} + \frac{1}{6} + \frac{19}{144} - \frac{3}{32} + \frac{1}{81} - \frac{1}{256} = \frac{9607}{20736}.$$

The zeta derivatives cancel, and the surviving gamma term is  $-\gamma/24$ .  $\square$

## 19 Localization on the whole coefficient circle

A sector expansion alone does not justify a coefficient asymptotic. We localize the actual finite function  $F_n$  in (13.5). Put  $\rho = e^{-\tau}$  and  $q = \rho e^{i\theta}$ . For part sizes  $j$  with  $1 \leq \tau j \leq 2$ , the normalized  $j$ th factor of  $P_m$  is the characteristic function of a geometric variable with its value  $m$  excluded.

When  $m \neq 1$ , its probabilities at 0 and 1 have a uniform positive lower bound. When  $m = 1$ , use instead the adjacent values 2 and 3. The normalization is at most  $(1 - e^{-1})^{-1}$ , so the bounds are uniform in both  $m$  and  $j$ . Squaring a characteristic function and retaining the contribution of this adjacent pair proves

$$|Z_{m,j}(\theta)| \leq \exp[-c_2 \sin^2(j\theta/2)]. \quad (19.1)$$

The same estimate holds for unrestricted geometric factors in  $P$ .

For the consecutive block of these  $j$ , we have

$$\sum_{j=\lceil 1/\tau \rceil}^{\lfloor 2/\tau \rfloor} \sin^2(j\theta/2) \geq \frac{c_3}{\tau} \min(1, (\theta/\tau)^2), \quad |\theta| \leq \pi. \quad (19.2)$$

For small  $|\theta|/\tau$  use the small-angle estimate. On a fixed compact range of  $|\theta|/\tau$  away from zero, the limiting integral over  $[1, 2]$  is strictly positive. For  $|\theta| \geq C\tau$ , the geometric-series bound gives  $\sum \cos(j\theta) = O(1/|\sin(\theta/2)|)$ , smaller than half the block length for large enough  $C$ . These ranges cover the whole circle, including rational angles.

Since  $P_m(\rho) \leq P(\rho)$  and  $\sum_{m \leq n} m = n(n+1)/2$ , we conclude that

$$|F_n(\rho e^{i\theta})| \leq n(n+1)P(\rho) \exp \left[ -\frac{c_4}{\tau} \min(1, (\theta/\tau)^2) \right]. \quad (19.3)$$

Take the central arc  $|\theta| \leq \tau^{3/2} \log(1/\tau)$ . Outside it, the integral is smaller than every fixed power of  $\tau$  relative to the partition saddle scale, since the factor  $n(n+1) = O(\tau^{-4})$  is only polynomial. The same unrestricted bound and the positive leading saddle term below show

$$p(n) \asymp e^{n\tau} P(e^{-\tau}) \tau^{3/2}.$$

On the central arc  $t = \tau - i\theta$  lies in our fixed sector. Extending the  $m$  sum from  $m \leq n$  to all  $m$  costs exponentially little: for  $m > n$ ,  $\sum_j |p_{jm}| \leq C e^{-\tau m}$ , whence

$$\sum_{m > n} m |1 - Q_m| \leq C(n/\tau + \tau^{-2}) e^{-\tau n}. \quad (19.4)$$

Here  $\tau n \sim A/\tau$ . We therefore use (18.8) only on the central arc and control its complement directly by (19.3).

**[write] Part I's counterpart, 7 October 2026.** This whole-circle argument is Part I's Section 6 (the adjacent-pair damping (6.2) and Lemma 6.1) applied to the weighted finite sum: the only change is the polynomial factor  $n(n+1)$  in (19.3) in place of  $2n$  in (6.4), and the weighted tail bound (19.4). It is kept because this Part's later sections cite it by its own labels.

## 20 Saddle transfer and analytic parameter differentiation

The modular transformation of the partition product gives, on the shrinking central arc,

$$P(e^{-t}) = \sqrt{\frac{t}{2\pi}} \exp(A/t - t/24) (1 + O(e^{-c_5/\tau})). \quad (20.1)$$

This classical partition saddle method, including logarithmic amplitudes, is substantial prior work; see Grabner, Knopfmacher, and Wagner [5]. We give the needed local calculation explicitly rather than assuming their global multiplier hypothesis for  $w$ .

Put  $\eta = \tau^{3/2} \log(1/\tau)$  and define the normalized central coefficient integral

$$J_s = \frac{e^{n\tau}}{2\pi} \int_{-\eta}^{\eta} P(e^{-\tau+i\theta}) (\tau - i\theta)^s e^{-in\theta} d\theta. \quad (20.2)$$

Here  $t = \tau - i\theta$  and  $t^s$  uses the sector logarithm. This integral is analytic in complex  $s$  on every fixed compact set. Write

$$t = \tau(1 + i\sqrt{\tau/A}u), \quad a = s + \frac{1}{2}.$$

The sign of  $u$  is immaterial. Since  $N = A/\tau^2$ ,

$$Nt + A/t = 2A/\tau - u^2 + i\sqrt{\tau/A}u^3 + (\tau/A)u^4 + \dots$$

Combining this expansion with  $(1 + i\sqrt{\tau/A}u)^a$ , the first relative integrated correction is

$$\frac{\tau}{A} \left[ -\frac{a(a-1)}{2} \langle u^2 \rangle + (1-a) \langle u^4 \rangle - \frac{1}{2} \langle u^6 \rangle \right] = -\frac{(s+1)(s+2)}{4A} \tau, \quad (20.3)$$

where the Gaussian moments are  $1/2, 3/4, 15/8$ . To control the remainder, expand through the odd third order; its integral vanishes on the symmetric arc. The fourth-order remainder is bounded by a fixed polynomial times  $e^{-cu^2}$  and has integrated size  $O(\tau^2)$ , uniformly on a slightly larger compact  $s$  set. Gaussian tails past logarithmic width are smaller than every power. The leading normalization evaluates to  $C_N$ , so

$$J_s = C_N \tau^s \left[ 1 - \frac{(s+1)(s+2)}{4A} \tau + O(\tau^2) \right]. \quad (20.4)$$

At  $s = 0$ , with the excluded partition arcs already controlled,

$$p(n) = C_N [1 - \tau/(2A) + O(\tau^2)]. \quad (20.5)$$

Dividing gives the local monomial transfer

$$\frac{J_s}{p(n)} = \tau^s \left[ 1 - \frac{s(s+3)}{4A} \tau + O(\tau^2) \right]. \quad (20.6)$$

No global definition of a multiplier  $t(q)^s$  is needed.

Analyticity and uniformity on the larger compact set permit Cauchy differentiation of the normalized remainder in  $s$ . Since  $t^s \log(1/t) = -\partial_s t^s$ , we obtain

$$\begin{aligned} J_{-1}/p(n) &= \tau^{-1} + 1/(2A) + O(\tau), \\ -\partial_s (J_s/p(n))|_{s=-1} &= \tau^{-1} \ell + \ell/(2A) + 1/(4A) + O(\tau \ell). \end{aligned} \quad (20.7)$$

Differentiating  $\tau^s$  accounts for the extra logarithm. The first transfer correction of  $t^{-1/2}$  is  $O(\sqrt{\tau})$ ; the omitted corrections for the remaining logarithm and constant are  $O(\tau \ell)$ .

The error in (18.8) needs no differentiated bound. On the central arc,  $|t| \asymp \tau$  and

$$|P(e^{-t})e^{nt}| \leq CP(e^{-\tau})e^{n\tau}e^{-c\theta^2/\tau^3}.$$

An absolute Gaussian integral therefore transfers its  $O(\sqrt{|t|})$  bound to  $O(\sqrt{\tau})p(n)$ . Combining this fact, (18.8), and (20.7) proves Theorem 12.1.

Finally multiplication by (20.5) cancels the canonical  $\ell/(4A)$  and  $c/(2A)$  contributions, leaving the constant  $1/(8A)$  in (12.5). The dropped relative  $O(\tau^2)$  partition factor contributes  $O(\tau \ell)$  inside that bracket, smaller than  $O(\sqrt{\tau})$ . This proves the absolute expansion. The logarithmic-width localization yields superpolynomially small omitted arcs; no exponentially small final coefficient remainder is asserted.

**[write] Part I's counterpart, 7 October 2026.** The normalized transfer (20.6) is the same formula as Part I's Lemma 7.1, (7.4). It is reached here by a second route: the absolute integral (20.4), with first correction  $-(s+1)(s+2)\tau/(4A)$ , is divided by the case  $s = 0$ , whereas Part I computes  $c(\beta)$  in (7.6) and subtracts  $c(1/2)$ ; and here it is proved uniformly for complex  $s$  on compact sets, which the logarithmic transfer (20.7) needs and Part I does not state. Both are kept.

## 21 Monotonicity and inversion with integer endpoints

Put  $x_n = \sqrt{n - 1/24}$  and  $B = 2k$ . Equation (12.5) says

$$a_n = \Phi(x_n) \left[ 1 + O\left(\frac{x_n^{-3/2}}{\log x_n}\right) \right], \quad (\log \Phi)'(x) = B + O(1/x). \quad (21.1)$$

The logarithm has the same error, because the relative error tends to zero. The template gap between successive integers is

$$\log \Phi(x_{n+1}) - \log \Phi(x_n) = \frac{B}{2x_n} + O(x_n^{-2}).$$

The sum of the two pointwise error bounds is  $O(x_n^{-3/2}/\log x_n) = o(x_n^{-1})$ . Thus  $a_{n+1} > a_n$  eventually. This does not differentiate an arbitrary error sequence.

Choose  $X > 1$ ,  $n_0$  with  $x_{n_0} \geq X$ , and  $M > 0$  large enough that

$$H_-(x_n) \leq \log a_n \leq H_+(x_n) \quad (n \geq n_0)$$

holds for (12.7). Increasing  $X, n_0$  if necessary makes both  $H_\pm$  strictly increasing: their derivatives equal  $B + O(1/x)$ , since the envelope derivative is  $O(x^{-5/2}/\log x)$ . Choose  $Y$  large enough that both roots in (12.8) lie beyond  $x_{n_0}$  and that

$$Y > \max\{a_n : 0 \leq n < n_0\}. \quad (21.2)$$

All finite early crossings are now excluded. Monotonicity gives  $x_-(Y) \leq x_Y \leq x_+(Y)$ , and the mean-value theorem gives

$$x_\pm(Y) - x_Y = O\left(\frac{x_Y^{-3/2}}{\log x_Y}\right), \quad x_\pm(Y)^2 + \frac{1}{24} = v_Y + O\left(\frac{x_Y^{-1/2}}{\log x_Y}\right). \quad (21.3)$$

For completeness check the ceiling endpoints directly. An integer below  $\lceil x_-^2 + 1/24 \rceil$  is strictly below  $x_-^2 + 1/24$ , even when the latter is an integer. Its upper envelope is strictly below  $\log Y$ , so it cannot cross the threshold. At the right ceiling, the integer is at least  $x_+^2 + 1/24$ ; its lower envelope is at least  $\log Y$ , including equality. Together with (21.2), this proves (12.9) for the first crossing, not just some later crossing.

The leading template is

$$\Phi(x) \sim C e^{Bx} \frac{\log x}{x}, \quad C = \frac{1}{8\sqrt{3}k}.$$

Thus  $x_Y \sim (\log Y)/B$ , and (21.3) proves (12.10). The unrounded interval tends to zero, so its endpoint ceilings are eventually equal or adjacent. Near integer boundaries it need not select one universal ceiling. An asymptotic initializer for solving the explicit template equation is

$$Bx_Y = \log Y + \log \log Y - \log \log \log Y + \log(4\sqrt{3}) + o(1). \quad (21.4)$$

It follows by taking logarithms of  $C e^{Bx} \log x/x$ ;  $BC = 1/(4\sqrt{3})$ . The finite initializer by itself does not establish the shrinking horizontal error. That statement uses the full equation (12.6) and the envelopes.

*Remark 21.1* ([write] The inverse and the transseries volume, 7 October 2026). Statement by statement, against the volume named in the Guide (its symbols carry the subscript “vol”).

- (a) *Instance.* For  $Y$  as in (21.2),  $\nu(Y)$  is the integer staircase  $N_*(Y)$  of `p0:def:three-inverses` with  $A_n = a_n$  and  $n_1 = n_0$ , so `p0:thm:staircase(1)` applies. In the shipped table  $a_{n+1} > a_n$  for  $1 \leq n < 2500$  (a finite observation).
- (b) *Not shown to be an instance.* The template  $\Phi$  of (12.6) is not of exponential–power type as `p0:def:model` defines it:  $\log \Phi(x) = Bx - \log x + \log \log x + O(1)$  contains  $\log \log x$ , which that model does not have, and the write knows no substitution that removes it.  $\log \Phi$  is an admissible core in the sense of `p0:def:core` (its slope tends to  $B$ ), and  $x_Y$  is its core solution, but that is only the definition; the volume’s exact Lambert solutions (`p0:thm:lambert-core`, `p0:prop:factorial-core`) do not apply, and the initializer (21.4), with its  $\log \log \log Y$ , is not shown to be an instance of `p0:thm:flattening`, whose terms carry no iterated logarithm.
- (c) *Instance of the residual step.* The estimate  $x_{\pm}(Y) - x_Y = O(x_Y^{-3/2}/\log x_Y)$  of (21.3) is `p0:thm:backward-error` applied to  $f = H_{\pm}$  at  $x_* = x_Y$ , with residual  $\pm Mx_Y^{-3/2}/\log x_Y$  and slopes  $B + O(1/x)$ .
- (d) *Analogue.* The bracket (12.9) is, as stated and proved, an analogue of `p0:thm:staircase(2)`: a two-ceiling bracket proved directly at the integers from smooth envelopes.

No novelty is claimed for any of these inversions.

## 22 Exact computations and reproducibility

The supplied source package implements (13.6) using exact integers through  $n = 2500$ . It checks the 43 terms listed in the official OEIS-maintained record, whose offset is 1, and sets  $a_0 = 0$  from the empty-partition definition. The first values are

$$a_0, \dots, a_{15} = 0, 1, 3, 5, 11, 18, 29, 48, 74, 107, 161, 237, 324, 471, 648, 876.$$

Independent positive allowed-occupancy dynamic programming agrees through 450. Independent Ferrers-gap partition enumeration agrees through 43. Partition numbers are cross-checked using Euler’s pentagonal recurrence. These are finite, exact statements; they do not prove the analytic remainders.

Table 1 compares exact means with the full right side of (12.3) without its error term, denoted  $T_n$ . Decimal values are diagnostics, not interval certificates.

The computation also checks the rational constant and Gaussian-moment identities using exact rational arithmetic. Floating evaluations of the smooth inverse and sector Boltzmann sums are explicitly separated from those exact tests. In particular, observed inverse errors are not used to choose a certified  $M$  or onset for Theorem 12.3.

The code archive contains the TeX source, the frozen official sequence record, a complete exact table, and programs that regenerate the arithmetic, diagnostic, and PDF outputs. Its builder requires a fresh destination and refuses to overwrite an existing path. It records runtime and tool versions, verifies its inputs, and writes SHA-256 manifests. The archive itself is deterministic in a fixed tool environment: timestamps and PDF metadata are normalized. The accompanying

| $n$  | $a_n/p(n)$     | $T_n$          | $(a_n/p(n) - T_n)/\sqrt{\tau}$ |
|------|----------------|----------------|--------------------------------|
| 50   | 10.7127593940  | 10.7724626585  | −0.1401564021                  |
| 100  | 16.5055108669  | 16.5632491277  | −0.1612060461                  |
| 200  | 25.3069546171  | 25.3611729887  | −0.1800297270                  |
| 500  | 44.1895677291  | 44.2349179157  | −0.1893544028                  |
| 1000 | 66.9921915478  | 67.0321177419  | −0.1982514012                  |
| 2000 | 101.1162136596 | 101.1516724945 | −0.2093835578                  |
| 2500 | 115.3459918779 | 115.3799955168 | −0.2123104961                  |

Table 1: Finite diagnostics for the five-scale canonical approximation.

README gives exact commands, dependencies, output meanings, and the distinction between finite tests and analytic proof. Normal and optimized Python runs are checked; validation uses explicit exceptions and is not removed by `python -O`.

## 23 Relation to prior work and the scope of the claims

The official A373273 data record [2], authored by Olivier Gérard in May 2024 and inspected at revision 14 dated 1 June 2024, directly supplies the definition and 43 terms. Its Mathematica formulation sums the union of positive partition run lengths. A373272 [3] groups the same weighted statistic by number of parts and lists A373273 as its row sums. The inspected A373273 record contains no formula or asymptotic section. This is a statement about that record, not the whole literature.

Ralaivaosaona [4] studies  $N_m$ , the number of part sizes with a prescribed multiplicity  $m$ , including a fixed- $m$  central limit theorem and a Poisson transition when  $m$  has order  $n^{1/4}$ . Those results motivate the transition scale in the present matching calculation, but do not directly yield the weighted occupancy functional  $\sum_m m 1_{\{N_m > 0\}}$ . Weighting makes a local-only tail treatment insufficient.

Grabner, Knopfmacher, and Wagner [5] provide a general asymptotic scheme for partition statistics, including Bessel and logarithmic-amplitude transfer. Their Section 1.5 concerns prescribed multiplicities; their general transfer theorem is substantial prior for the method. We claim no new general saddle-transfer theorem. Section 19 supplies a separate whole-circle bound for the present weighted positive-product sum, instead of assuming their global multiplier condition.

The inspected material of Archibald et al. [8] concerns multiplicities and sums associated with largest parts. Fill, Janson, and Ward [9] count partitions whose nonzero part multiplicities are pairwise different, a different restriction from averaging (12.1) over all ordinary partitions. These comparisons distinguish the objects; they are not exhaustive theorem-by-theorem exclusion certificates. The related Corteel–Pittel–Savage–Wilf article [6] was checked at the publisher-abstract level, but its full text was not obtained in this source screen. Lugo’s thesis [7], Section 6.1, was checked at targeted passages concerning fixed multiplicities and finite allowed multiplicity sets. Neither limited check excludes overlap with every theorem in those works.

Targeted sequence, phrase, and related-statistic searches, and bounded checks of available report and artifact records, located no exact earlier instance of this expansion. Their coverage was finite and cannot establish worldwide novelty or absence from every previous artifact. The mathematical conclusions of this report are the stated finite-order asymptotics, eventual increase, and existential two-ceiling inverse. No distributional conclusion, arbitrary-order expansion, explicit numerical onset, certified error constant, or historical priority claim follows from the computation or the bounded

source search.

## 24 Further questions

The finite-order proof leaves several concrete questions open.

1. **The next matched order.** Can the  $O(\sqrt{t})$  sector remainder be expanded with a uniform smaller error? Additional collision terms and the nonlocal delta corrections must be retained together; possible logarithmic resonances require a fresh convergence analysis. The numerical residuals are not a proof of a next coefficient or an arbitrary-order scheme.
2. **Other multiplicity weights.** For  $\sum_m m^\alpha \mathbf{1}_{\{N_m > 0\}}$ , how do the leading scale and the matching constants change as  $\alpha$  passes the logarithmic transition represented here by  $\alpha = 1$ ? This asks for a new family of estimates, rather than extrapolating the present coefficients.
3. **Explicit inverse certification.** Can the sector, localization, and saddle bounds be made numerical with a practical onset, so that  $M$  and the finite-prefix threshold in Section 21 are computable and useful? The current floating checks do not answer this.
4. **Weighted fluctuations.** What are the variance and limiting law of  $W$  under the uniform partition measure? Marginal results for one prescribed multiplicity do not alone settle the dependence in this weighted occupied-value sum.
5. **A fuller source comparison.** A complete reading of the unavailable or partially inspected related sources and their successors is needed for a stronger assessment of priority, independently of the correctness of this proof.

[write] **7 October 2026.** Question 2 contains Part I's case:  $\alpha = 0$  is the unweighted count of Part I (Theorem 1.1, leading scale  $n^{1/4}$ ) and  $\alpha = 1$  is this Part (leading scale  $\sqrt{n} \log n$ ); so this Part answers Part I's Question 4 for  $w(m) = m$  only, and the family  $m^\alpha$  remains open. Under Vladimir's standing rule of 4 October 2026, the write found no claim of this Part that is unproved as stated and none that is wrong, so nothing was moved here or refuted. For Question 3 note only that the shipped table increases strictly for  $1 \leq n < 2500$ , which proves no onset.

[write] **Part III, 7 October 2026.** Part III answers Question 4 and part of Question 2; the note above, which calls the family  $m^\alpha$  open, was written before Part III arrived and is kept. *Question 4:* Part III's Corollary 26.3 gives, for  $W = D_1$ ,  $\tau_n W - \frac{1}{2} \log \frac{1}{\tau_n} - \frac{3-3\gamma}{2} \rightarrow G$  in law,  $G$  standard Gumbel ( $\tau_n = \pi/\sqrt{6n}$ ), and  $\text{Var}_n W \sim n$ ; the dependence that this question names is handled by its collision estimate (Lemma 31.1). *Question 2:* for  $\sum_m m^\alpha \mathbf{1}_{\{N_m > 0\}}$  the leading scale is  $M_\alpha \tau_n^{-(\alpha+1)/2}$  with next term  $\Gamma(\alpha+1)\zeta(\alpha)\tau_n^{-\alpha}$  for  $0 < \alpha < 1$  (Proposition 30.1),  $\frac{1}{2\tau_n} \log \frac{1}{\tau_n}$  at  $\alpha = 1$  (this Part's leading term), and  $\Gamma(\alpha+1)\zeta(\alpha)\tau_n^{-\alpha}$  for  $\alpha > 1$  ((31.8)): the occupancy scale  $\tau_n^{-(\alpha+1)/2}$  and the small-part scale  $\tau_n^{-\alpha}$  cross at  $\alpha = 1$ , where the logarithm appears. The matching constants beyond these two terms, an expansion of this Part's precision for  $\alpha \neq 1$ , remain open. Questions 1, 3 and 5 are untouched.

[write] **Independent check, 7 October 2026.** For  $\alpha > 1$  the note above gives one term only: Part III proves the leading term  $\Gamma(\alpha+1)\zeta(\alpha)\tau_n^{-\alpha}$  there and no second term. A second term is proved for  $0 < \alpha \leq 1$ ; for  $\alpha > 1$  it is open (precision in Section 25.2).

## Part III

# Fluctuations of distinct multiplicities in random integer partitions: exact variance, Gaussian and non-Gaussian limits, and a logarithmic transition window

A continuation manuscript of 7 October 2026, archive `ProveIt_Research_2026-10-07.zip`, delivered as `partitions/article.tex` with ten section files (35 pages). Delivered title block “PROVEIT RESEARCH CONTINUATION / Fluctuations of Distinct Multiplicities in Random Integer Partitions / Exact variance, Gaussian and non-Gaussian limits, and a logarithmic transition window / 7 October 2026 / Research draft prepared with ChatGPT”. Printed as delivered after Section 25, which the write added, with [write] notes; the manuscript’s Section  $k$  is Section  $k + 25$  here, its statements move with their sections, its equation  $(k.m)$  is equation  $(k+25.m)$ , and its Figures 1 and 2 keep their numbers. Its labels carry the prefix `dmv:fl:`, and its citation key `oeis` is `oeisA373271` and `oeisA373273` here. This Part is self-contained: it uses no theorem of Parts I and II as a premise.

**Abstract of this Part (as delivered).** Let  $M_j$  be the multiplicity of the part size  $j$  in a uniform partition of  $n$ , and let  $D_\alpha = \sum_{m \geq 1} m^\alpha \mathbf{1}_{\{M_j=m \text{ for some } j\}}$ . We determine the fluctuation law for every fixed  $\alpha \geq 0$ . For  $0 \leq \alpha < 1/2$  the limit is Gaussian, with an explicit positive covariance kernel and variance of order  $n^{\alpha/2+1/4}$ . At  $\alpha = 1/2$  the variance acquires a factor  $\log n$ . For  $\alpha > 1/2$  the centered limit is an explicit convergent series of powers of independent exponential random variables and is non-Gaussian. In particular, the number  $D_0$  of distinct multiplicities has variance  $0.3080002444 \dots n^{1/4}$  and a central limit theorem, whereas the sum  $D_1$  of distinct multiplicities has a Gumbel limit and variance asymptotic to  $n$ . We also determine a Gaussian covariance field in the moving window  $\alpha = 1/2 + \theta/\log(1/t)$ ,  $t \sim \pi/\sqrt{6n}$ . The three limiting families are mutually independent.

The proof treats the negative dependence between occupied multiplicity values explicitly. A localization theorem shows that the entire occupied set is determined, with arbitrarily small polynomial failure probability, by part sizes at most  $O(t^{-1/2} \log^{3/2}(1/t))$ . A separately proved local limit theorem then transfers both laws and second moments from the independent geometric model to uniform partitions. This answers the variance and fluctuation question in the inspected ProveIt report on OEIS A373271 and develops its proposed weighted extension. The present proofs are written mathematical proofs, accompanied by exact finite checks; they have not been formalized in a proof assistant.

## 25 Provenance, scope and notation of Part III

[Added 7 October 2026.] Part III was added in a second write of this report, after Parts I and II had been written (0064d11e6) and independently checked (a03243ec3). It is the manuscript *Fluctuations of Distinct Multiplicities in Random Integer Partitions*, which starts from Part I’s Questions 2 and 4 (Section 11) and answers them for the weights  $m^\alpha$ . Everything of the manuscript is printed in this Part except its title page (recorded below), its table of contents and its reference list (merged into this report’s bibliography): its abstract above, and its Sections 1–9 as Sections 26–34. Its symbols are kept (Section 25.5). This section and the notes marked [write] are the write’s; all other text of Sections 26–34 is the manuscript’s.

## 25.1 The manuscript

The archive `ProveIt_Research_2026-10-07.zip` (1,527,416 bytes, SHA-256 49c8000eea95...07b2b5b4c07; 32 files in one wrapper directory, 2,225,214 bytes unpacked) arrived in commit 7b8ed30e7 (7 October 2026, 10:50). It holds two independent manuscripts: `partitions/`, this Part, and `sensitivity/`, placed as the separate report `Combinatorics/BooleanFunctions/Research/sensitivity-block-sensitivity` by commit 5476940ed, which also retired the archive. This Part’s files were placed in this directory by commit 9a10a617f (7 October 2026, 13:16; the placement records of the batch name it by its side-branch copy 119d8d35d, whose tree for this directory is the same): 12 files, all byte-identical to the delivery (confirmed at the write), with the prefix `03-fluct-`, the Part number (the prefixes `197-count-` and `198-sum-` are bundle report numbers, and `199-` is bundle Report 199’s prefix in another report). The package’s SHA256SUMS (31 entries) verified at the write.

The manuscript is `partitions/article.tex` with `sections/01-results.tex` to `sections/10-references.tex` (2,346 lines, 99,736 bytes; a 35-page PDF, which a build of the delivered text reproduces), dated 7 October 2026. Its title page reads “PROVEIT RESEARCH CONTINUATION”, “Research draft prepared with ChatGPT” and “Prepared for Vladimir Reshetnikov’s research programme. Source snapshots and attribution are recorded in Section 27”; its PDF author field reads “ChatGPT”, and the package README says it was “Prepared with ChatGPT for the requested continuation of research in VladimirReshetnikov/ProveIt and VladimirReshetnikov/math”. It names ChatGPT as the tool and no human author. Its pins (Section 27 and the bibliography entries [10, 11, 12]) are ProveIt commit 1210e9d3ac88a573f98d7ef8626ba3d2c30e9749 (7 October 2026, 16:29:40 UTC, as quoted) and commit adc7f1241b42 of Vladimir’s `math` repository, which is outside this repository and was not checked. At the ProveIt pin this `article.tex` was Report 197 alone, Git blob 77a379cfb, as the manuscript’s bibliography records; Part II was written into it three hours later. The manuscript therefore knows Part I as Report 197 and does not know Part II (Section 27, dated note).

*How the manuscript was merged.* Its 118 labels carry the prefix `dmv:fl:`, and its 127 references to them (83 `\eqref`, 44 `\ref`) were updated. Its citation keys `proveit`, `prior197`, `mathrepo`, `math132`, `fristedt` and `erdoslehner` are new entries of the merged bibliography, marked [Part III]; its entries for Ralaivaosaona and for Corteel, Pittel, Savage and Wilf are printed once with Parts I and II, with its own note added; its single entry `oeis` for A373271 and A373273 is cited as the two OEIS entries of this report, its note added to both. The two figure files are included under their prefixed names. The title page and table of contents are not printed. No other text was changed. The numbering, against a build of the delivered text: the manuscript’s Sections 1–9 are Sections 26–34 here, its Theorem 1.1 is Theorem 26.1, its Corollaries 1.2–1.4 are Corollaries 26.2–26.4, its Theorem 1.5 is Theorem 26.5, its Proposition 3.4 is Proposition 28.4, its Theorem 8.1 is Theorem 33.1, every statement and equation  $k.m$  is  $(k+25).m$ , and Figures 1 and 2 are unchanged (all 118 labels compared). The manuscript cites no number literally; its package README (not shipped) names Theorem 1.1, Corollaries 1.2–1.4, Theorem 1.5, Theorem 8.1 and Proposition 3.4 in the delivered numbering.

*What the manuscript says about the repository.* Its account of Part I (the three-term mean, with the constant  $K$  of Theorem 1.1, and the variance, conditioning and weighted questions of its final section) is correct. Two of its statements were true at its pin and are stale now; dated notes in Section 27 record them: that this file “contains the unweighted report” (it now contains Part II as well), and that the Gowers–Szemerédi source ledger records 113 exact companions and seven open statements (now 114 and six). Its account of the other threads it surveyed is context, not a premise of any proof.

## 25.2 What Part III answers in Parts I and II

| Question                                                                              | After Part III                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Part I, Question 2 (variance and fluctuations; conditioning on the total size)        | <b>Answered:</b> Corollary 26.2, $\text{Var}_n D \sim v_0 n^{1/4}$ with $v_0 = 0.3080002444\dots$ , and a central limit theorem, from Theorem 26.1(i) at $\alpha = 0$ with the covariance kernel of Section 29; the conditioning is Lemmas 28.2–28.3 and Proposition 28.4.                                                                         |
| Part I, Question 4 (weighted occupied values)                                         | <b>Answered for the weights <math>m^\alpha</math>, <math>\alpha \geq 0</math>:</b> the fluctuation laws (Theorems 26.1, 26.5 and 33.1) and the leading means (Proposition 30.1, (31.8)). Not answered for other weights (Part III’s Question 4 proposes regularly varying ones), nor with the precision of Parts I and II for $\alpha \neq 0, 1$ . |
| Part II, Question 2 (other weights $m^\alpha$ ; leading scale and matching constants) | <b>Partly answered:</b> the leading scale and the next term for every $\alpha$ (Proposition 30.1, (31.8)); the matched expansion to Part II’s precision is not extended to $\alpha \neq 1$ .                                                                                                                                                       |
| Part II, Question 4 (variance and law of $W$ )                                        | <b>Answered:</b> Corollary 26.3, a Gumbel law with explicit centering and $\text{Var}_n W \sim n$ ( $W = D_1$ ).                                                                                                                                                                                                                                   |
| Part I, Questions 1, 3, 5;<br>Part II, Questions 1, 3, 5                              | Untouched. Part III’s Question 8 (the historical comparison) is the same task as Questions 5 of both Parts.                                                                                                                                                                                                                                        |

Dated notes after the two question lists (Sections 11 and 24) say the same; the earlier notes there, which call the weights  $m^\alpha$  open, are kept as written.

[write] **Independent check, 7 October 2026.** A precision to the row for Part II’s Question 2 and to the note after Part II’s questions. Part III proves a second term of the mean only for  $0 < \alpha \leq 1$  (Proposition 30.1). At  $\alpha = 0$  its remainder is  $O(1)$ , so the second term  $-\frac{23}{16}$  is Part I’s. For  $\alpha > 1$  it states only the leading term, as the asymptotic equivalence (31.8). “The leading scale and the next term for every  $\alpha$ ” therefore holds for  $0 < \alpha \leq 1$ ; for  $\alpha > 1$  the second term is open. The check’s geometric-law means at  $\alpha = \frac{3}{2}$  suggest  $\mathbb{E}_t D_{3/2} = \Gamma(\frac{5}{2})\zeta(\frac{3}{2})t^{-3/2} + M_{3/2}t^{-5/4} + \dots$ , with  $M_\alpha = \Gamma(\frac{1-\alpha}{2})/(1+\alpha)$  continued to  $\alpha = \frac{3}{2}$ :  $t^{5/4}(\mathbb{E}_t D_{3/2} - \Gamma(\frac{5}{2})\zeta(\frac{3}{2})t^{-3/2}) = -2.083, -2.004, -1.975$  at  $t = 10^{-2}, 10^{-3}, 10^{-4}$ , against  $M_{3/2} = -1.9607\dots$ . This is an observation, not a theorem. For  $1 < \alpha < 3$  the collision mean (31.7) has the same order  $t^{-(\alpha+1)/2}$ , so a proof must evaluate it as well.

*What Part III proves again.* Its means at  $\alpha = 0$  and  $\alpha = 1$  re-derive, by a different route and with weaker remainders, the leading term of Part I’s mean (Theorem 1.1) and the first two terms of Part II’s (Theorem 12.1, with the same constant  $c = \frac{3}{2} - \frac{\gamma}{2} = \frac{3-\gamma}{2}$ ); its identity (34.1)–(34.2) extends Part I’s forbidden-value products (Proposition 2.1) to pairs of values. These are kept, with a dated note at the end of Section 30, because Part III’s later statements cite them and its proofs are not Parts I and II’s. *What is new to the report:* every distributional statement; the localization of the whole occupied set to part sizes  $O(t^{-1/2} \log^{3/2}(1/t))$ ; the reservoir local limit theorem with second-moment transfer; the covariance kernel and its strict positivity; the collision estimates; the critical window and its kernel  $K_{\text{crit}}$ ; the mutual independence of the three regimes; the quadratic weight; the sequence-level variances  $p(n)B_\alpha(n) - A_\alpha(n)^2$ ; and the second moments of A373271 and A373273 in the exact tables.

## 25.3 What the write read and checked

*At placement* (the intake dossier, 7 October 2026): the exact part of the delivered suite rerun on a copy without simulations (moments CSV byte-identical); fixed- $n$  moments to  $n = 300$  by the

intake’s own code;  $v_0$ ,  $V_0$ , the Gumbel centering and the constants of Corollary 26.4 re-derived; no claim found wrong.

*At the write* (7 October 2026; Python 3.14.4, mpmath 1.3.0, NumPy 2.4.4; own code, kept with the intake records and not shipped). Every proof of Sections 28–33 was read line by line, with no error found; rechecked by hand in particular: the localization bounds (28.11)–(28.12); the moments (28.13) ( $\text{Var}_t N \sim 2\zeta(2)t^{-3}$ ) and the likelihood ratio and its  $L^2$  rate; the pair intensity (29.2) and the kernel (29.4); the Cauchy–Schwarz positivity (29.10); the diagonal integral of (26.5) (by  $e^{-s^2} - e^{-2s^2} = \int_1^2 s^2 e^{-cs^2} dc$ ); the leading constant  $M_\alpha$  (one integration by parts); the identity  $H(u) - 2H(2u) = 1/(e^u + 1)$ ; the singular part and  $C_{\text{reg}} = 1$  at  $\alpha = 1$ ; the Gumbel identification and its centering  $\frac{3-\gamma}{2} - \gamma = \frac{3-3\gamma}{2}$ ; the critical row covariance  $c_{1/2,1/2} = 1 - \frac{\pi}{4}$  and the window integral (32.6) giving (26.15); and every mean-shift order in Section 30. With 40-digit mpmath it confirmed:  $V_0$  and  $v_0$  by the closed form, by the one-dimensional formula (26.5) (to all 40 digits) and by a two-dimensional quadrature of (26.4) in the variables  $1/y, 1/z$  (to  $10^{-14}$ );  $V_{1/4} = 0.600519279618463641\dots$  (one-dimensional; to  $4 \cdot 10^{-7}$ , relatively, by the two-dimensional quadrature);  $(1-2\alpha)V_\alpha \rightarrow c_*$ ;  $M_\alpha$  and  $\Gamma(\alpha+1)\zeta(\alpha)$  as integrals at  $\alpha = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$ ;  $C_{\text{reg}} = 1$ ; the Gumbel characteristic function (the finite products at  $k = 2000$  and  $10^6$  approach  $e^{-i\gamma u}\Gamma(1-iu)$ , which equals the Gumbel Fourier integral times  $e^{-i\gamma u}$ ); the constants 2, 8,  $\frac{2\pi^4}{9}$  and 1 of Corollaries 26.3 and 26.4;  $K_{\text{crit}}$  against its integral; and Part I’s  $K$  in the form quoted in Section 27. It enumerated all 215,308 partitions of  $n \leq 40$  by multiplicity vectors (all 41 rows and nine columns of the shipped moments table agree), extended the exact moments to  $n = 300$  by forbidden-value products (the same identity as (34.2), so a range extension, not a second route), recomputed the covariance and sample tables of Section 34, evaluated the geometric-law means against Parts I–III, and reran the delivered `verify.py` without simulations on a copy (notes in Sections 26, 27, 30 and 34; the README has the details). It read the live OEIS entries A373271 and A373273 again, and the repository at the manuscript’s pin. *Not read by the write*: Fristedt, Erdős and Lehner, the `math` repository and its family-132 manuscript; for Ralaivaosaona and Corteel, Pittel, Savage and Wilf see Parts I and II. The manuscript’s accounts of them are reported, not confirmed.

[write] **Independent check of the Part III write, 7 October 2026.** An independent adversarial check of the write `b0ad11d15` re-derived its statements with its own code. *Confirmed*: the provenance (archive bytes and SHA-256, 32 files and 2,225,214 bytes unpacked, SHA256SUMS 31/31, the 12 staged files byte-identical, 2,346 lines and 99,736 bytes, 35 delivered pages, the title-page and PDF-author quotations, the pins and their times, blob `77a379cfb`, the side-branch tree, the not-shipped sizes); the numbering (all 174 earlier labels unchanged, all 118 delivered labels shifted by 25 sections, Figures 1 and 2 unchanged, 7 added, 299 in all);  $V_0$ ,  $v_0$  (the printed (26.11) is rounded and the truncation is as stated),  $V_{1/4}$  (also by a second two-dimensional reduction, to  $10^{-14}$ ), the  $r$ -integral,  $(1-2\alpha)V_\alpha \rightarrow c_*$ ,  $M_\alpha$ ,  $\Gamma(\alpha+1)\zeta(\alpha)$ ,  $C_{\text{reg}} = 1$ , the singular part, the mean constant  $\frac{3-\gamma}{2}$ , equal to Part II’s  $c$ ; the identification  $Z_1 \stackrel{d}{=} G - \gamma$ , which makes the centering  $\frac{3-3\gamma}{2} = 0.634176502647700709\dots$  of (26.12) Part II’s constant minus  $\gamma$ , consistent with it; the constants 2, 8,  $\frac{2\pi^4}{9}$  and 1;  $K_{\text{crit}}$  and its Brownian form; Part I’s  $K$ ; the exact tables (its own brute force to  $n = 45$ , all 41 rows of the moments table, its own forbidden-pair products to  $n = 300$ , which reproduce the write’s variances); the covariance and sample tables; the geometric-law means of the note at the end of Section 30 (the same digits); the rerun of the delivered verifier; A373271 #15 and A373273 #14 live. *Independent of the kernel*: geometric-law variances from the exact absence products approach  $V_0$ ,  $V_{1/4}$  ( $t^{3/4} \text{Var}_t D_{1/4} = 0.561, 0.581, 0.590$  at  $t = 10^{-2}, 10^{-3}, 10^{-4}$ , against 0.6005),  $[\Gamma(\frac{5}{2}) - \Gamma(\frac{7}{4})^2]\zeta(\frac{3}{2})$  at  $\alpha = \frac{3}{4}$ ,  $\pi^2/6$  at  $\alpha = 1$  and  $20\zeta(4)$  at  $\alpha = 2$ ; at  $\alpha = \frac{1}{2}$ ,  $(t/H) \text{Var}_t D_{1/2} = 0.267, \dots, 0.190$  for  $t = 10^{-2}, \dots, 10^{-4}$ , consistent with  $c_*/2$  plus a term of order  $1/H$ ; and the fixed- $n$  gap to  $v_0$  is the size-conditioning correction (note after Corollary 26.2). *Precisions and one correction*, by dated notes: a second term of the mean is proved for  $0 < \alpha \leq 1$  only (after the table of Section 25.2 and after Part II’s questions); the parent of the write is `2bac1ea07` (Section 27). The 127 updated references are those of the manuscript’s Sections 1–9;

its reference list has one more \eqref (the Erdős–Lehner entry), also updated, and the title page’s \ref is printed in the quotation of Section 25.1. No claim of Part III was found wrong or unproved as stated.

## 25.4 What is not claimed

From the manuscript: the finite checks and the high-precision constants are evidence for the implementation, not proofs, and are not interval-certified; the samples are unconditioned finite-row Boltzmann samples, not uniform partitions of a fixed integer; the critical finite-size discrepancy of the samples is “deliberately retained” and no onset or finite- $t$  error is claimed; only finite-dimensional convergence is asserted in the window, “No tightness statement in a space of functions of  $\theta$  is used or claimed”; no effective (Kolmogorov or Wasserstein) bound; the variance formula for regularly varying weights in its Question 4 “is a proposed extension, not a theorem”; nothing is formalized; and, since the full text of Corteel, Pittel, Savage and Wilf was not obtained, no global priority (“No assertion that every potentially overlapping paper has been excluded is needed”). The review was “not an audit of every repository theorem or every manuscript”. The write adds: its checks are floating, finite or hand computations; the fixed- $n$  variances to  $n = 300$  are consistent with  $v_0$  but do not decide it; nothing was submitted to the OEIS.

## 25.5 Reading conventions

Part III keeps its letters. Where they clash with Parts I and II, read them as follows (the write’s own text in this Part uses Part III’s letters).

| Symbol             | Part III                                                                                                                                                                            | Parts I and II                                                            |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| $D_\alpha$         | $\sum_{m \in \mathcal{M}} m^\alpha$ ; $D_0$ is Part I’s $D$ , $D_1$ is Part II’s $W$                                                                                                | $D(\lambda)$ (Part I); $D(t)$ a delta correction (Part II)                |
| $\tau_n, t$        | $\tau_n = \pi/\sqrt{6n}$ ; $t = t_n$ the exact saddle (26.3), also the free parameter of the law $\mathbb{Q}_t$                                                                     | $\tau = \sqrt{A/(n - 1/24)}$ ; $t$ a complex sector variable              |
| $g$                | $g(u) = e^u/(e^u - 1)^2$                                                                                                                                                            | Part I: $g(t) = \mathbb{E}_t D$ ; Part II writes $F(y)$ for this function |
| $M_\alpha, M_j, M$ | $M_\alpha = \Gamma(\frac{1-\alpha}{2})/(1+\alpha)$ ; $M_j$ multiplicities; $M_s$ a geometric variable and $M = t^{-1/2}$ in Section 32; $M$ a constant in (31.6)                    | $M_j$ ; $M$ a cutoff or envelope constant                                 |
| $W(\theta)$        | the window variance (26.17)                                                                                                                                                         | Part II’s statistic $W$ ; Lambert’s $W_{-1}$ (Part I)                     |
| $c_*, c$           | $c_* = 1 - \pi/4$ ; $c, C$ generic constants                                                                                                                                        | Part II: $c = \frac{3}{2} - \frac{\gamma}{2}$                             |
| $H$                | $H_n = \log(1/\tau_n)$ , $H = \log(1/t)$ ; $H(u)$ in Section 30; $H_k$ harmonic numbers in Section 33                                                                               | $H_\pm$ envelopes; $H_2, H_3, H_{22}$ matched sums                        |
| $K, \mathcal{K}$   | $K_{\text{crit}}$ the window kernel; $\mathcal{K}$ the occupancy kernel (29.4); $K$ an integer cutoff in Section 33                                                                 | the constants $K$ of the two mean expansions                              |
| $A, B$             | $A$ the error exponent of Lemma 28.2; $A_J, B_J$ row sums; $A_\alpha(n), B_\alpha(n)$ total sequences, $A_0 = a(n)$ of Part I, $A_1 = a_n$ of Part II; $B$ a constant in that lemma | $A = \pi^2/6$ ; $B = 2\sqrt{A}$ or $B(z)$                                 |
| $G, \mathcal{G}$   | $G$ a standard Gumbel variable; $\mathcal{G}(\theta)$ the window field                                                                                                              | $G(t)$ (Part I); $G = F^2 - 2F/y^2$ (Part II)                             |

| Symbol                 | Part III                                                                                                                                 | Parts I and II                                                                  |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| $E, \mathcal{E}$       | $E_j$ independent exponentials; $\mathcal{E}$ an edge set                                                                                | $E_{2,m}, E(t), E_1$                                                            |
| $N$                    | $N = \sum_j jM_j$ the total size; $N_m, N_{m,J}$ counts of part sizes with multiplicity $m$                                              | $N_m$ the same; $N$ a cutoff (Part I), $N = n - \frac{1}{24}$ (Part II); $N(Y)$ |
| $Q, \mathbb{Q}_t$      | $Q_m, Q_S$ absence probabilities, $Q_m = Q_m(e^{-t})$ of Parts I–II; $\mathbb{Q}_t$ the geometric law                                    | $Q_m(q)$ the product                                                            |
| $\lambda$              | $\lambda_m$ the expected number of part sizes with multiplicity $m$ ; $\lambda(y) = y^{-2}$ in Section 29; $\lambda$ also a partition    | $\lambda$ a partition                                                           |
| $Z, Y$                 | $Z_\alpha$ the series (26.8), $Y_\alpha$ its uncentered form; $Z_m$ absence indicators                                                   | $Z(Y), Z(u), Z_m; Y$ a threshold                                                |
| $\varepsilon, J, L, R$ | $\varepsilon = \sqrt{t}$ (Parts I–II: $h$ ); $J$ the row cutoff (28.9); $L = C_A H^{3/2}$ ; $R$ a block size; $R_t$ the likelihood ratio | $J_s; L, L_m; R(q); R_D$                                                        |
| $T, C, S$              | $T_f$ occupancy tests; $T$ a parameter bound; $S_{\alpha,J}, C_{\alpha,J}$ additive sums and collisions                                  | $T(n), T_n; C, C_N$                                                             |
| $P_S, p$               | $P_S(q)$ forbids every value in $S$ , $P_{\{m\}} = P_m$ ; $p_{jm} = \mathbb{Q}_t(M_j = m)$                                               | $P_m = PQ_m; p(n)$                                                              |

## 26 The statistic and the main results

Write a partition as  $\lambda = (1^{M_1} 2^{M_2} \dots)$ , with  $\sum_{j \geq 1} jM_j = n$ . Its set of positive multiplicity values is

$$\mathcal{M}(\lambda) = \{m \geq 1 : M_j = m \text{ for at least one } j \geq 1\}.$$

For a real parameter  $\alpha \geq 0$ , define

$$D_\alpha(\lambda) = \sum_{m \in \mathcal{M}(\lambda)} m^\alpha. \quad (26.1)$$

The value 0 is excluded, and a multiplicity appearing at several part sizes is counted only once. For example,  $\lambda = (5, 5, 3, 3, 1)$  has  $\mathcal{M} = \{1, 2\}$ , and hence  $D_0 = 2$ ,  $D_1 = 3$ , and  $D_2 = 5$ .

Let  $\mathbb{P}_n$  denote the uniform law on the  $p(n)$  ordinary partitions of  $n$ ; expectations under this law carry subscript  $n$ . Put

$$\tau_n = \frac{\pi}{\sqrt{6n}}, \quad c_* = 1 - \frac{\pi}{4}, \quad M_\alpha = \frac{\Gamma((1-\alpha)/2)}{1+\alpha} \quad (0 \leq \alpha < 1). \quad (26.2)$$

All limits below are as  $n \rightarrow \infty$ . In the proofs,  $t = t_n$  is the exact positive solution of

$$\sum_{j \geq 1} \frac{j}{e^{tj} - 1} = n. \quad (26.3)$$

We prove  $t_n = \tau_n + O(\tau_n^2)$ , so either parameter can be used in the fluctuation statements.

For  $0 \leq \alpha < 1/2$  define

$$\begin{aligned} V_\alpha &= \int_0^\infty y^{2\alpha} (e^{-1/y^2} - e^{-2/y^2}) dy \\ &\quad - 2 \int_0^\infty \int_0^\infty \frac{y^\alpha z^\alpha e^{-1/y^2 - 1/z^2}}{(y+z)^3} dy dz. \end{aligned} \quad (26.4)$$

These integrals are finite and  $V_\alpha > 0$ . A one-dimensional formula is

$$V_\alpha = \Gamma(\tfrac{1}{2} - \alpha) \left\{ \frac{2^{\alpha+1/2} - 1}{2\alpha + 1} - \int_0^1 [r(1-r)]^{1-\alpha} [r^2 + (1-r)^2]^{\alpha-1/2} dr \right\}. \quad (26.5)$$

**Theorem 26.1** (The three fluctuation regimes). *For every fixed  $\alpha \geq 0$  the following statements hold.*

(i) *If  $0 \leq \alpha < 1/2$ , then*

$$\begin{aligned} \text{Var}_n D_\alpha &\sim V_\alpha \tau_n^{-\alpha-1/2}, \\ \tau_n^{\alpha/2+1/4} (D_\alpha - \mathbb{E}_n D_\alpha) &\xrightarrow{d} N(0, V_\alpha). \end{aligned} \quad (26.6)$$

*The same normal limit holds when  $\mathbb{E}_n D_\alpha$  is replaced by  $M_\alpha \tau_n^{-(\alpha+1)/2}$ .*

(ii) *At  $\alpha = 1/2$ ,*

$$\begin{aligned} \text{Var}_n D_{1/2} &\sim \frac{c_*}{2\tau_n} \log \frac{1}{\tau_n}, \\ \sqrt{\frac{\tau_n}{\log(1/\tau_n)}} (D_{1/2} - \mathbb{E}_n D_{1/2}) &\xrightarrow{d} N(0, c_*/2). \end{aligned} \quad (26.7)$$

*Again the mean may be replaced by  $M_{1/2} \tau_n^{-3/4}$ .*

(iii) *If  $\alpha > 1/2$ , let  $E_1, E_2, \dots$  be independent exponential variables of mean 1, and set*

$$Z_\alpha = \sum_{j \geq 1} \frac{E_j^\alpha - \Gamma(\alpha + 1)}{j^\alpha}. \quad (26.8)$$

*The series converges in  $L^2$ , and*

$$\begin{aligned} \tau_n^\alpha (D_\alpha - \mathbb{E}_n D_\alpha) &\xrightarrow{d} Z_\alpha, \\ \tau_n^{2\alpha} \text{Var}_n D_\alpha &\longrightarrow [\Gamma(2\alpha + 1) - \Gamma(\alpha + 1)^2] \zeta(2\alpha). \end{aligned} \quad (26.9)$$

*The limit  $Z_\alpha$  is non-Gaussian. For  $\alpha > 1$  there is also the uncentered limit, with convergence of the first two moments,*

$$\tau_n^\alpha D_\alpha \xrightarrow{d} Y_\alpha := \sum_{j \geq 1} \frac{E_j^\alpha}{j^\alpha}. \quad (26.10)$$

In particular, the order of the standard deviation changes from  $n^{\alpha/4+1/8}$  below  $1/2$ , to  $n^{1/4} \sqrt{\log n}$  at  $1/2$ , to  $n^{\alpha/2}$  above  $1/2$ . The change comes from two different portions of the partition. The subcritical statistic aggregates occupied values  $m \asymp n^{1/4}$  across many part sizes. Above the threshold, the few smallest part sizes, whose multiplicities are of order  $\sqrt{n}$ , control the fluctuation. The critical case accumulates a comparable contribution over logarithmically many intermediate scales.

**Corollary 26.2** (The unweighted open question). *Put*

$$v_0 = 6^{1/4} \left( \sqrt{2} - \frac{3}{4} - \frac{3\sqrt{2}}{8} \log(1 + \sqrt{2}) \right) = 0.308000244403717958 \dots \quad (26.11)$$

Then

$$\mathrm{Var}_n D_0 \sim v_0 n^{1/4}, \quad \frac{D_0 - (6n)^{1/4}}{\sqrt{v_0} n^{1/8}} \xrightarrow{d} N(0, 1).$$

Thus the variance and central limit questions stated in the inspected ProveIt report on A373271 are answered.

[write] **7 October 2026.** The digits of  $v_0$  in (26.11) are rounded in the last place: the truncation is  $v_0 = 0.308000244403717957\dots$  (next digits 652...; recomputed by the write from the closed form, which the package’s `data/03-fluct-checks.json` also records to 48 digits). The abstract’s 0.3080002444... and  $V_0 = 0.348809403800813564\dots$  in (29.14) are truncations. “The inspected ProveIt report on A373271” is Part I, and the questions are its Question 2 (Section 11); a dated note there records the answer. The centering  $(6n)^{1/4}$  is the leading term of Part I’s Theorem 1.1, whose next term  $-\frac{23}{16}$  is  $o(n^{1/8})$ . The write’s exact fixed- $n$  variances, from forbidden-value products, give  $\mathrm{Var}_n D_0/n^{1/4} = 0.2119, 0.2343, 0.2458, 0.2516$  at  $n = 40, 100, 200, 300$ , increasing towards  $v_0$  without deciding it; the corresponding  $\mathrm{Var}_n D_1/n$  are 0.751, 0.828, 0.872, 0.894 (Corollary 26.3: limit 1).

[write] **Independent check, 7 October 2026.** The digits of  $v_0$  and their rounding are confirmed: the closed form at 50 digits is 0.308000244403717957652..., and the package’s `checks.json` records the same digits, so only the display in (26.11) rounds. The check also located the gap between the fixed- $n$  variances above and  $v_0$ . Under the geometric law  $\mathbb{Q}_t$ , computed from the exact one- and two-value absence products (not from the kernel),  $t^{1/2} \mathrm{Var}_t D_0 = 0.348901, 0.349050, 0.349008, 0.348939, 0.348890$  at  $t = 10^{-2}, 3 \cdot 10^{-3}, 10^{-3}, 3 \cdot 10^{-4}, 10^{-4}$ , against  $V_0 = 0.348809\dots$ ; at the exact saddle  $t_n$ ,  $\mathrm{Var}_t D_0/n^{1/4} = 0.3063, 0.3072, 0.3074$  for  $n = 100, 200, 300$ . Subtracting the Gaussian size-conditioning term  $\mathrm{Cov}_t(D_0, N)^2 / \mathrm{Var}_t N$ , also computed exactly, gives 0.2323, 0.2445, 0.2505, against the exact 0.2343, 0.2458, 0.2516. That term is 24%, 20%, 19% of  $\mathrm{Var}_t D_0$  at these  $n$ , and 7.7%, 2.4%, 0.78% at  $n = 10^4, 10^6, 10^8$ , decreasing like  $t_n^{1/2}$ , within the  $L^2$  rate of Proposition 28.4. The same holds for  $D_1$ : 0.834, 0.876, 0.896 against the exact 0.828, 0.872, 0.894, with a term of 0.59% at  $n = 10^6$ . These are floating computations, not proofs; they make the slow approach of the exact variances to  $v_0$  and to 1 quantitative.

**Corollary 26.3** (An explicit Gumbel law). *Let  $G$  have the standard Gumbel distribution  $\mathbb{P}(G \leq x) = \exp(-e^{-x})$ , and let  $\gamma$  denote Euler’s constant. For the sum  $D_1$  of distinct positive multiplicities,*

$$\tau_n D_1 - \frac{1}{2} \log \frac{1}{\tau_n} - \frac{3 - 3\gamma}{2} \xrightarrow{d} G. \quad (26.12)$$

Furthermore,

$$\mathbb{E}_n D_1 = \frac{1}{2\tau_n} \log \frac{1}{\tau_n} + \frac{3 - \gamma}{2\tau_n} + o(\tau_n^{-1}), \quad \mathrm{Var}_n D_1 \sim n. \quad (26.13)$$

[write] **7 October 2026.**  $D_1$  is Part II’s statistic  $W$ , and (26.12) answers Part II’s Question 4 (Section 24). The mean in (26.13) is the first two terms of Part II’s Theorem 12.1: with  $c = \frac{3}{2} - \frac{\gamma}{2} = \frac{3-\gamma}{2}$  there, Part II’s  $(\ell/2 + c)/\tau$  is  $\frac{1}{2\tau} \log \frac{1}{\tau} + \frac{3-\gamma}{2\tau}$ , and its  $\tau = \sqrt{A/(n-1/24)}$  differs from  $\tau_n$  by  $O(\tau_n^3)$ ; Part II’s remainder  $O(\tau^{-1/2})$  is sharper than  $o(\tau_n^{-1})$ . Part III proves the mean without Part II (Section 30, dated note at its end).

**Corollary 26.4** (A quadratic-weight example). *One has*

$$\frac{D_2}{n} \xrightarrow{d} \frac{6}{\pi^2} \sum_{j \geq 1} \frac{E_j^2}{j^2}, \quad \frac{\mathbb{E}_n D_2}{n} \longrightarrow 2, \quad \frac{\mathrm{Var}_n D_2}{n^2} \longrightarrow 8.$$

In particular, the quadratic-weight statistic does not concentrate about its mean at its natural scale.

## 26.1 The moving transition window

The abrupt distinction between fixed exponents has a precise window. For  $\theta \in \mathbb{R}$  put

$$H_n = \log(1/\tau_n), \quad \alpha_n(\theta) = \frac{1}{2} + \frac{\theta}{H_n}.$$

These exponents are positive for all sufficiently large  $n$ .

**Theorem 26.5** (The logarithmic window). *For any fixed finite collection of real parameters  $\theta$ , the variables*

$$X_n(\theta) = \sqrt{\frac{\tau_n}{H_n}} \left( D_{\alpha_n(\theta)} - \mathbb{E}_n D_{\alpha_n(\theta)} \right) \quad (26.14)$$

*converge jointly to centered Gaussian variables  $\mathcal{G}(\theta)$  with covariance*

$$K_{\text{crit}}(\theta, \phi) = c_* \frac{e^{\theta+\phi} - e^{(\theta+\phi)/2}}{\theta + \phi}, \quad (26.15)$$

*interpreted as  $c_*/2$  when  $\theta + \phi = 0$ . Their second moments converge. An equivalent realization of this covariance is*

$$\mathcal{G}(\theta) = \sqrt{c_*} \int_0^{1/2} e^{\theta(1-u)} dB(u), \quad (26.16)$$

*where  $B$  is standard Brownian motion. In particular,*

$$\text{Var}_n D_{\alpha_n(\theta)} \sim \frac{H_n}{\tau_n} W(\theta), \quad W(\theta) = c_* \frac{e^{2\theta} - e^\theta}{2\theta}, \quad W(0) = c_*/2. \quad (26.17)$$

*The centering in (26.14) may be replaced by  $M_{\alpha_n(\theta)} \tau_n^{-(\alpha_n(\theta)+1)/2}$ .*

Only finite-dimensional convergence is asserted in this theorem. No tightness statement in a space of functions of  $\theta$  is used or claimed. The proof is uniform when the finitely many parameters range over a fixed compact interval. Theorem 33.1 further proves that the subcritical Gaussian vector, critical-window Gaussian vector, and fixed-supercritical series vector are mutually independent in their joint limit. In particular the limits for  $D_0$ ,  $D_{1/2}$ , and  $D_1$  are mutually independent.

## 26.2 Proof structure

The main proof has four logically separate tasks. We first localize the occupied multiplicity set and justify conditioning on total size. We then establish a Gaussian covariance field on the scale  $m\sqrt{t}$  by finite-order cumulants. A collision estimate compares weighted occupied values with additive powers of geometric variables. That comparison gives the critical and supercritical laws. Finally, matched integral estimates supply the explicit centerings. Every limiting claim used in this chain is proved below; the earlier report's refined mean expansion is contextual information rather than a required premise.

## 27 Repository review, source boundary, and selection

### 27.1 What was inspected

This work began with a targeted review of the two repositories requested by the user. The pinned snapshots were:

ProveIt: 1210e9d3ac88a573f98d7ef8626ba3d2c30e9749  
 7 October 2026, 16:29:40 UTC.  
 math: adc7f1241b42e322a6451854ab7e4b4c146bf78a  
 6 October 2026, 21:58:50 UTC.

The latter snapshot has one initial commit; its dates in manuscript directory names do not constitute a longer repository commit history. Its README describes 722 manuscripts in 372 families and explicitly distinguishes levels of verification [12].

The review used recent commit descriptions to identify candidates, then read mathematical source for the selected candidates. It was not an audit of every repository theorem or every manuscript. The most useful comparisons were the following.

| Thread                            | Inspected result or claim                                                                                                                                                                                                 | Research decision                                                                                                                      |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| ProveIt:<br>Gowers/Szemerédi      | Recent structural-cover and inverse-induction work retains explicit conditional dimensions; the source ledger reports 113 exact companions and seven open statements.                                                     | Promising, but the remaining structural boundary requires a separate substantial project.                                              |
| ProveIt: Keller maps              | The displayed three-variable map has constant Jacobian $-2$ and explicit integral collisions; the determinant and one collision were checked independently. Written follow-ups treat weighted rigidity and degree growth. | Interesting geometry; an exact second dynamical-degree theorem was not established in this investigation.                              |
| ProveIt: partition multiplicities | The A373271 manuscript proves a refined mean and explicitly asks for variance, a central limit theorem, and weighted occupied-value statistics.                                                                           | Selected: a definite unanswered question and a nontrivial dependence mechanism amenable to a complete proof.                           |
| math, family 159                  | The source states quasipolynomial bounds for fixed-length arithmetic progressions and the reciprocal-sum consequence.                                                                                                     | Its weighted harmonic-tail consequences already appear in the manuscript; they should not be presented as new deductions here.         |
| math, family 132                  | A total-Boolean-function construction gives a superquadratic sensitivity separation, with associated formalization material.                                                                                              | A separate companion note sharpens a certified exponent to $65/32$ , by retaining more of the construction's quantitative information. |

The repository statements in the table are drawn from the pinned sources [10, 12, 13]; the conclusions proved in this article are distinguished from them. No proof in the following sections assumes the large Ramsey or Keller-map claims. The companion sensitivity note is mathematically independent of the partition argument.

[write] **7 October 2026.** Checked against the repository at the pin 1210e9d3a. *The Gowers–Szemerédi row:* the source ledger `Combinatorics/Ramsey/gowers-proof-status.json` then counted 113 exact companions and 7 open statements, as stated; this is stale: since the repair of Lemma 16.10 (commit 456b6e3ad, merged into the main line by c05c2f73e, 7 October 2026) it counts 114 and 6, and so it does at the parent 76eca179c of this write. *The Keller row* agrees with `Algebra/JacobianConjecture/README.md` at the pin (constant Jacobian determinant  $-2$ ; the integral collision  $F(-1, 1, 5) = F(0, -2, -16)$ ). *The partition row* is Part I, as Section 27 quotes it. The two `math` rows and the snapshot facts of the `math` repository were not checked: that repository is outside this one. The sensitivity companion named above is the report

Combinatorics/BooleanFunctions/Research/sensitivity-block-sensitivity. None of these threads is a premise of Part III.

[write] **Independent check, 7 October 2026.** Correction: the parent of the write b0ad11d15 is 2bac1ea07, not 76eca179c, which is an earlier merge and an ancestor of it. The ledger counts 114 and 6 at both, as at 456b6e3ad, c05c2f73e and the current main line, and 113 and 7 at the pin and at 4d60e6063; the Keller lines quoted are lines 22 and 36 of that README at the pin.

## 27.2 The exact ProveIt question answered

The inspected file is

```
SetTheory/Cardinals/docs/reports/
generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a373271-distinct-multiplicity-values/article.tex.
```

It identifies  $\sum_{\lambda \vdash n} D_0(\lambda)$  with OEIS A373271 and proves the three-term mean

$$\mathbb{E}_n D_0 = (6n)^{1/4} - \frac{23}{16} + \frac{\pi}{6^{1/4}} \left( \frac{12365}{82944} + \frac{15}{8\pi^2} \right) n^{-1/4} + O(n^{-1/2}).$$

Its section on further questions asks for a covariance or cumulant analysis and a separate justification of conditioning before stating a variance or central limit theorem. It also proposes replacing the unit weight by general weights [11]. These are precisely the two gaps addressed here.

There is a source-packaging nuance. A placement commit discusses both Reports 197 and 198, and auxiliary files for the latter are present, but the inspected main manuscript contains the unweighted report. We therefore use the actual text of Report 197 for the open-question attribution. The weighted mean formulas needed here are proved afresh, rather than inferred from a placement description. All statements in this article remain valid independently of that packaging issue.

[write] **7 October 2026.** True at the pin and stale now. At 1210e9d3a this `article.tex` was Report 197 alone (blob 77a379cfb), and Report 198 was present only as programs and data with the prefix 198-sum-; the write 0064d11e6 (7 October 2026, 12:43) made Report 198 Part II of this file. The quoted three-term mean is Part I’s Theorem 1.1 with  $K$  of (1.4) ( $K = 0.680582149566835918851\dots$ ; the write recomputed it from the form quoted here), and the questions are Part I’s Questions 2 and 4. Part II proves the five-scale mean of  $D_1$ ; Part III’s weight-one mean (30.2) is therefore a second route to its first two terms (dated notes after Corollary 26.3 and at the end of Section 30). Part III does not use Part II, so its statements stand as written.

## 27.3 Relation to earlier partition literature

Ralaivaosaona’s theorem concerns  $N_m = \#\{j : M_j = m\}$  for one prescribed multiplicity  $m$ . It gives a Gaussian–Poisson–degenerate transition as  $m$  crosses the scale  $n^{1/4}$  [4]. Our statistic sums the indicators  $\mathbf{1}_{\{N_m > 0\}}$  over all  $m$ , so a one-bin limit theorem does not determine its variance: weak pairwise correlations accumulate over quadratically many pairs. The explicit covariance term in (26.4) is a quantitative expression of that obstruction.

The independent geometric model and conditioning strategy belong to the classical probabilistic theory of partitions, notably Fristedt’s work [14]. We prove the particular localization and local limit estimates needed here, including the moment-transfer statement; we do not assume that

convergence of unconditioned characteristic functions alone justifies conditioning on an event of probability tending to zero.

The Gumbel law for the number of summands is classical [15]. The new point here is that deleting repeated *multiplicity values* changes the expectation of  $D_1$  at leading order but changes its centered fluctuation by  $o(\sqrt{n})$ . The resulting centering in (26.12) therefore differs from the classical number-of-summands centering.

Earlier work of Corteel, Pittel, Savage, and Wilf concerns prescribed multiplicities and sampling a part size [6]. Its bibliographic record and abstract were located, but the attempted author PDF returned an access error. This limitation is recorded because a finite source search cannot establish global historical priority. The precise claims made here are complete proofs of the displayed results and progress beyond the inspected repository statements. No assertion that every potentially overlapping paper has been excluded is needed.

## 27.4 Verification status

The infinite limit theorems below are proved in ordinary mathematics. The accompanying program checks exact finite generating functions against independent partition enumeration and evaluates the stated constants. These checks can detect normalization and implementation errors; they are not substitutes for the proofs and are not proof-assistant certificates. No Lean or Rocq formalization of the new partition results is supplied in this package.

# 28 The geometric model and conditioning on total size

## 28.1 Independent geometric coordinates

For  $t > 0$  let  $\mathbb{Q}_t$  be the law of independent random variables  $M_j$  with

$$\mathbb{Q}_t(M_j = m) = (1 - e^{-tj})e^{-tjm}, \quad m = 0, 1, 2, \dots \quad (28.1)$$

Write

$$N = \sum_{j \geq 1} j M_j, \quad p_{jm} = (1 - e^{-tj})e^{-tjm}, \quad N_m = \sum_{j \geq 1} \mathbf{1}_{\{M_j = m\}}, \quad I_m = \mathbf{1}_{\{N_m > 0\}}.$$

Since  $\sum_j \mathbb{Q}_t(M_j > 0) < \infty$ , only finitely many  $M_j$  are positive almost surely. For a partition  $\lambda$  of size  $n$ ,

$$\mathbb{Q}_t(\lambda) = e^{-tn} \prod_{j \geq 1} (1 - e^{-tj}).$$

Consequently the conditional law given  $N = n$  is exactly  $\mathbb{P}_n$ .

We use  $\varepsilon = \sqrt{t}$  throughout. The expected number of occurrences of a multiplicity value has the exact telescoping form

$$\lambda_m = \sum_j p_{jm} = \frac{1}{e^{tm} - 1} - \frac{1}{e^{t(m+1)} - 1} = \int_{tm}^{t(m+1)} g(u) du, \quad g(u) = \frac{e^u}{(e^u - 1)^2}. \quad (28.2)$$

For a finite set  $S$  of positive multiplicities, the absence probability is

$$Q_S = \mathbb{Q}_t(N_m = 0 \text{ for every } m \in S) = \prod_{j \geq 1} \left( 1 - \sum_{m \in S} p_{jm} \right). \quad (28.3)$$

In particular  $Q_m := Q_{\{m\}} \leq e^{-\lambda_m}$ . For  $m \neq \ell$ , each factor satisfies  $1 - p_{jm} - p_{j\ell} \leq (1 - p_{jm})(1 - p_{j\ell})$ , whence

$$\text{Cov}_t(I_m, I_\ell) = Q_{m,\ell} - Q_m Q_\ell \leq 0. \quad (28.4)$$

Only this explicitly proved pairwise fact is needed for the weighted tail variances below.

**Lemma 28.1** (Elementary uniform bounds). *For all  $m \geq 1$  and  $0 < t \leq 1$ ,*

$$\lambda_m \leq \frac{1}{tm(m+1)} \leq \frac{1}{tm^2}, \quad \lambda_m \leq Cte^{-tm} \quad (tm \geq 1). \quad (28.5)$$

*If  $tm \leq 1$ , then  $\lambda_m \geq c/(tm(m+1))$ . For any finite  $J$ ,*

$$\lambda_m^{(J)} := \sum_{j \leq J} p_{jm} \leq \min \left\{ \frac{tJ(J+1)}{2}, \frac{1}{tm^2} \right\}. \quad (28.6)$$

*For each fixed  $r > 0$ ,*

$$\mathbb{E}_t M_j^r \leq C_r (tj)^{-r}. \quad (28.7)$$

*The constants in the last bound may be chosen uniformly for  $r$  in a fixed compact subinterval of  $(0, \infty)$ .*

*Proof.* The identity  $g(u) = [2 \sinh(u/2)]^{-2}$  gives  $g(u) \leq u^{-2}$ ; on  $(0, 2]$  it also gives a positive constant times  $u^{-2}$  as a lower bound. If  $tm \leq 1$  and  $t \leq 1$ , the integration interval lies in  $(0, 2]$ . On  $[1, \infty)$ ,  $g(u) \leq Ce^{-u}$ . Integrating proves (28.5). The estimate  $p_{jm} \leq tj$  proves the first part of (28.6). Finally, on a common probability space we may set

$$M_j = \left\lfloor \frac{E_j}{tj} \right\rfloor, \quad E_j \text{ independent } \text{Exp}(1). \quad (28.8)$$

This has exactly the tail probabilities in (28.1), and  $M_j^r \leq E_j^r / (tj)^r$  proves the moment bound.  $\square$

## 28.2 The entire occupied set is determined by small part sizes

Let  $\mathcal{M}_J = \{M_j : j \leq J, M_j > 0\}$  and  $D_\alpha^{(J)} = \sum_{m \in \mathcal{M}_J} m^\alpha$ .

**Lemma 28.2** (Localization of the occupied set). *For every fixed  $A > 0$  there is a constant  $C_A$  such that, with*

$$J = \left\lceil C_A t^{-1/2} \log^{3/2}(1/t) \right\rceil, \quad (28.9)$$

*one has*

$$\mathbb{Q}_t(\mathcal{M} \neq \mathcal{M}_J) = O_A(t^A). \quad (28.10)$$

*On the complementary event,  $D_\alpha = D_\alpha^{(J)}$  simultaneously for all  $\alpha \geq 0$ . The constant  $C_A$  is allowed to depend on  $A$ .*

*Proof.* Write  $H = \log(1/t)$  and choose constants  $B, C > 0$  below. Put  $m_0 = \lfloor t^{-1/2} / \sqrt{BH} \rfloor$  and  $J = \lceil Ct^{-1/2} H^{3/2} \rceil$ . For all sufficiently small  $t$ , the indices

$$\frac{1}{2tm_0} \leq j \leq \frac{1}{tm_0}$$

lie below  $J$ . On this interval,  $(1 - e^{-tj})e^{-tm_0} \geq ctj$ , with an absolute  $c > 0$ . Summing gives  $\lambda_{m_0}^{(J)} \geq c'/(tm_0^2) \geq c'BH$ . Since  $p_{jm}$  decreases in  $m$ , this lower bound holds for every  $1 \leq m \leq m_0$ . A union bound therefore gives

$$\mathbb{Q}_t(\text{some } m \leq m_0 \text{ is absent from } \mathcal{M}_J) \leq m_0 e^{-c'BH}. \quad (28.11)$$

For the large values, sum the geometric tail exactly:

$$\begin{aligned} \mathbb{Q}_t(\exists j > J : M_j > m_0) &\leq \sum_{j>J} e^{-tj(m_0+1)} \\ &= \frac{e^{-t(J+1)(m_0+1)}}{1 - e^{-t(m_0+1)}} \leq C't^{-1/2} \sqrt{BH} e^{-CH/\sqrt{B}}. \end{aligned} \quad (28.12)$$

First choose  $B$  so  $c'B > A + 2$ , and then choose  $C$  so  $C/\sqrt{B} > A + 2$ . Both displayed bounds are  $O(t^A)$ . Outside these two events, every value supplied by a row  $j > J$  already appears in a row  $j \leq J$ . This proves equality of the occupied sets.  $\square$

This localization concerns the set of multiplicities, not the size of the partition. The rows  $j > J$  carry most of the variance of  $N$  and provide the reservoir that makes conditioning asymptotically harmless.

### 28.3 A uniform local limit estimate

Put

$$A_J = \sum_{j \leq J} jM_j, \quad B_J = \sum_{j > J} jM_j, \quad N = A_J + B_J.$$

For the next lemma it suffices that  $J \geq 1$  and  $Jt \rightarrow 0$ .

**Lemma 28.3** (Reservoir local limit theorem). *Let  $\mu = \mathbb{E}_t N$ ,  $\sigma^2 = \text{Var}_t N$ , and let  $\mu_B, \sigma_B^2$  be the corresponding parameters of  $B_J$ . Then*

$$\begin{aligned} \mu &= \frac{\pi^2}{6t^2} + O(t^{-1}), & \sigma^2 &\sim \frac{\pi^2}{3t^3}, & \sigma_B^2 &\sim \sigma^2, \\ \text{Var}_t A_J &= O(Jt^{-2}), & \mathbb{E}_t(A_J - \mathbb{E}_t A_J)^4 &= O((J^2 + J)t^{-4}). \end{aligned} \quad (28.13)$$

Writing  $\varphi(x) = (2\pi)^{-1/2} e^{-x^2/2}$ , uniformly in integer  $k$ ,

$$\sigma_B \mathbb{Q}_t(B_J = k) = \varphi\left(\frac{k - \mu_B}{\sigma_B}\right) + O(\sqrt{t}). \quad (28.14)$$

The identical estimate holds for  $N$  with its own mean and variance.

*Proof.* The mean estimate follows by a Riemann sum for  $u/(e^u - 1)$ . For the variance, use the integrable function  $u^2 e^{-u}/(1 - e^{-u})^2$  and

$$\int_0^\infty \frac{u^2 e^{-u}}{(1 - e^{-u})^2} du = \sum_{k \geq 1} k \int_0^\infty u^2 e^{-ku} du = 2\zeta(2) = \frac{\pi^2}{3}.$$

Each summand  $jM_j$  has variance  $O(t^{-2})$  and centered fourth moment  $O(t^{-4})$ , by (28.7). Independence gives the two bounds for  $A_J$  and hence  $\sigma^2 - \sigma_B^2 = O(Jt^{-2})$ . The summed absolute third

centered moments of all  $jM_j$  are  $O(t^{-4})$ : for  $tj \leq 1$  use  $O(t^{-3})$  per row, and for  $tj > 1$  use  $O(j^3 e^{-tj})$  and a Riemann sum.

Here are the Fourier estimates needed for a local, rather than merely distributional, limit. For  $q_j = e^{-tj}$ ,

$$\left| \mathbb{E}_t e^{i\theta j M_j} \right|^2 = \frac{(1 - q_j)^2}{(1 - q_j)^2 + 4q_j \sin^2(j\theta/2)}.$$

For  $j \in [1/t, 2/t] \cap \mathbb{N}$  the parameter  $q_j$  stays in a compact subinterval of  $(0, 1)$ . All these rows belong to  $B_J$  eventually. Thus

$$|\mathbb{E}_t e^{i\theta B_J}| \leq \exp \left[ -c \min\{\theta^2/t^3, 1/t\} \right], \quad -\pi \leq \theta \leq \pi. \quad (28.15)$$

For completeness, the needed trigonometric bound is

$$\sum_{1/t \leq j \leq 2/t} \sin^2(j\theta/2) \geq c \min\{\theta^2/t^3, 1/t\}.$$

When  $|\theta| \leq c_0 t$ , apply the elementary lower bound for sine near zero. When  $|\theta| \geq C_0 t$ , use  $|\sum_{a \leq j \leq b} e^{ij\theta}| \leq 1/|\sin(\theta/2)|$  and  $\sin^2(u/2) = (1 - \cos u)/2$ . On the remaining compact range  $c_0 \leq |\theta|/t \leq C_0$ , a Riemann sum for  $\int_1^2 \sin^2(s\theta/(2t)) ds$  is bounded below by a positive constant. This proves (28.15) on the entire circle.

For the centered characteristic function at  $\theta = u/\sigma_B$ , Taylor expansion of its logarithm gives

$$\log \mathbb{E}_t e^{iu(B_J - \mu_B)/\sigma_B} = -u^2/2 + O(\sqrt{t}|u|^3).$$

For an explicit uniform derivative bound, the geometric logarithm series is

$$\log \mathbb{E}_t e^{i\theta B_J} = \sum_{j>J} \sum_{k \geq 1} \frac{e^{-tjk}}{k} (e^{ijk\theta} - 1).$$

Its third derivative in absolute value is at most  $\sum_{j>J} j^3 q_j (1 + q_j)/(1 - q_j)^3 = O(t^{-4})$ , uniformly in real  $\theta$ . On  $|u| \leq c_1 t^{-1/2}$  choose  $c_1$  small enough that the remainder is at most a fixed small multiple of  $u^2$ . The difference from  $e^{-u^2/2}$  is bounded there by  $C\sqrt{t}|u|^3 e^{-c_2 u^2}$ . On the rest of the Fourier interval, (28.15) gives an exponentially small integral, including the range  $|\theta| \geq t$ . Consequently the  $L^1$  distance of the rescaled characteristic function from  $e^{-u^2/2}$  is  $O(\sqrt{t})$ . Fourier inversion proves (28.14), uniformly in  $k$ . The proof for  $N$  is the same.  $\square$

The saddle equation (26.3) has a unique positive solution: its left side is continuous and strictly decreasing from infinity to zero. The mean estimate in (28.13) gives  $t_n = \tau_n + O(\tau_n^2)$  as promised.

**Proposition 28.4** (Likelihood ratio and moment transfer). *Take  $t = t_n$ , and let  $J$  satisfy (28.9). The conditional law of  $(M_j)_{j \leq J}$  given  $N = n$  has likelihood ratio, relative to its unconditional law,*

$$R_t = \frac{\mathbb{Q}_t(B_J = n - A_J)}{\mathbb{Q}_t(N = n)}. \quad (28.16)$$

Uniformly over all low-row configurations,  $0 \leq R_t \leq C$ , and

$$\|R_t - 1\|_{L^2(\mathbb{Q}_t)} = O(Jt + \sqrt{t}) = o(1). \quad (28.17)$$

Thus the total variation distance of the two low-row laws tends to zero. If a sequence of low-row random variables  $Y_t$  has a limiting law and uniformly integrable squares under  $\mathbb{Q}_t$ , its first and second moments and its law have the same limits after conditioning. For any square integrable low-row statistic  $T$ ,

$$|\mathbb{E}_n T - \mathbb{E}_t T| \leq C(Jt + \sqrt{t})\sqrt{\text{Var}_t T}. \quad (28.18)$$

*Proof.* Independence proves (28.16). Apply Lemma 28.3 at the saddle  $\mu = n$ . With  $z = (A_J - \mathbb{E}_t A_J)/\sigma_B$ , it gives

$$R_t = \frac{\sigma}{\sigma_B} \frac{e^{-z^2/2} + O(\sqrt{t})}{1 + O(\sqrt{t})}.$$

The uniform upper bound follows. Moreover  $\sigma/\sigma_B = 1 + O(Jt)$ , and

$$\mathbb{E}_t z^4 = O((J^2 + J)t^2), \quad |1 - e^{-z^2/2}| \leq \min\{z^2/2, 1\}.$$

These prove (28.17). Since  $\mathbb{E}_t R_t = 1$ , the mean difference equals  $\mathbb{E}_t[(T - \mathbb{E}_t T)(R_t - 1)]$ , and Cauchy–Schwarz proves (28.18). Finally, boundedness of  $R_t$  and convergence to 1 in probability allow truncation of any uniformly integrable family  $Y_t^2$ , which proves the moment-transfer assertion. Total variation and the same truncation prove the law and first-moment assertions.  $\square$

## 28.4 Removing the row cutoff after conditioning

The support-localization event involves both low and high rows, so it must not be transferred using a likelihood ratio for low rows alone. Instead, at  $t = t_n$ , Lemma 28.3 gives

$$\mathbb{P}_n(\mathcal{M} \neq \mathcal{M}_J) \leq \frac{\mathbb{Q}_t(\mathcal{M} \neq \mathcal{M}_J)}{\mathbb{Q}_t(N = n)} = O_A(t^{A-3/2}). \quad (28.19)$$

For each fixed  $\alpha \geq 0$ ,

$$0 \leq D_\alpha \leq N^{\max(1, \alpha)}. \quad (28.20)$$

Indeed  $m^\alpha \leq m$  for  $0 \leq \alpha \leq 1$ , whereas for  $\alpha \geq 1$  the sum of the  $\alpha$ th powers is at most the  $\alpha$ th power of the sum. Also  $\mathbb{E}_t N^r = O_r(t^{-2r})$  for every fixed positive  $r$ , by the geometric moment or cumulant bounds (integer moments suffice, followed by Hölder).

Choosing  $A$  sufficiently large for the finitely many exponents and moments under consideration, (28.19), (28.20), and Cauchy–Schwarz show that  $D_\alpha - D_\alpha^{(J)}$  is negligible in  $L^2$  at every scale used in this article, under both  $\mathbb{Q}_t$  and  $\mathbb{P}_n$ . The same holds for any specified finite moment. This explains why the quantifier “for every  $A$ ” in Lemma 28.2 matters.

## 29 A Gaussian field of occupied multiplicities

### 29.1 The covariance on the occupancy scale

On the scale  $y = \varepsilon m$ , a bin has expected occupancy of order one. Uniformly for  $a \leq \varepsilon m \leq b$ , where  $0 < a < b < \infty$  are fixed,

$$\lambda_m = y^{-2} + O(\varepsilon), \quad Q_m = e^{-y^{-2}}(1 + O(\varepsilon)). \quad (29.1)$$

Indeed, (28.2) and  $g(u) = u^{-2} + O(1)$  prove the first estimate. For the second, use  $\max_j p_{jm} = O_a(\varepsilon)$  and  $\sum_j p_{jm}^2 = O_a(\varepsilon)$  in the logarithm of (28.3).

For distinct  $m, \ell$  in this range, put  $z = \varepsilon \ell$ . The Riemann sum with  $x = \varepsilon j$  gives

$$\sum_j p_{jm} p_{j\ell} = \varepsilon \int_0^\infty x^2 e^{-(y+z)x} dx + O(\varepsilon^2) = \frac{2\varepsilon}{(y+z)^3} + O(\varepsilon^2). \quad (29.2)$$

Here all derivatives of the exponentially decaying integrands are uniformly integrable when  $y, z$  stay in the compact interval. Expanding the logarithm of the pair absence probability yields

$$\log \frac{Q_{m,\ell}}{Q_m Q_\ell} = - \sum_j p_{jm} p_{j\ell} + O(\varepsilon^2),$$

because the cubic error is at most a constant times  $\sum_j p_{jm} p_{j\ell} (p_{jm} + p_{j\ell}) = O(\varepsilon^2)$ . Consequently

$$\text{Cov}_t(I_m, I_\ell) = - \frac{2\varepsilon e^{-1/y^2 - 1/z^2}}{(y+z)^3} + O(\varepsilon^2). \quad (29.3)$$

Each correlation is small, but there are order  $\varepsilon^{-2}$  pairs. Their total is of the same order as the sum of the diagonal variances.

For compactly supported, continuous real functions  $f, g$  on  $(0, \infty)$  define

$$\begin{aligned} \mathcal{K}(f, g) &= \int_0^\infty f(y)g(y)(e^{-1/y^2} - e^{-2/y^2}) dy \\ &\quad - 2 \int_0^\infty \int_0^\infty \frac{f(y)g(z)e^{-1/y^2 - 1/z^2}}{(y+z)^3} dy dz. \end{aligned} \quad (29.4)$$

The same formula applies to bounded piecewise continuous functions supported inside a compact subinterval. With  $T_f = \sum_m f(\varepsilon m) I_m$ , the diagonal formula and (29.3) prove

$$\varepsilon \text{Cov}_t(T_f, T_g) \longrightarrow \mathcal{K}(f, g). \quad (29.5)$$

The error  $O(\varepsilon^2)$  per off-diagonal pair contributes only  $O(\varepsilon)$  after this normalization.

## 29.2 A finite-order cumulant lemma

The following argument treats repeated indices as well as distinct ones. This distinction is necessary when summing ordinary cumulants. Let  $Z_m = 1 - I_m$  be the absence indicator.

**Lemma 29.1** (Connected cumulant bound). *Fix  $r \geq 1$  and  $0 < a < b < \infty$ . If an ordered  $r$ -tuple of indices in  $[a/\varepsilon, b/\varepsilon]$  contains  $d$  distinct indices, its joint cumulant of absence indicators, and hence of presence indicators for  $r \geq 2$ , is  $O_{a,b,r}(\varepsilon^{d-1})$ . Consequently, for uniformly bounded weights  $f_m$  supported in this interval,*

$$\kappa_r \left( \sum_m f_m I_m \right) = O_{a,b,r, \sup |f_m|}(\varepsilon^{-1}). \quad (29.6)$$

*Proof.* Label the  $d$  distinct bins by  $[d]$ . In the compact range, for every nonempty subset  $B \subseteq [d]$  of size at most  $r$ ,

$$\sum_j \prod_{i \in B} p_{ji} = O_{a,r}(\varepsilon^{|B|-1}). \quad (29.7)$$

To prove this, set  $x = \varepsilon j$  and use  $p_{ji} \leq \varepsilon x e^{-ax}$ , followed by an exponentially convergent Riemann sum. In particular  $\sum_{i=1}^d p_{ji} \leq 1/2$  for sufficiently small  $\varepsilon$ .

For nonempty  $B \subseteq [d]$  take the Boolean differences

$$c_B = \sum_{S \subseteq B} (-1)^{|B|-|S|} \log Q_S, \quad Q_\emptyset = 1.$$

In the absolutely convergent expansion of  $\log(1 - \sum_{i \in S} p_{ji})$ , this difference kills every monomial that omits an index of  $B$ . The sum of the remaining monomials is bounded by  $C_d \prod_{i \in B} p_{ji}$ : after removing one occurrence of each index, the remaining series is dominated by a geometric series with argument at most  $1/2$ . Thus

$$c_B = O(\varepsilon^{|B|-1}). \quad (29.8)$$

Put  $a_i = Q_{\{i\}}$  and  $w_B = e^{c_B} - 1$  for  $|B| \geq 2$ . Boolean inversion gives

$$Q_S = \prod_{i \in S} a_i \prod_{\substack{B \subseteq S \\ |B| \geq 2}} (1 + w_B).$$

For complex  $s_i$  near zero, set  $z_i = e^{s_i} - 1$ ,  $h_i = 1 + a_i z_i$ , and  $b_i = a_i z_i / h_i$ . Expanding the moment generating function exactly, with  $\mathcal{E} = \{B \subseteq [d] : |B| \geq 2\}$ , gives

$$\mathbb{E}_t e^{\sum_i s_i Z_i} = \prod_i h_i R(w, b), \quad R(w, b) = \sum_{F \subseteq \mathcal{E}} \left( \prod_{B \in F} w_B \right) \left( \prod_{i \in \bigcup F} b_i \right). \quad (29.9)$$

This is a finite polynomial identity before taking the logarithm. The function  $\log R$  is analytic in common small polydisks in  $w, s$ , uniformly for  $0 \leq a_i \leq 1$ , because  $R = 1$  at  $w = 0$ .

Consider a Taylor monomial of  $\log R$  in the edge variables  $w_B$ . If its edge multiset splits into vertex-disjoint components, its coefficient is zero unless there is only one component: setting all other edge variables to zero factorizes  $R$  across those components, and its logarithm becomes a sum. A derivative with positive order in every  $s_i$  also vanishes unless the union of the edges covers  $[d]$ . Every surviving term therefore has a connected spanning hypergraph. If edge  $B$  occurs  $k_B$  times, connectivity implies

$$\sum_B k_B (|B| - 1) \geq d - 1.$$

By (29.8), such a term is  $O(\varepsilon^{d-1})$ . To justify the bound for the entire analytic series, separate it at total  $w$ -degree  $d - 1$ . The tail and each fixed  $s$  derivative of the tail are  $O(\varepsilon^{d-1})$ , since every  $w_B = O(\varepsilon)$ . Below that degree there are finitely many terms, to which the connected-spanning argument applies individually. When  $d = 1$ , every fixed cumulant of a Bernoulli variable is bounded.

Differentiating with possibly repeated  $s_i$  now proves the claimed ordinary joint cumulant bound. There are  $O_r(\varepsilon^{-d})$  ordered  $r$ -tuples with  $d$  distinct indices. Summing over  $d = 1, \dots, r$  proves (29.6); bounded weights do not affect its order.  $\square$

**Theorem 29.2** (Compact occupancy Gaussian field). *For any fixed finite family of bounded piecewise continuous test functions supported inside compact subintervals of  $(0, \infty)$ ,*

$$\sqrt{\varepsilon} \left( \sum_m f(\varepsilon m) I_m - \mathbb{E}_t \sum_m f(\varepsilon m) I_m \right)$$

*converges jointly under  $\mathbb{Q}_t$  to a centered Gaussian vector with covariance  $\mathcal{K}$ . Every fixed moment of each compact test statistic converges. Under  $\mathbb{P}_n$  the same finite-dimensional law and second-moment conclusion hold, with  $t = t_n$ .*

*Proof.* After centering, the first cumulant vanishes, the second converges by (29.5), and for  $r \geq 3$  Lemma 29.1 gives  $O(\varepsilon^{r/2-1}) \rightarrow 0$ . The finite moment–cumulant formulas yield convergence of every fixed moment to the Gaussian moment. Bounded second moments give tightness, and the Gaussian

distribution is determined by its moments. Apply this scalar argument to every linear combination to obtain the joint assertion. The fourth-moment bound gives uniform integrability of squares. Localization and Proposition 28.4 give the conditioned assertion.  $\square$

### 29.3 Strict positivity of the covariance

The negative double integral in (29.4) cannot cancel the whole diagonal. This can be seen without numerical computation. Write  $\lambda(y) = y^{-2}$  and  $h(x, y) = xe^{-xy}$  for  $x, y > 0$ . Then

$$\int_0^\infty h(x, y) dy = 1, \quad \int_0^\infty h(x, y) dx = \lambda(y), \quad \int_0^\infty h(x, y)h(x, z) dx = \frac{2}{(y+z)^3}.$$

For each fixed  $x$ , apply Cauchy–Schwarz with probability density  $h(x, y) dy$ . Integrating in  $x$  gives

$$2 \iint \frac{f(y)f(z)e^{-\lambda(y)-\lambda(z)}}{(y+z)^3} dy dz \leq \int f(y)^2 \lambda(y) e^{-2\lambda(y)} dy.$$

Therefore

$$\mathcal{K}(f, f) \geq \int_0^\infty f(y)^2 e^{-2\lambda(y)} [e^{\lambda(y)} - 1 - \lambda(y)] dy > 0 \quad (29.10)$$

for every nonzero real test function modulo equality almost everywhere. The same proof applies whenever the integrals converge, including  $f(y) = y^\alpha$ ,  $0 \leq \alpha < 1/2$ .

### 29.4 Removing the multiplicity cutoff

For nonnegative weights  $w_m$  and a set of bins  $S$ , the exact sign (28.4) implies

$$\text{Var}_t \left( \sum_{m \in S} w_m I_m \right) \leq \sum_{m \in S} w_m^2 \min\{Q_m, 1 - Q_m\}. \quad (29.11)$$

Fix  $0 \leq \alpha < 1/2$ . On  $m < a/\varepsilon$ , the lower bound for  $\lambda_m$  in Lemma 28.1 gives

$$\varepsilon^{2\alpha+1} \text{Var}_t \sum_{m < a/\varepsilon} m^\alpha I_m \leq C_\alpha \int_0^{2a} y^{2\alpha} e^{-c/y^2} dy + o(1). \quad (29.12)$$

A direct maximum bound also proves this for sufficiently small  $a$ ; the displayed Riemann-integral version makes its limit zero explicit. For  $m > b/\varepsilon$ , (28.5) gives

$$\varepsilon^{2\alpha+1} \text{Var}_t \sum_{m > b/\varepsilon} m^\alpha I_m \leq \varepsilon^{2\alpha-1} \sum_{m > b/\varepsilon} m^{2\alpha-2} \leq C_\alpha b^{2\alpha-1}. \quad (29.13)$$

The low and high tails have nonpositive covariance. Their centered sum is consequently small in  $L^2$ , after multiplication by  $\varepsilon^{\alpha+1/2}$ , as first  $t \rightarrow 0$  and then  $a \downarrow 0$ ,  $b \uparrow \infty$ .

Apply Theorem 29.2 to  $f(y) = y^\alpha \mathbf{1}_{[a,b]}(y)$ . The two tail bounds extend the Gaussian law to all bins and extend the limiting variance by Cauchy–Schwarz. The resulting constant is  $\mathcal{K}(y^\alpha, y^\alpha) = V_\alpha$ . They also prove uniform integrability of the normalized squares: the compact statistic has bounded fourth moments and the tails are arbitrarily small in  $L^2$ . Proposition 28.4 and the support-localization lemma therefore transfer the full Gaussian and variance statements to uniform partitions.

This proves the centered subcritical assertion of Theorem 26.1. It also proves joint Gaussian limits for any finite list  $0 \leq \alpha_i < 1/2$ , with covariance  $\mathcal{K}(y^{\alpha_i}, y^{\alpha_j})$  under their respective normalizations. Linear combinations are treated using the individual  $L^2$  tail bounds, so no sign assumption on those combinations is needed.

## 29.5 Evaluation and the critical pole

For  $0 \leq \alpha < 1/2$ , integration by parts gives

$$\int_0^\infty y^{2\alpha} (e^{-1/y^2} - e^{-2/y^2}) dy = \Gamma(\tfrac{1}{2} - \alpha) \frac{2^{\alpha+1/2} - 1}{2\alpha + 1}.$$

In the double integral of (26.4), substitute  $y = sr$ ,  $z = s(1 - r)$ . The Jacobian is  $s$ , and integration in  $s$  gives the integral subtracted in (26.5). This proves that formula. At  $\alpha = 0$ ,

$$\int_0^1 \frac{r(1-r)}{\sqrt{r^2 + (1-r)^2}} dr = \frac{3\sqrt{2}}{8} \log(1 + \sqrt{2}) - \frac{1}{4},$$

by  $u = 2r - 1$  and elementary antiderivatives. Hence

$$V_0 = \sqrt{\pi} \left( \sqrt{2} - \frac{3}{4} - \frac{3\sqrt{2}}{8} \log(1 + \sqrt{2}) \right) = 0.348809403800813564 \dots \quad (29.14)$$

Using  $\tau_n^{-1/2} = 6^{1/4} n^{1/4} / \sqrt{\pi}$  gives (26.11).

As  $\alpha \uparrow 1/2$ , the bracket in (26.5) tends to

$$\frac{1}{2} - \int_0^1 \sqrt{r(1-r)} dr = \frac{1}{2} - \frac{\pi}{8} = \frac{c_*}{2}.$$

Since  $\Gamma(1/2 - \alpha) \sim (1/2 - \alpha)^{-1}$ ,

$$V_\alpha \sim \frac{c_*}{1 - 2\alpha}. \quad (29.15)$$

The pole indicates the failure of a uniformly integrable high-bin variance tail at the endpoint. Section 32 resolves that failure directly and yields the logarithmic coefficient  $c_*/2$ .

## 30 Explicit centerings and the logarithmic mean

The centered limit theorems do not require a high-order expansion of the mean. Nevertheless, obtaining a genuinely explicit limit law requires controlling the mean to better than the fluctuation scale. This section does so without importing the earlier report's refined unweighted expansion.

**Proposition 30.1** (Weighted means). *For every fixed  $0 < \alpha < 1$ , under both the geometric law and the uniform partition law at the exact saddle,*

$$\mathbb{E}D_\alpha = M_\alpha t^{-(\alpha+1)/2} + \Gamma(\alpha + 1) \zeta(\alpha) t^{-\alpha} + o(t^{-\alpha}). \quad (30.1)$$

At  $\alpha = 1$ ,

$$\mathbb{E}D_1 = \frac{1}{2t} \log \frac{1}{t} + \frac{3 - \gamma}{2t} + o(t^{-1}). \quad (30.2)$$

For  $\alpha = 0$  the geometric mean is

$$\mathbb{E}_t D_0 = \sqrt{\pi} t^{-1/2} + O(1), \quad (30.3)$$

and the same bounded-error formula holds under  $\mathbb{P}_n$  at  $t = t_n$ . All these statements remain valid with  $t_n$  replaced by  $\tau_n$ . The estimates used for (30.1) are uniform for  $\alpha$  in a compact subinterval of  $(0, 1)$ .

### 30.1 Reduction to one integral

For  $0 < \alpha < 2$ , define

$$F_t(u) = u^\alpha(1 - e^{-tg(u)}), \quad J_\alpha(t) = \int_0^\infty F_t(u) du.$$

We first prove the geometric estimate

$$\mathbb{E}_t D_\alpha = t^{-\alpha-1} J_\alpha(t) + O_\alpha(t^{-\alpha/2}). \quad (30.4)$$

There are three approximations, and their errors must all be summed over the full range of multiplicities.

First, the maximum of  $(1-q)q^m$  on  $0 \leq q \leq 1$  is at most  $1/(m+1) \leq 1/2$ . Expanding  $-\log(1-p)$  gives

$$0 \leq e^{-\lambda_m} - Q_m \leq e^{-\lambda_m} \sum_j p_{jm}^2 \leq e^{-\lambda_m} \lambda_m / m. \quad (30.5)$$

For  $tm \leq 1$ , Lemma 28.1 bounds the weighted sum by

$$Ct^{-1} \sum_m m^{\alpha-3} e^{-c/(tm^2)} = O(t^{-\alpha/2}).$$

For  $tm > 1$ , use  $\lambda_m \leq Cte^{-tm}$  to obtain  $O(t^{1-\alpha})$ , which is no larger when  $\alpha < 2$ .

Second,  $\lambda_m \leq tg(tm)$ , and

$$0 \leq tg(tm) - \lambda_m \leq \begin{cases} C/(tm^3), & tm \leq 1, \\ Ct^2 e^{-tm}, & tm > 1. \end{cases}$$

The mean value theorem for the exponential, with the lower bound for  $\lambda_m$ , yields the same weighted error  $O(t^{-\alpha/2})$ .

Third, the elementary estimates for  $g, g'$  imply

$$\int_0^\infty |F'_t(u)| du = O_\alpha(t^{\alpha/2}). \quad (30.6)$$

For the derivative of  $u^\alpha$ , split at  $u = \sqrt{t}$  and use  $1 - e^{-tg(u)} \leq \min(1, Ct/u^2)$  on  $(0, 1]$ . For the other term use  $Ctu^{\alpha-3}e^{-ct/u^2}$ . Beyond 1, exponential decay gives  $O(t)$ . These estimates prove (30.6) for  $0 < \alpha < 2$ . The bounded-variation Riemann-sum inequality is

$$\left| t \sum_{m \geq 1} F_t(tm) - \int_0^\infty F_t(u) du \right| \leq t \int_0^\infty |F'_t(u)| du.$$

Multiplication by  $t^{-\alpha-1}$  gives the third error in (30.4). Each argument is uniform on compact exponent intervals inside  $(0, 2)$ .

### 30.2 Two terms below the mean threshold

For  $0 < \alpha < 1$ , integration by parts gives

$$\int_0^\infty u^\alpha(1 - e^{-t/u^2}) du = M_\alpha t^{(\alpha+1)/2}.$$

Moreover,  $0 < g(u) < u^{-2}$  and  $u^\alpha(u^{-2} - g(u))$  is integrable at both endpoints. Dominated convergence therefore gives

$$J_\alpha(t) = M_\alpha t^{(\alpha+1)/2} + t \int_0^\infty u^\alpha(g(u) - u^{-2}) du + o(t). \quad (30.7)$$

The regularized Mellin integral equals

$$\int_0^\infty u^\alpha(g(u) - u^{-2}) du = \Gamma(\alpha + 1)\zeta(\alpha). \quad (30.8)$$

Here is a direct convergent-integral justification. Let  $H(u) = (e^u - 1)^{-1} - u^{-1}$ . Boundary terms vanish in integration by parts, so the left side is  $\alpha \int_0^\infty u^{\alpha-1} H(u) du$ . The identity

$$H(u) - 2H(2u) = \frac{1}{e^u + 1}$$

implies that this last Mellin integral, multiplied by  $1 - 2^{1-\alpha}$ , is  $\Gamma(\alpha)\eta(\alpha)$ . This follows by integrating the alternating geometric expansion of the right side; the integrated remainder tends to zero for  $\alpha > 0$ . Using  $\eta(\alpha) = (1 - 2^{1-\alpha})\zeta(\alpha)$  proves (30.8). Substitution in (30.4) proves (30.1) under  $\mathbb{Q}_t$ .

For  $\alpha < 1/2$ , the second term in (30.1) is smaller than the standard deviation because  $\alpha < \alpha/2 + 1/4$ . At  $\alpha = 1/2$  it is smaller by a factor  $\sqrt{\log(1/t)}$ . At  $1/2 < \alpha < 1$  it is exactly on the limiting fluctuation scale and must be retained when an explicit centering is wanted.

### 30.3 The mean at weight one

At  $\alpha = 1$  the integral in (30.8) is no longer convergent at infinity. Split at  $u = 1$  instead. Dominated convergence gives

$$J_1(t) = \int_0^1 u(1 - e^{-t/u^2}) du + tC_{\text{reg}} + o(t), \quad (30.9)$$

where

$$\begin{aligned} C_{\text{reg}} &= \int_0^1 u(g(u) - u^{-2}) du + \int_1^\infty ug(u) du \\ &= \lim_{\delta \downarrow 0} \left\{ \frac{\delta}{e^\delta - 1} - \log(1 - e^{-\delta}) + \log \delta \right\} = 1. \end{aligned}$$

The singular part can be evaluated exactly:

$$\begin{aligned} \int_0^1 u(1 - e^{-t/u^2}) du &= \frac{1 - e^{-t}}{2} + \frac{t}{2} \int_t^\infty \frac{e^{-v}}{v} dv \\ &= \frac{t}{2} \left( \log \frac{1}{t} + 1 - \gamma \right) + O(t^2). \end{aligned}$$

Equations (30.4) and (30.9) prove (30.2) under the geometric law. The mean constant  $(3 - \gamma)/2$  and the standard-Gumbel shift  $(3 - 3\gamma)/2$  in Corollary 26.3 have different roles; confusing them would shift the limiting distribution by  $\gamma$ .

### 30.4 A bounded error for the unweighted mean

Set  $a_m = [tm(m+1)]^{-1}$ . Then  $\lambda_m \leq a_m$ , and the exact telescoping identity gives

$$\sum_{m \geq 1} (a_m - \lambda_m) = \frac{1}{t} - \frac{1}{e^t - 1} = O(1).$$

The exponential function is Lipschitz on the nonnegative half-line. Equation (30.5), summed with  $\alpha = 0$ , contributes only  $O(1)$  by the same small- and large- $tm$  split. Thus

$$\mathbb{E}_t D_0 = \sum_{m \geq 1} (1 - e^{-a_m}) + O(1).$$

The inequalities  $(m+1)^{-2} \leq [m(m+1)]^{-1} \leq m^{-2}$  sandwich this sum between two shifted Riemann sums. Monotonicity gives

$$\sum_{m \geq 1} (1 - e^{-1/(tm^2)}) = t^{-1/2} \int_0^\infty (1 - e^{-1/y^2}) dy + O(1) = \sqrt{\pi} t^{-1/2} + O(1),$$

proving (30.3).

### 30.5 Transfer of these centerings to fixed size

The centered variance estimates proved in Sections 29, 31, and 32 do not use the mean asymptotics. Apply (28.18) to  $D_\alpha^{(J)}$ , and use localization to replace it by  $D_\alpha$ . Since  $Jt = O(\sqrt{t} H^{3/2})$ , the mean shift for fixed  $0 \leq \alpha < 1/2$  is

$$O(t^{1/4-\alpha/2} H^{3/2}) = o(1).$$

At  $\alpha = 1/2$ , and uniformly in the moving window, it is  $O(H^2)$ . For fixed  $\alpha > 1/2$  it is  $O(t^{1/2-\alpha} H^{3/2})$ . Each error is  $o(t^{-\alpha})$  for  $0 < \alpha \leq 1$ , and is  $o(1)$  for  $\alpha = 0$ . This proves all conditioned mean statements. The same bounds and the compact-uniform integral estimates prove the explicit centering claim in Theorem 26.5.

Finally, replacing  $t_n$  by  $\tau_n$  changes the leading mean for  $\alpha < 1$  by  $O(t_n^{(1-\alpha)/2})$ , and changes the weight-one mean by  $O(\log(1/t_n))$ . These are smaller than the stated mean remainders and every relevant fluctuation scale. This completes the proof of Proposition 30.1 and the centering assertions in Theorem 26.1.

**[write] Parts I and II's counterparts, 7 October 2026.** At  $\alpha = 0$  and  $\alpha = 1$  this section proves weaker forms of results of Parts I and II, by another route (Riemann sums for the geometric law and dominated convergence, against matched sector expansions and saddle transfer there): (30.3) is the leading term of Part I's radial expansion, Theorem 5.1 ( $\mathbb{E}_t D = \sqrt{\pi} t^{-1/2} - \frac{23}{16} + \sqrt{\pi} d t^{1/2} + O(|t|)$ ), and (30.2) is the first two terms of Part II's, Proposition 18.1, whose constant  $c = \frac{3}{2} - \frac{\gamma}{2}$  equals  $\frac{3-\gamma}{2}$ . Both are kept: Part III's later statements cite them, and Part III proves them without Parts I and II. For  $0 < \alpha < 1$  and  $\alpha > 1$  the means are new to this report. As  $\alpha \downarrow 0$  the second coefficient of (30.1) tends to  $\zeta(0) = -\frac{1}{2}$ , not to Part I's  $-\frac{23}{16}$ ; there is no conflict, since (30.1) is stated for fixed  $\alpha > 0$  and the remainder  $O(t^{-\alpha/2})$  of (30.4) is of the order of that constant at  $\alpha = 0$ . The write evaluated  $\mathbb{E}_t D_\alpha = \sum_m m^\alpha (1 - Q_m)$  under  $\mathbb{Q}_t$  in binary64 (every row and multiplicity with  $tjm < 60$ ) at  $t = 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}$ :  $(\mathbb{E}_t D_0 - \text{Part I's three terms})/t = -0.0709, -0.0718, -0.0720, -0.0721$ ;  $t^{1/2}(\mathbb{E}_t D_{1/2} - \text{the two terms of (30.1)}) = -0.180, -0.100, -0.055, -0.031$ ;  $t(\mathbb{E}_t D_1 - \text{the two terms of (30.2)}) = -0.0714, -0.0238, -0.0077, -0.0024$ ; and  $t^{-1/2}(\mathbb{E}_t D_1 - \text{Part II's five terms}) = -0.122, -0.126, -0.128, -0.128$ . All four are as the three Parts require; they are observations, not proofs.

## 31 Collision removal and non-Gaussian limits

### 31.1 An Efron–Stein estimate for repeated values

For  $\alpha > 0$  and a finite row set  $j \leq J$ , write

$$S_{\alpha,J} = \sum_{j \leq J} M_j^\alpha, \quad C_{\alpha,J} = S_{\alpha,J} - D_\alpha^{(J)} = \sum_{m \geq 1} m^\alpha (N_{m,J} - 1)_+,$$

where  $N_{m,J} = \sum_{j \leq J} \mathbf{1}_{\{M_j=m\}}$ . Thus  $C$  is the contribution discarded when repeated multiplicity values are counted only once.

**Lemma 31.1** (Collision variance). *For every  $\alpha > 0$  and finite  $J$ ,*

$$\text{Var}_t C_{\alpha,J} \leq 2 \sum_{m \geq 1} m^{2\alpha} \lambda_m^{(J)} \min\{1, \lambda_m^{(J)}\}. \quad (31.1)$$

*For every fixed  $\alpha > 1/2$ , uniformly in  $J$ ,*

$$\text{Var}_t C_{\alpha,J} \leq \begin{cases} O_\alpha(t^{-\alpha-1/2}), & 1/2 < \alpha < 3/2, \\ O(t^{-2} \log(1/t)), & \alpha = 3/2, \\ O_\alpha(t^{1-2\alpha}), & \alpha > 3/2. \end{cases} \quad (31.2)$$

*In every case this is  $o(t^{-2\alpha})$ .*

*Proof.* Replace row  $M_j$  by an independent copy  $M'_j$ . Removing its old positive value  $m$  changes  $C$  only if another row has value  $m$ ; in that case the change has magnitude  $m^\alpha$ . The corresponding statement holds for insertion of the new value. The square of the net change is bounded by twice the sum of the squared removal and insertion changes. Efron–Stein, with its factor  $1/2$ , therefore gives

$$\text{Var}_t C_{\alpha,J} \leq 2 \sum_{j \leq J} \sum_{m \geq 1} m^{2\alpha} p_{jm} \mathbb{Q}_t(\text{another row has value } m).$$

The last probability is at most  $\min(1, \lambda_m^{(J)})$ , proving (31.1). All variables have finite moments, so the variance inequality applies directly; infinite  $J$  may also be obtained by  $L^2$  approximation when needed.

To prove (31.2), split at  $m = t^{-1/2}$  and  $m = t^{-1}$ . On the first range the sum is bounded by

$$Ct^{-1} \sum_{m \leq t^{-1/2}} m^{2\alpha-2} = O_\alpha(t^{-\alpha-1/2}),$$

since  $2\alpha - 2 > -1$ . On the middle range it is at most  $Ct^{-2} \sum m^{2\alpha-4}$ . This gives the three cases in (31.2), with a logarithm when  $2\alpha - 4 = -1$ . Beyond  $t^{-1}$  use the sharper exponential bound in (28.5); the contribution is

$$Ct^2 \sum_{m > t^{-1}} m^{2\alpha} e^{-2tm} = O_\alpha(t^{1-2\alpha}).$$

Comparison with  $t^{-2\alpha}$  proves the final assertion.  $\square$

The estimate concerns a centered error. Its expectation need not be small at the fluctuation scale. The explicit means in the preceding section are what allow that distinction to be handled correctly.

### 31.2 The limiting series

For fixed  $\alpha > 1/2$ , the exponential coupling (28.8) implies, for every fixed  $j$ ,

$$t^\alpha(M_j^\alpha - \mathbb{E}_t M_j^\alpha) \xrightarrow{L^2} j^{-\alpha}(E_j^\alpha - \Gamma(\alpha + 1)).$$

Moreover, independence and (28.7) give

$$\text{Var}_t \left[ t^\alpha \sum_{K < j \leq J} (M_j^\alpha - \mathbb{E}_t M_j^\alpha) \right] \leq C_\alpha \sum_{j > K} j^{-2\alpha} \rightarrow 0 \quad (K \rightarrow \infty). \quad (31.3)$$

The same bound holds for the tail of (26.8). Taking first  $t \rightarrow 0$  and then  $K \rightarrow \infty$  proves

$$t^\alpha(S_{\alpha,J} - \mathbb{E}_t S_{\alpha,J}) \xrightarrow{L^2} Z_\alpha \quad (31.4)$$

on the common coupling, for  $J$  in (28.9).

By Lemma 31.1,

$$t^\alpha(C_{\alpha,J} - \mathbb{E}_t C_{\alpha,J}) \xrightarrow{L^2} 0.$$

Subtracting proves the law and variance limit for  $D_\alpha^{(J)}$  under  $\mathbb{Q}_t$ . It also proves uniform integrability of the normalized squares. Alternatively, the additive sums have uniformly bounded fourth centered moments, by independence and  $\sum_j j^{-2\alpha} < \infty$ , and the collision error tends to zero in  $L^2$ . Proposition 28.4 transfers these statements to the conditioned low-row law, including the small difference between its own mean and the geometric mean. Section 28 removes the cutoff. This proves (26.9).

For a finite list  $\alpha_1, \dots, \alpha_r > 1/2$  the same coupling gives joint convergence to the series built from the *same* exponentials  $E_j$ . Their limiting covariance is

$$\text{Cov}(Z_\alpha, Z_\beta) = [\Gamma(\alpha + \beta + 1) - \Gamma(\alpha + 1)\Gamma(\beta + 1)]\zeta(\alpha + \beta). \quad (31.5)$$

This is generally nonzero; the supercritical limits are not independent of one another.

### 31.3 Why these limits are not Gaussian

Write  $Z_\alpha = E_1^\alpha - \Gamma(\alpha + 1) + R_\alpha$ , with an independent almost surely finite remainder  $R_\alpha$ . Choose  $M$  so that  $p := \mathbb{P}(R_\alpha \geq -M) > 0$ . For all sufficiently large  $x$ ,

$$\mathbb{P}(Z_\alpha > x) \geq p \exp \left[ -(x + M + \Gamma(\alpha + 1))^{1/\alpha} \right]. \quad (31.6)$$

Since  $1/\alpha < 2$ , this lower bound decays more slowly than the right tail of every nondegenerate Gaussian distribution. The variance in (26.9) is positive, so a degenerate Gaussian is also excluded. This proves the non-Gaussian assertion without a uniqueness theorem for independent decompositions.

### 31.4 Uncentered limits for $\alpha > 1$

For  $\alpha > 1$ , the additive sums converge uncentered in  $L^2$ : the centered tails satisfy (31.3) and their means are bounded by  $C_\alpha \sum_{j > K} j^{-\alpha} \rightarrow 0$ . To show that collisions have negligible *mean* in this range, use

$$\mathbb{E}_t(N_{m,J} - 1)_+ \leq \min\{\lambda_m^{(J)}, (\lambda_m^{(J)})^2/2\}.$$

The first bound is immediate; the second follows from  $(k-1)_+ \leq k(k-1)/2$  and independence of rows. Splitting as in Lemma 31.1, now with exponent  $\alpha$  rather than  $2\alpha$ , gives

$$\mathbb{E}_t C_{\alpha,J} \leq \begin{cases} O_\alpha(t^{-(\alpha+1)/2}), & 1 < \alpha < 3, \\ O(t^{-2} \log(1/t)), & \alpha = 3, \\ O_\alpha(t^{1-\alpha}), & \alpha > 3. \end{cases} \quad (31.7)$$

Each is  $o(t^{-\alpha})$ . Along with the variance estimate, this shows  $t^\alpha C_{\alpha,J} \rightarrow 0$  in  $L^2$ . The conditioned likelihood is bounded, so the same negligible uncentered collision error transfers. The additive normalized squares are uniformly integrable, and the cutoff is harmless. Thus (26.10) holds with convergence of its first two moments. In particular,

$$\mathbb{E}_n D_\alpha \sim \Gamma(\alpha+1) \zeta(\alpha) \tau_n^{-\alpha} \quad (\alpha > 1). \quad (31.8)$$

## 32 The critical exponent and its logarithmic window

The fixed-exponent arguments on either side of  $1/2$  cease to be uniform at the boundary. The Gaussian variance integral develops a pole, whereas the variance of the exponential series diverges. A moving exponent resolves the transition directly.

In this section set

$$H = \log(1/t), \quad \alpha_t(\theta) = \frac{1}{2} + \frac{\theta}{H}, \quad L = C_A H^{3/2}, \quad J = \lceil t^{-1/2} L \rceil.$$

For a fixed finite bound  $T$ , all exponents with  $|\theta| \leq T$  lie in  $[2/5, 3/5]$  once  $t$  is sufficiently small. The constant  $C_A$  is chosen as in Lemma 28.2, with  $A$  large enough for all the moment errors used below. No assertion is made that one fixed  $C_A$  works for every error power  $A$ .

### 32.1 A uniform bound for the collision fluctuation

The global estimate in Lemma 31.1 is insufficient at  $\alpha = 1/2$ : it would retain the full logarithmic variance. The additional cutoff bound in (28.6) is decisive.

**Lemma 32.1** (Critical collision error). *Uniformly for  $|\theta| \leq T$ ,*

$$\text{Var}_t C_{\alpha_t(\theta),J} = O_T(t^{-1}(1 + \log L)) = o(H/t). \quad (32.1)$$

*Proof.* First take  $\alpha = 1/2$  and put  $M = t^{-1/2}$ . In (31.1), split the sum at  $M/L$ ,  $M$ , and  $1/t$ . On the first range,  $\lambda_m^{(J)} \leq CL^2$ , so its contribution is

$$O\left(L^2 \sum_{m \leq M/L} m\right) = O(t^{-1}).$$

On the second range, use  $\lambda_m^{(J)} \leq 1/(tm^2)$  and  $\min(1, \lambda_m^{(J)}) \leq 1$  to obtain  $O(t^{-1} \log L)$ . On the third range, use the squared intensity bound to obtain

$$O\left(t^{-2} \sum_{m > M} m^{-3}\right) = O(t^{-1}).$$

Finally, the exponential range  $m > 1/t$  contributes  $O(t^2 \sum m e^{-2tm}) = O(1)$ .

For  $|\theta| \leq T$  and  $m \leq 1/t$ , the moving weight satisfies

$$m^{2\alpha_t(\theta)} = m \exp\left(\frac{2\theta \log m}{H}\right) \leq e^{2T} m.$$

Thus all preceding bounds remain uniform. Beyond  $1/t$ , the bound is

$$C_T t^2 \sum_{m>1/t} m^{2\alpha_t(\theta)} e^{-2tm} = O_T(t^{1-2\alpha_t(\theta)}) = O_T(1).$$

Since  $\log L = O(\log H) = o(H)$ , the result follows.  $\square$

The expectation of  $C_{\alpha,J}$  has not been discarded. Only its centered contribution is negligible; this is why the limit theorem first uses the exact expectation before applying the mean expansion.

### 32.2 A triangular array of independent row contributions

For a geometric variable  $M_s = \lfloor E/s \rfloor$ , with  $E \sim \text{Exp}(1)$ , concavity of  $x^\alpha$  gives, for  $0 < a \leq \alpha \leq b < 1$  and  $s \leq 1$ ,

$$|(sM_s)^\alpha - E^\alpha| \leq s^\alpha \leq s^a. \quad (32.2)$$

Exponential moments of fixed polynomial orders are uniformly bounded on compact exponent intervals. Consequently, uniformly for  $\alpha, \beta \in [a, b]$ ,

$$\begin{aligned} \text{Cov}(M_s^\alpha, M_s^\beta) &= s^{-\alpha-\beta} (c_{\alpha,\beta} + O(s^a)), \\ c_{\alpha,\beta} &= \Gamma(\alpha + \beta + 1) - \Gamma(\alpha + 1)\Gamma(\beta + 1), \end{aligned} \quad (32.3)$$

and

$$\mathbb{E}|M_s^\alpha - \mathbb{E}M_s^\alpha|^4 \leq C_{a,b} s^{-4\alpha}. \quad (32.4)$$

In particular  $c_{1/2,1/2} = 1 - \pi/4 = c_*$ .

**Lemma 32.2** (Additive critical-window limit). *For each fixed finite list of parameters, the centered variables*

$$\sqrt{t/H} (S_{\alpha_t(\theta),J} - \mathbb{E}_t S_{\alpha_t(\theta),J})$$

*converge jointly to a centered Gaussian vector with covariance  $K_{\text{crit}}$  in (26.15). Their covariances converge, and their fourth moments are uniformly bounded. The covariance estimates are uniform on compact parameter sets and the conclusions remain valid for parameter sequences tending to fixed values.*

*Proof.* Put  $s_0 = \theta + \phi$  and  $\delta = s_0/H$ . Since  $tJ \rightarrow 0$ , independence and (32.3) give

$$\text{Cov}_t(S_{\alpha_t(\theta),J}, S_{\alpha_t(\phi),J}) = (c_* + o(1)) t^{-1-\delta} \sum_{j \leq J} j^{-1-\delta}. \quad (32.5)$$

The row-approximation error relative to the sum is  $O((tJ)^{2/5}) = o(1)$ , uniformly for bounded parameters. Integral comparison, together with  $\log J = H/2 + \log L + o(1)$ , gives

$$\frac{1}{H} \sum_{j \leq J} j^{-1-s_0/H} \longrightarrow \int_0^{1/2} e^{-s_0 u} du. \quad (32.6)$$

For bounded  $s_0$ , the sum–integral error is  $O(1/H)$  and the endpoint error is  $O_T(\log L/H)$ . Since  $t^{-\delta} = e^{s_0}$ , equations (32.5)–(32.6) yield

$$\frac{t}{H} \text{Cov}_t(S_{\alpha_t(\theta),J}, S_{\alpha_t(\phi),J}) \longrightarrow c_* \int_0^{1/2} e^{(\theta+\phi)(1-u)} du.$$

This is exactly (26.15).

For the central limit theorem, (32.4) implies

$$\sum_{j \leq J} \mathbb{E}_t |M_j^{\alpha_t(\theta)} - \mathbb{E}_t M_j^{\alpha_t(\theta)}|^4 \leq C_T t^{-4\alpha_t(\theta)} \sum_{j \geq 1} j^{-4\alpha_t(\theta)} = O_T(t^{-2}).$$

Here  $4\alpha_t(\theta) \geq 8/5 > 1$ , uniformly for small  $t$ , and  $t^{-4\alpha_t(\theta)} = t^{-2}e^{4\theta}$ . After division by  $(H/t)^2$ , the fourth-moment sum is  $O_T(H^{-2})$ . The same estimate holds for every fixed linear combination, using  $|\sum_{i=1}^r z_i|^4 \leq r^3 \sum_{i=1}^r |z_i|^4$ . If its limiting variance is positive, Lyapunov’s theorem applies; if it is zero, convergence to zero holds in  $L^2$ . The Cramér–Wold criterion gives the joint Gaussian limit.

For independent centered row contributions  $U_j$ , the fourth-moment identity

$$\mathbb{E} \left( \sum_j U_j \right)^4 = \sum_j \mathbb{E} U_j^4 + 6 \sum_{i < j} \mathbb{E} U_i^2 \mathbb{E} U_j^2$$

and the covariance bound show that every normalized fourth moment is bounded. All estimates were compact-uniform, which also proves the claim for converging parameter sequences.  $\square$

### 32.3 Returning to occupied values and uniform partitions

*Proof of Theorem 26.5 and the critical case.* Under  $\mathbb{Q}_t$  the identity

$$D_{\alpha,J} - \mathbb{E}_t D_{\alpha,J} = S_{\alpha,J} - \mathbb{E}_t S_{\alpha,J} - (C_{\alpha,J} - \mathbb{E}_t C_{\alpha,J})$$

holds, where  $D_{\alpha,J}$  here denotes  $D_{\alpha}^{(J)}$ . Lemma 32.1 makes the last term negligible in  $L^2$  after multiplication by  $\sqrt{t/H}$ . The additive lemma therefore proves the joint geometric limits and the convergence of covariances. Localization removes the row cutoff with negligible first and second moment errors, as established in Section 28.

For the uniform law, first retain the cutoff. The normalized additive terms have bounded fourth moments, so Proposition 28.4 transfers their laws and first two moments. Its bounded-likelihood consequence also transfers the centered collision errors, whose  $L^2$  norms tend to zero. In particular the normalized conditional mean differs from the normalized geometric mean by  $o(1)$ , so the centering can be changed to the conditional expectation. The rare-event bound (28.19) then removes the cutoff under conditioning as well. It is applied separately because the support mismatch event depends on both the initial block and the reservoir.

These arguments use the exact saddle  $t = t_n$ . To recover the statement with  $\tau_n$ , observe the exact identity

$$\frac{1}{2} + \frac{\theta}{\log(1/\tau_n)} = \frac{1}{2} + \frac{\theta_t}{\log(1/t_n)}, \quad \theta_t = \theta \frac{\log(1/t_n)}{\log(1/\tau_n)} \longrightarrow \theta.$$

The compact-uniform estimates just proved apply to  $\theta_t$ , while  $t_n/\tau_n \rightarrow 1$ . Continuity of the covariance kernel gives the stated normalization and exponent without requiring a derivative estimate for  $D_{\alpha}$ . Proposition 30.1 supplies the explicit leading-mean centering, uniformly in this window.

Finally, the deterministic Brownian integrands in (26.16) have covariance  $c_* \int_0^{1/2} e^{(\theta+\phi)(1-u)} du$  by the Itô isometry. This proves that representation for the finite-dimensional limits. Taking  $\theta = 0$  gives (26.7). No tightness result for the prelimit processes is invoked.  $\square$

The variable  $u = \log j/H$  in (32.6) identifies the scales responsible for the critical fluctuation. Part sizes from order one up to order  $t^{-1/2}$  occupy an interval of length  $1/2$  in this logarithmic coordinate. Their independent contributions accumulate to the Brownian integral. The additional factor  $L$  needed to localize the entire occupied set changes that interval only by  $o(1)$ .

### 33 Explicit consequences and separation of fluctuation sources

#### 33.1 The Gumbel distribution at weight one

The series  $Z_1$  has a classical closed distribution. For real  $u$ , independence and  $L^2$  convergence give

$$\mathbb{E} e^{iuZ_1} = \prod_{j \geq 1} \frac{e^{-iu/j}}{1 - iu/j} = e^{-i\gamma u} \Gamma(1 - iu). \quad (33.1)$$

The last equality is Euler's convergent product for the gamma function; it follows as well by taking the limit in

$$\prod_{j=1}^k \frac{e^{-iu/j}}{1 - iu/j} = e^{-iuH_k} \frac{\Gamma(k+1)\Gamma(1-iu)}{\Gamma(k+1-iu)}, \quad H_k - \log k \rightarrow \gamma.$$

The standard Gumbel density is  $e^{-x} \exp(-e^{-x})$ . Substituting  $v = e^{-x}$  in its Fourier integral yields the characteristic function  $\Gamma(1 - iu)$ . Thus  $Z_1$  has the law  $G - \gamma$ . Proposition 30.1 then gives the location in (26.12) exactly.

Also,  $\text{Var } Z_1 = \zeta(2) = \pi^2/6$ , so

$$\text{Var}_n D_1 \sim \frac{\pi^2}{6} \tau_n^{-2} = n.$$

This proves every assertion of Corollary 26.3, including the variance, which does not follow merely from weak convergence. The statistic  $\sum_j M_j$ , counting all parts, is much larger in mean than  $D_1$ . The collision argument explains why removing repeated multiplicity values nonetheless leaves the same centered limiting random series.

#### 33.2 Explicit centerings above the Gaussian range

For  $1/2 < \alpha < 1$ , the two-term mean formula gives the useful form

$$\tau_n^\alpha \left( D_\alpha - M_\alpha \tau_n^{-(\alpha+1)/2} \right) \xrightarrow{d} Z_\alpha + \Gamma(\alpha+1)\zeta(\alpha). \quad (33.2)$$

Thus no unknown expectation is needed to state a limiting distribution in any regime: use the leading mean below and at  $1/2$ , the displayed deterministic shift for  $1/2 < \alpha < 1$ , the Gumbel centering for  $\alpha = 1$ , and the uncentered limit for  $\alpha > 1$ .

At  $\alpha = 2$ ,  $\mathbb{E} E_j^2 = 2$  and  $\text{Var}(E_j^2) = 24 - 4 = 20$ . Therefore

$$\mathbb{E} Y_2 = 2\zeta(2) = \frac{\pi^2}{3}, \quad \text{Var } Y_2 = 20\zeta(4) = \frac{2\pi^4}{9}.$$

Multiplying by  $\tau_n^{-2}/n = 6/\pi^2$  proves Corollary 26.4. More generally, for  $\alpha > 1$ ,

$$\frac{\text{Var}_n D_\alpha}{(\mathbb{E}_n D_\alpha)^2} \longrightarrow \frac{[\Gamma(2\alpha + 1) - \Gamma(\alpha + 1)^2]\zeta(2\alpha)}{\Gamma(\alpha + 1)^2\zeta(\alpha)^2} > 0. \quad (33.3)$$

For every fixed  $0 \leq \alpha \leq 1$ , in contrast, the same ratio tends to zero by the mean and variance formulas already proved. The fluctuation transition at  $1/2$  and the loss of relative concentration above 1 are therefore distinct thresholds.

### 33.3 Mutual independence of the three fluctuation regimes

The separation between the three regimes is stronger than a comparison of variances. Their limiting vectors are mutually independent, although the coordinates within each vector generally remain dependent.

**Theorem 33.1** (Mutual independence across row scales). *Fix any finite lists of exponents and parameters*

$$0 \leq \alpha_1, \dots, \alpha_r < \frac{1}{2}, \quad \theta_1, \dots, \theta_s \in \mathbb{R}, \quad \beta_1, \dots, \beta_v > \frac{1}{2}.$$

*Take the normalized centered subcritical variables from Theorem 26.1, the moving-window variables from Theorem 26.5, and the normalized centered fixed supercritical variables from Theorem 26.1. They converge jointly. Their three limiting vectors are mutually independent:*

1. *the subcritical Gaussian vector with covariance  $\mathcal{K}(y^{\alpha_i}, y^{\alpha_j})$ ;*
2. *the critical-window Gaussian vector with covariance  $K_{\text{crit}}(\theta_i, \theta_j)$ ;*
3. *the vector  $(Z_{\beta_1}, \dots, Z_{\beta_v})$  built from the same exponential sequence within that vector.*

*The assertions hold under both  $\mathbb{Q}_t$  and  $\mathbb{P}_n$ , with their respective expectations and the normalizations already specified. All cross-covariances between the three normalized vectors converge to zero.*

*The same conclusion holds if the first vector consists instead of a finite family of normalized bounded piecewise continuous compactly supported occupancy tests on  $(0, \infty)$ . These are finite-dimensional assertions; no convergence or tightness in a function space is claimed.*

We first prove a row-deletion estimate for bounded signed weights. It is essential to center the deletion error: its expectation need not be small.

For a finite row set  $A \subseteq [J]$ , write

$$I_m^A = \mathbf{1}_{\{\exists j \in A: M_j = m\}}, \quad T_f^A = \sum_{m \geq 1} f(\varepsilon m) I_m^A.$$

Here  $\varepsilon = \sqrt{t}$ . For the next lemma,  $f$  is any bounded real function supported in a fixed compact interval  $[a, b] \subset (0, \infty)$ ; no continuity or sign assumption is required.

**Lemma 33.2** (Deleting the smallest growing block of rows). *Let  $0 < \eta \leq 1$ , put  $R = \lfloor \eta/\varepsilon \rfloor$ , and assume  $1 \leq R < J$ . Set  $A = [R]$  and  $B = \{R + 1, \dots, J\}$ . There is a constant  $C_f$ , depending only on  $a, b$  and  $\sup |f|$ , such that*

$$\text{Var}_t(T_f^{[J]} - T_f^B) \leq C_f \eta^2 \varepsilon^{-1}. \quad (33.4)$$

In particular, the centered normalized statistics satisfy

$$\left\| \sqrt{\varepsilon} \left[ T_f^{[J]} - \mathbb{E}_t T_f^{[J]} - (T_f^B - \mathbb{E}_t T_f^B) \right] \right\|_2 \leq C_f^{1/2} \eta. \quad (33.5)$$

The constants are uniform in  $J$ .

*Proof.* Put  $f_m = f(\varepsilon m)$  and  $\mathcal{B}_f = \{m \geq 1 : a \leq \varepsilon m \leq b\}$ . The exact difference is

$$\Delta_f := T_f^{[J]} - T_f^B = \sum_{m \in \mathcal{B}_f} f_m I_m^A (1 - I_m^B).$$

This identity remains valid for signed  $f$ . There are  $O_{a,b}(\varepsilon^{-1})$  relevant bins, and

$$\lambda_m^A := \sum_{j \leq R} p_{jm} \leq \frac{tR(R+1)}{2} \leq \eta^2, \quad (33.6)$$

$$\lambda_m^B := \sum_{R < j \leq J} p_{jm} \leq \lambda_m \leq \frac{1}{tm^2} \leq a^{-2} \quad (m \in \mathcal{B}_f). \quad (33.7)$$

The bound in (33.6) uses  $R \geq 1$ , so  $R+1 \leq 2R$ , and  $\varepsilon R \leq \eta$ .

Apply Efron–Stein to  $\Delta_f$ , resampling one independent row at a time. Suppose first that  $j \in A$ . Resampling  $M_j$  can change  $I_m^A$  only at its old and new values. The factor  $1 - I_m^B$  is unchanged and is at most one. Thus the squared change is bounded by twice  $\sup |f|^2$  times the sum of the indicators that the old and new values belong to  $\mathcal{B}_f$ . Summing the resulting Efron–Stein contributions over  $j \in A$  gives at most

$$C \sup |f|^2 \sum_{j \leq R} \sum_{m \in \mathcal{B}_f} p_{jm} = C \sup |f|^2 \sum_{m \in \mathcal{B}_f} \lambda_m^A \leq C_f \eta^2 \varepsilon^{-1}.$$

Now suppose that  $j \in B$ . A changed factor  $1 - I_m^B$  matters only if  $I_m^A = 1$ . The old and new values of row  $j$  are independent of all rows in  $A$ . The analogous squared-change estimate therefore gives a total contribution at most

$$C \sup |f|^2 \sum_{j \in B} \sum_{m \in \mathcal{B}_f} p_{jm} \mathbb{Q}_t(I_m^A = 1) \leq C \sup |f|^2 \sum_{m \in \mathcal{B}_f} \lambda_m^B \lambda_m^A \leq C_f \eta^2 \varepsilon^{-1},$$

by (33.6)–(33.7). We have only enlarged the estimate by dropping any condition on the other retained rows. Hence it is valid regardless of repeated values among those rows. Adding the two row-block contributions proves (33.4). Centering and multiplying by  $\sqrt{\varepsilon}$  proves (33.5).  $\square$

*Proof of Theorem 33.1.* We first work under  $\mathbb{Q}_t$  with bounded piecewise continuous compactly supported tests  $f_1, \dots, f_r$  in the first vector. Use the same row cutoff

$$J = \lceil t^{-1/2} L \rceil, \quad L = C_A H^{3/2}, \quad H = \log(1/t),$$

in every occupied statistic. Choose  $A$  large enough for the localization errors in all the finitely many moments and exponents under consideration. Fix an integer  $K \geq 1$  and a number  $0 < \eta \leq 1$ . Put  $R = \lfloor \eta/\varepsilon \rfloor$ . For all sufficiently small  $t$ ,

$$K < R < J.$$

Partition the retained rows into the three disjoint sets

$$[K], \quad \{K+1, \dots, R\}, \quad \{R+1, \dots, J\}. \quad (33.8)$$

The next approximations use these three sets, respectively, for the supercritical, critical-window, and compact occupancy vectors.

**The compact occupancy vector.** Replace its coordinate  $\sqrt{\varepsilon}(T_{f_i}^{[J]} - \mathbb{E}_t T_{f_i}^{[J]})$  by

$$\sqrt{\varepsilon} \left( T_{f_i}^{\{R+1, \dots, J\}} - \mathbb{E}_t T_{f_i}^{\{R+1, \dots, J\}} \right).$$

Lemma 33.2 bounds each error in  $L^2$  by  $C_{f_i}^{1/2} \eta$ . This estimate does not require the deleted statistic itself to have a limit for a fixed  $\eta$ .

**The critical-window vector.** For  $\alpha_t(\theta) = 1/2 + \theta/H$ , first replace the occupied statistic by its centered additive approximation, as proved in Section 32. Next retain only the middle rows:

$$\sqrt{\frac{t}{H}} \sum_{K < j \leq R} \left( M_j^{\alpha_t(\theta)} - \mathbb{E}_t M_j^{\alpha_t(\theta)} \right).$$

For parameters in a fixed compact interval, the variance of the discarded first  $K$  centered rows is, after normalization,

$$\frac{t}{H} \text{Var}_t \sum_{j \leq K} M_j^{\alpha_t(\theta)} \leq \frac{C_T}{H} \sum_{j \leq K} j^{-1-2\theta/H} = O_T((1 + \log K)/H). \quad (33.9)$$

For the discarded rows above  $R$ , use  $j^{-2\theta/H} \leq C_T$  for  $j \leq J$  and small  $t$  to obtain

$$\begin{aligned} \frac{t}{H} \text{Var}_t \sum_{R < j \leq J} M_j^{\alpha_t(\theta)} &\leq \frac{C_T}{H} \sum_{R < j \leq J} j^{-1-2\theta/H} \\ &\leq \frac{C_T}{H} (1 + \log(J/R)) = O_T\left(\frac{1 + \log(L/\eta)}{H}\right). \end{aligned} \quad (33.10)$$

Both quantities tend to zero for fixed  $K, \eta$ , since  $\log L = O(\log H)$ . Together with the centered collision bound, these estimates make the middle-row approximation asymptotically exact in  $L^2$ .

**The fixed supercritical vector.** For each fixed  $\beta > 1/2$ , retain the first  $K$  centered additive rows:

$$t^\beta \sum_{j \leq K} (M_j^\beta - \mathbb{E}_t M_j^\beta).$$

The centered collision estimate and (31.3) imply

$$\limsup_{t \downarrow 0} \left\| t^\beta (D_\beta^{(J)} - \mathbb{E}_t D_\beta^{(J)}) - t^\beta \sum_{j \leq K} (M_j^\beta - \mathbb{E}_t M_j^\beta) \right\|_2 \leq C_\beta \left( \sum_{j > K} j^{-2\beta} \right)^{1/2}. \quad (33.11)$$

This bound tends to zero as  $K \rightarrow \infty$ . The retained finite sums converge jointly to the first  $K$  terms of the corresponding exponential series.

**Independence and the order of limits.** For every  $t$  with  $K < R < J$ , the three approximating vectors just defined are independent under  $\mathbb{Q}_t$ : they depend on the three disjoint row sets in (33.8). The marginal limits of the original vectors are already known. To identify their joint limit without assuming any additional limit for the deleted compact statistic, compare characteristic functions directly. For any fixed frequency vectors, the inequality  $|e^{ix} - e^{iy}| \leq |x - y|$  and the preceding  $L^2$

bounds show that the difference between the original joint characteristic function and the product of its original marginal characteristic functions has limiting upper bound

$$C\eta + C \sum_{\ell=1}^v \left( \sum_{j>K} j^{-2\beta_\ell} \right)^{1/2}.$$

One obtains this bound by replacing all three vectors by their independent approximations, then replacing the marginal characteristic functions back; each replacement has the same type of bound.

First let  $t \downarrow 0$  with  $K, \eta$  fixed. Then let  $\eta \downarrow 0$  and  $K \rightarrow \infty$ . The displayed error vanishes. The known marginal characteristic functions therefore multiply in the joint limit. This proves mutual independence for compact occupancy tests.

**Removing the multiplicity cutoffs.** For a subcritical exponent  $0 \leq \alpha < 1/2$ , use the compact test

$$f_{\alpha,a,b}(y) = y^\alpha \mathbf{1}_{[a,b]}(y).$$

The normalized power statistic has the exact form

$$\varepsilon^{\alpha+1/2} (D_\alpha^{(J)} - \mathbb{E}_t D_\alpha^{(J)}) = \sqrt{\varepsilon} \sum_{m \geq 1} (\varepsilon m)^\alpha (I_m^{[J]} - \mathbb{E}_t I_m^{[J]}).$$

The centered low- and high-multiplicity bounds in Section 29 show that replacing this by the compact test statistic has arbitrarily small limiting  $L^2$  error as  $a \downarrow 0$  and  $b \uparrow \infty$ . The same statement with row cutoff  $J$  follows by support localization and its moment bounds; no localization assertion after deleting rows is needed. For a finite list of exponents, apply the individual bounds and the triangle inequality to every linear combination.

More explicitly, fix the compact bin cutoffs first, carry out the limit  $t \downarrow 0$ , then  $\eta \downarrow 0$  and  $K \rightarrow \infty$ , and finally remove the compact bin cutoffs. A second characteristic function approximation gives the claimed product limit for the subcritical power statistics.

**Conditioning and centerings.** All original truncated statistics in this proof are measurable with respect to  $(M_j)_{j \leq J}$ . Proposition 28.4 therefore transfers their entire joint law by total variation. The marginal normalized squares are uniformly integrable by the already proved fluctuation theorems and their moment estimates. Thus changing each geometric centering to its own conditional centering changes the normalized variable by  $o(1)$ . The support-localization event and its separately proved conditional moment bounds then replace the original statistics with cutoff  $J$  by the full statistics.

In particular, the argument never assumes that  $\mathcal{M} = \mathcal{M}_J$  remains true after deleting rows. Deletion is performed inside the common finite block  $[J]$ ; localization is used only for the original, undeleted statistics. The exact-saddle to  $\tau_n$  replacement, including the window reparameterization, is the one already proved in Section 32.

Finally, uniform integrability of the individual normalized squares implies uniform integrability of every pairwise product. For example, split the event  $|XY| > M$  into  $|X| > \sqrt{M}$  or  $|Y| > \sqrt{M}$  and apply Cauchy–Schwarz. Joint convergence therefore transfers all pairwise moments. The limiting cross-covariances between the independent centered vectors are zero, as asserted.  $\square$

For example, the Gaussian limit for  $D_0$ , the critical Gaussian limit for  $D_{1/2}$ , and the standard Gumbel limit for  $D_1$  are mutually independent after the stated deterministic centerings. Within

the supercritical vector, the covariance (31.5) remains strictly positive. Within the two Gaussian vectors, the covariance kernels  $\mathcal{K}$  and  $K_{\text{crit}}$  describe their dependence. The independence theorem concerns the separation between these three families.

### 33.4 Consequences for total sequences

Let  $A_\alpha(n) = \sum_{\lambda \vdash n} D_\alpha(\lambda)$ , and let  $B_\alpha(n) = \sum_{\lambda \vdash n} D_\alpha(\lambda)^2$ . The exact relation to the uniform law is

$$\frac{A_\alpha(n)}{p(n)} = \mathbb{E}_n D_\alpha, \quad \frac{B_\alpha(n)}{p(n)} - \left( \frac{A_\alpha(n)}{p(n)} \right)^2 = \text{Var}_n D_\alpha. \quad (33.12)$$

The unweighted total  $A_0$  is OEIS A373271;  $A_1$  is its linked weight-one counterpart A373273 [1, 2]. Thus, for example,

$$p(n)B_0(n) - A_0(n)^2 \sim v_0 p(n)^2 n^{1/4}, \quad p(n)B_1(n) - A_1(n)^2 \sim p(n)^2 n.$$

These are sequence-level forms of the variance theorems, with the centering cancellation kept explicit. The first equality in (33.12) also converts every weighted mean formula into a total-sequence asymptotic without another analytic transfer theorem.

[write] **7 October 2026.**  $A_0(n)$  is Part I’s  $a(n)$  and  $A_1(n)$  is Part II’s  $a_n$ . The cited OEIS entries are A373271 and A373273 (Remarks 1.4 and 12.4); their names, read again by the write on 7 October 2026 at revisions #15 and #14, are “ $a(n)$  = sum for all integer partitions of  $n$  of the number of distinct multiplicities in each partition.” and “ $a(n)$  = sum of all distinct multiplicities in every integer partition of  $n$ .”; the name quoted in the bibliography for A373271 is the paraphrase that Part I also uses. A373273, which the manuscript could not retrieve, was read directly at the write. Nothing was submitted to the OEIS.

## 34 Exact checks, numerical illustrations, and further research

### 34.1 Independent exact finite calculations

The finite check uses two different descriptions of a partition. The first directly enumerates every partition through  $n = 40$ , computes its multiplicity set, and accumulates the first and second raw moments of  $D_0, D_1, D_2$  and the mixed moment  $D_0 D_1$ . The second computes the same totals by generating functions that forbid one or two multiplicity values. Its state and recurrence do not enumerate partitions or reuse their occupied sets.

Let  $P(q) = \prod_{j \geq 1} (1 - q^j)^{-1}$  and, for a finite set  $S$  of positive integers, define

$$P_S(q) = \prod_{j \geq 1} \left( \frac{1}{1 - q^j} - \sum_{m \in S} q^{jm} \right). \quad (34.1)$$

Each factor has coefficients 0 or 1 and permits precisely the multiplicities outside  $S$ . Therefore  $[q^n]P_S(q)$  counts the partitions of  $n$  in which every value in  $S$  is absent. Inclusion–exclusion gives

$$\begin{aligned} \sum_{n \geq 0} A_\alpha(n) q^n &= \sum_{m \geq 1} m^\alpha (P - P_{\{m\}}), \\ \sum_{n \geq 0} B_\alpha(n) q^n &= \sum_{m \geq 1} m^{2\alpha} (P - P_{\{m\}}) \\ &\quad + 2 \sum_{m < \ell} m^\alpha \ell^\alpha (P - P_{\{m\}} - P_{\{\ell\}} + P_{\{m, \ell\}}). \end{aligned} \quad (34.2)$$

These are formal power-series identities: to compute through degree  $n$ , only  $j, m, \ell \leq n$  can contribute. The analogous mixed-moment identity has off-diagonal coefficient  $m^\alpha \ell^\beta + m^\beta \ell^\alpha$  and diagonal coefficient  $m^{\alpha+\beta}$ . Euler’s pentagonal recurrence independently recomputes  $p(n)$ .

The two methods agree exactly for every  $0 \leq n \leq 40$ , encompassing 215,308 directly enumerated partitions. The first ten listed values of A373271 and its linked A373273 counterpart also agree with the recorded source prefixes [1, 2]. A few exact totals are:

| $n$ | $p(n)$ | $A_0(n)$ | $B_0(n)$ | $A_1(n)$ | $B_1(n)$  |
|-----|--------|----------|----------|----------|-----------|
| 10  | 42     | 70       | 128      | 161      | 861       |
| 20  | 627    | 1,358    | 3,192    | 3,754    | 30,928    |
| 30  | 5,604  | 13,998   | 37,660   | 43,416   | 457,512   |
| 40  | 37,338 | 103,077  | 304,455  | 347,571  | 4,357,827 |

The complete exact CSV includes the quadratic-weight totals and cross moments as well. All comparisons use arbitrary-precision integers; rational means and variances are derived only after the integer identities have been checked.

[write] **7 October 2026.** The write enumerated all 215,308 partitions of  $n \leq 40$  with its own code (multiplicity vectors, not the manuscript’s descending parts): all 41 rows of `data/03-fluct-exact_partition_moments.csv`, in all nine columns, agree, and so do the four rows above. The first ten values agree with the live entries A373271 (revision #15) and A373273 (#14), read on 7 October 2026. The identity (34.1) is Part I’s  $P_m = PQ_m$  (Proposition 2.1) for  $S = \{m\}$ , and  $P_S = P \prod_j (1 - (1 - q^j) \sum_{m \in S} q^{jm})$  in general; with it the write extended the exact moments to  $n = 300$  (note after Corollary 26.2). A rerun of the delivered `verify.py` without simulations, on a copy, reproduced the moments CSV byte for byte; its `checks.json` differs from the shipped one only in the skipped simulation block, the software versions and seven floating values in the last digit.

## 34.2 Constants and the covariance kernel

The program separately evaluates the one-dimensional covariance integral, its unweighted closed form, the regularized Mellin constants, and the Gumbel moments with 55 working decimal digits. In particular, the integral and closed formula for  $V_0$  agree to more than 45 digits. This is high-precision numerical evidence for implementation consistency; the exact equality was proved in Section 29. The computation uses no interval arithmetic and is not described as a directed-rounding certificate.

A separate covariance check uses the compact multiplicity interval  $0.75 \leq \varepsilon m \leq 2.5$ . Finite products for one-bin and two-bin absence probabilities give the following comparison:

| $\varepsilon$ | $\varepsilon \text{Var}_t \sum_{0.75 \leq \varepsilon m \leq 2.5} I_m$ | limiting $\mathcal{K}(\mathbf{1}_{[0.75, 2.5]}, \mathbf{1}_{[0.75, 2.5]})$ |
|---------------|------------------------------------------------------------------------|----------------------------------------------------------------------------|
| 0.0500        | 0.2805135573                                                           | 0.2755896124                                                               |
| 0.0250        | 0.2780882612                                                           | 0.2755896124                                                               |
| 0.0125        | 0.2768478709                                                           | 0.2755896124                                                               |

These finite products use a documented part-size cutoff; the omitted one-bin intensity sums have analytically derived floating-point bounds below  $6.4 \cdot 10^{-14}$  in these three runs. The computations are binary64 diagnostics, not exact evaluations of the infinite products. In particular this check is of a moving *multiplicity interval*; it should not be confused with the moving *exponent* window of Theorem 26.5.

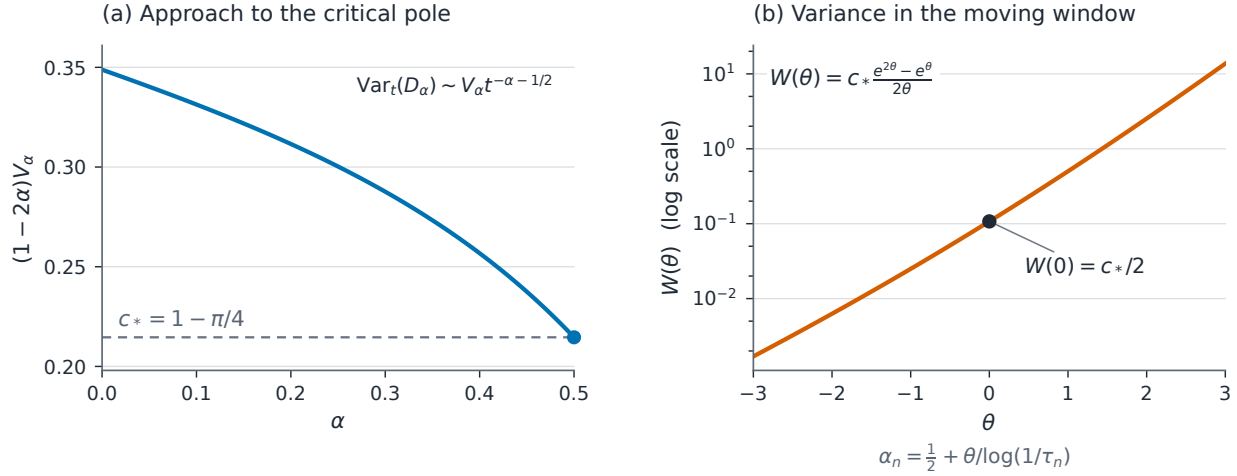


Figure 1: Variance constants across the transition. Left:  $(1 - 2\alpha)V_\alpha$  extends continuously to  $c_* = 1 - \pi/4$  at  $\alpha = 1/2$ . Right: the logarithmic-window variance  $W(\theta)$  on a logarithmic vertical scale, with  $W(0) = c_*/2$  marked. The curves evaluate proved formulas and contain no fitted parameters.

### 34.3 Seeded experiments and slow critical convergence

The package includes 2,000 independent samples at each of  $t = 10^{-4}, 10^{-5}, 10^{-6}$ , using all rows  $j \leq \lceil 40/\sqrt{t} \rceil$ . They are finite-row, unconditioned Boltzmann experiments. They are not uniform partitions of a fixed integer, and omitted rows have not been silently sampled or corrected. The generator is NumPy PCG64 with the three seeds 2026100701, 2026100702, and 2026100703. All 6,000 records, including their normalized statistics, are supplied in the CSV.

For comparison with the limiting laws, the unweighted normalization is  $t^{1/4}(D_0 - \sqrt{\pi} t^{-1/2})$ . For the critical diagnostic the two-term mean in (30.1) is subtracted before multiplying by  $\sqrt{t/H}$ . A deterministic centering does not affect the sample variance. The resulting unbiased sample variances are:

| $t$       | normalized $D_0$ | critical $D_{1/2}$ | Gumbel $D_1$ | $D_2/(\pi^2/(6t^2))$ |
|-----------|------------------|--------------------|--------------|----------------------|
| $10^{-4}$ | 0.347428         | 0.188401           | 1.637908     | 7.915215             |
| $10^{-5}$ | 0.340051         | 0.172240           | 1.662296     | 6.838436             |
| $10^{-6}$ | 0.352854         | 0.168793           | 1.830369     | 9.404014             |
| Limit     | 0.348809         | 0.107301           | 1.644934     | 8                    |

The critical discrepancy is substantial at these values of  $t$  and has been retained. The proof provides only a leading logarithmic asymptotic; the collision bound loses an  $O(\log H/H)$  fraction on that scale, and no explicit onset or finite- $t$  error is claimed. The quadratic statistic has a broad limiting law, so 2,000 samples also leave visible variability in its sample variance. These features are reasons to develop effective estimates, not grounds to alter or select the sample data.

**[write] 7 October 2026.** Recomputed by the write with its own code: the three rows of the covariance table, by the one-bin and two-bin products over all rows with  $tjm < 80$  (all ten printed digits agree, as rounded values); the limit  $\mathcal{K}(\mathbf{1}_{[0.75, 2.5]}, \mathbf{1}_{[0.75, 2.5]}) = 0.2755896124337\dots$  by quadrature of (29.4); and the twelve sample variances from the raw columns  $D_0, D_{1/2}, D_1, D_2$  of `data/03-fluct-boltzmann_samples.csv`

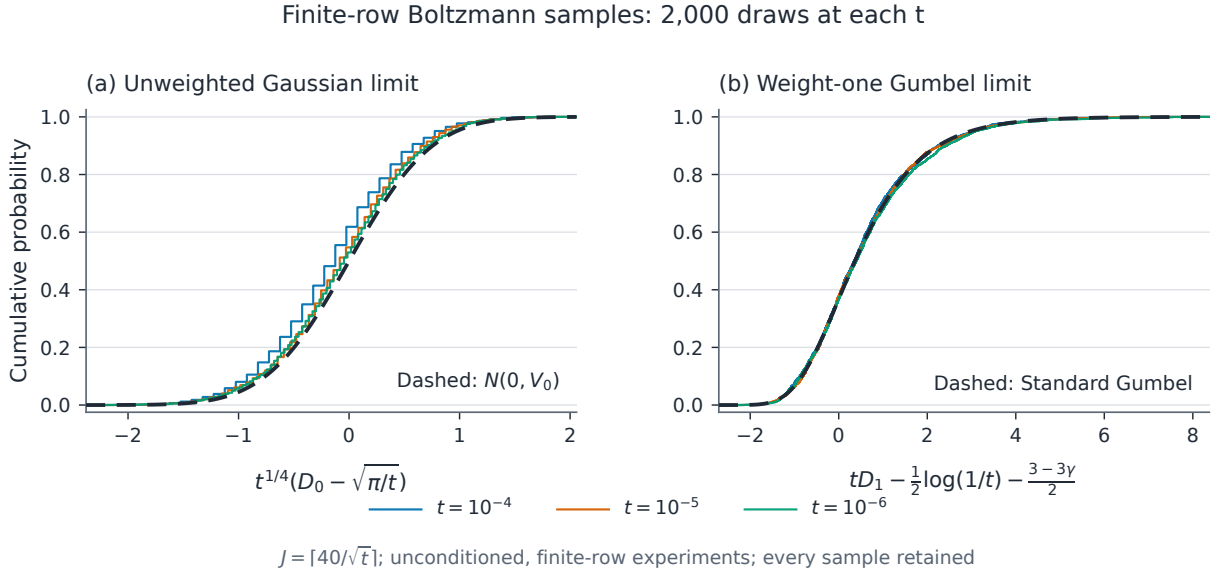


Figure 2: Empirical CDFs for the normalized unweighted and weight-one statistics. Each colored curve uses all 2,000 samples at the indicated  $t$ ; dashed curves are the limiting  $N(0, V_0)$  and standard Gumbel CDFs. The finite-row geometric samples are unconditioned, with  $J = \lceil 40/\sqrt{t} \rceil$ , and no samples were selected for agreement. The unweighted statistic retains its theoretical leading centering rather than an empirical recentering.

with the stated normalizations (all printed digits agree). The “Limit” row is rounded:  $V_0 = 0.3488094\dots$ ,  $c_*/2 = 0.10730091\dots$ ,  $\pi^2/6 = 1.6449340\dots$  and 8.

### 34.4 Reproduction and the boundary of verification

From the article directory, run

```
python3 code/verify.py
python3 code/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The first command regenerates the exact checks, constants, finite-window diagnostics, and sample records. The second creates the vector figure components from those records and the proved formulas. Dependencies and versions are listed in `code/requirements.txt`; the included PDFs can be rebuilt from the TeX and existing figure components without running the simulations. A quicker check that skips the simulations is documented in `code/README.txt`.

[write] **7 October 2026.** The paths in these commands are the delivery’s. Here the programs are `code/03-fluct-verify.py` and `code/03-fluct-make_figures.py`, their data and requirements are under `data/` with the prefix `03-fluct-`, and the figures under `figures/`; the programs resolve `../data` and `../figures` from their own location, so run them on a copy with the delivered names (the README has a recipe). The manuscript’s TeX is Part III of this file. The package builder `code/03-fluct-build.py` builds the delivered directories `partitions/` and `sensitivity/` and does not run here.

Every verification condition uses an explicit exception and remains active under optimized Python. An independent optimized replay gave the same exact and analytic records and a byte-identical

exact CSV. Numerical byte reproduction otherwise requires compatible versions and floating-point behavior. File hashes in the package manifest establish which files were delivered; they do not establish mathematical truth.

The difficult proof steps were separately reviewed: the connected cumulant bound including repeated indices; whole-circle Fourier decay for the growing reservoir; quantitative likelihood and second-moment transfer; weighted mean constants; collision estimates; and the critical-window triangular array. This was an independent derivation and internal review exercise, not external peer review or a formal proof-assistant run. A later formalization should preserve that distinction in its theorem and artifact metadata.

### 34.5 A proposed research programme

The following questions are ordered by their connection to the results proved here. They are open within this article; a literature claim of longstanding global openness is not implied.

1. **Effective Gaussian approximation.** Obtain a quantitative bound in Kolmogorov or Wasserstein distance for the normalized  $D_0$ , with computable constants and an explicit onset. The compact cumulant calculation suggests the natural small parameter  $\sqrt{\varepsilon} = n^{-1/8}$  up to constants, but it does not by itself prove a bound of that order. A successful argument must make the cumulant order dependence explicit, control both multiplicity tails, and retain the conditioning error. Quantitative normal approximation for the negatively dependent occupation indicators may be more efficient than an all-order expansion.
2. **The next variance term and critical finite-size correction.** Determine whether  $\text{Var}_n D_0 - v_0 n^{1/4}$  has a finite limit, or whether a logarithmic or other intermediate correction occurs. Independently, determine the coefficient after  $c_* H/(2t)$  in  $\text{Var} D_{1/2}$ . The present  $O(t^{-1} \log H)$  collision bound may be nonsharp and does not identify such a coefficient. This problem has immediate numerical value because the critical approximation is visibly slow. It requires matching pair occupancies and collision covariances, not only improving the first-moment expansion.
3. **Functional convergence of the critical window.** Strengthen finite-dimensional convergence of  $X_n(\theta)$  to weak convergence in  $C([-T, T])$ , or in a topology that also controls derivatives in  $\theta$ . The proposed limit is the smooth Gaussian field in (26.16). A route is to estimate increments and derivatives, whose weights introduce factors  $\log m/H$ , and obtain uniform control of the collision process. Pointwise collision variance alone is insufficient to claim process tightness. This would permit simultaneous confidence bands over a range of exponents.
4. **Regularly varying weights.** Replace  $m^\alpha$  by  $m^\alpha L(m)$  for a specified slowly varying function  $L$ . Below the threshold the relevant argument of  $L$  should be  $t^{-1/2}$ , whereas above it the smallest rows sample  $t^{-1}$ . At the boundary, a test case is  $w(m) = \sqrt{m}(\log(m+1))^\beta$ . The row calculation suggests a variance scale  $t^{-1} H^{2\beta+1}$  and coefficient

$$c_* \int_0^{1/2} (1-u)^{2\beta} du.$$

This formula is a proposed extension, not a theorem of this paper. The task is to specify regularity conditions that justify the collision estimate, whole-range mean approximation, and conditioning uniformly for these weights.

5. **Largest occupied values and a joint point-process theorem.** Study the random point measure formed by  $tM_j$  for positive multiplicities, with repeated values removed. Fixed rows converge to  $E_j/j$ , and the high-weight moment series suggests a common limiting point measure away from zero. Prove a topology and tightness statement that makes this precise, identify the largest occupied values, and couple this limit with the Gaussian occupancy field. A moment theorem for a few power weights does not automatically imply a point-process theorem near its accumulation point at zero.
6. **Conditioning corrections and universality.** The likelihood estimate proves that total-size conditioning does not change the leading laws. Determine the first correction it contributes to covariance and moderate deviations. Then treat specified restricted partition models, such as colored partitions or a prescribed allowed set of part sizes. The occupancy intensity and reservoir may change; the threshold  $1/2$  should be derived from their scaling instead of assumed universal. Partitions into distinct parts require a different observable because their positive multiplicity set is essentially trivial.
7. **Formalization and exact second-moment computation.** Formalize first the exact finite identities (34.1)–(34.2), the pairwise covariance sign, and the finite-row resampling inequalities. These supply independently reusable lemmas before the analytic limit theory is formalized. On the computational side, develop a faster exact algorithm for  $B_0(n)$  and  $B_1(n)$  so that variance asymptotics can be tested well beyond the enumeration range. The quadratic family of forbidden pairs is a concrete algorithmic obstruction.
8. **Complete the historical comparison.** Inspect the full 1999 multiplicity paper and follow its citations, including statistics of distinct gaps in conjugate Ferrers diagrams. The conjugate partition has successive positive gaps equal to the original multiplicities, so the present observable may appear under that terminology. This bibliographic task changes the appropriate novelty claim and attribution, even though it does not change the proofs. A priority statement should be strengthened only after that comparison has actually been completed.

For immediate continuation, the strongest mathematical targets are an effective error bound for  $D_0$  and the critical variance correction. They address a limitation exposed by the present proof and the retained data. Functional and point-process results would then organize the individual limit laws into a fuller description of the random multiplicity set. The independent sensitivity companion supplies a different research track: optimize the certified exponent, improve the local recurrences, or replace the probabilistic label construction.

[write] **7 October 2026.** Under Vladimir’s standing rule of 4 October 2026 (unproved claims are moved to questions, wrong ones are refuted on the record), the write found no claim of this Part that is wrong, and none that is unproved as stated: the proposed variance formula of Question 4 and the suggested small parameter of Question 1 are already marked as unproved here. The eight questions stay open. Question 8 is the same task as Questions 5 of Parts I and II. Question 7’s faster exact algorithm for  $B_0(n)$  and  $B_1(n)$  is a separate matter from the write’s range extension to  $n = 300$ , which used the quadratic family of pairs as stated.

[write] **The merged bibliography, 7 October 2026.** The two delivered bibliographies are merged, and every detail of the delivered entries is kept; where both Parts cite the same work, the entry is printed once and each Part’s own note on what it inspected is kept, marked “Part I” or “Part II”. The two OEIS entries, both delivered under the key `oeis`, are `oeisA373271` (Part I) and `oeisA373273` (Part II).

[write] **Part III, 7 October 2026.** Part III’s reference list is merged the same way, every detail kept. Its entries for Ralaivaosaona and for Corteel, Pittel, Savage and Wilf are the entries above, with Part III’s own note added; its single entry for A373271 and A373273 (key `oeis`) is cited as the two OEIS entries, its note added to both; its other six entries are new and marked [Part III].

## References

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- [2] O. Gérard, OEIS A373273, *Sum of all distinct multiplicities in every integer partition of  $n$* , OEIS Foundation, 2024. Official maintained record, inspected 4 October 2026: <https://github.com/oeis/oeisdata/blob/main/seq/A373/A373273.seq>. [Part II] Part III: see the previous entry (“direct retrieval of A373273 was unavailable in this source screen”).
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